MATH 1071 Full Course Notes

Cole Durham

2026

Contents

1 Week 1 **4**
1.1 Day 1 . . . 4
1.1.1 Intro to Exponents . . . 4
1.1.2 Exponents, Continued . . . 5
1.2 Day 2 . . . 8
1.2.1 Introduction to Functions . . . 8
1.2.2 Interval Notation and Sets in $\mathbb{R}$ . . . 9
1.3 Day 3 . . . 11
1.3.1 Function Shapes . . . 11

2 Week 2 **13**
2.1 Day 1 . . . 13
2.1.1 Linear Functions . . . 13
2.2 Day 2 . . . 14
2.2.1 End Behavior of Power Functions and Polynomials . . . 15
2.2.2 Polynomials . . . 18
2.2.3 Zeros and Factoring Polynomials . . . 19
2.3 Day 3 . . . 22
2.3.1 Rational Functions . . . 22
2.3.2 Piecewise Functions . . . 22
2.3.3 Models of Cost, Revenue, and Profit . . . 23
2.3.4 Break-Even Analysis . . . 23
2.3.5 Supply and Demand Models . . . 24
2.3.6 Quadratic Models . . . 25

3 Week 3 **27**
3.1 Day 1 . . . 27
3.1.1 Quadratic Models, Continued . . . 27
3.2 Day 2 . . . 28
3.2.1 Exponential Functions . . . 28
3.2.2 Practical Application: Compounded Interest . . . 29
3.2.3 Continuously Compounded Interest . . . 30
3.3 Day 3 . . . 31
3.3.1 An Overview of Inverse Functions . . . 31
3.3.2 Logarithms as Inverses . . . 32

4 Week 4 **35**
4.1 Day 1 . . . 35
4.1.1 Exam 1 Topics . . . 35
4.1.2 Review Problems . . . 36
4.2 Day 2 . . . 39

4.3 Day 3 . . . 39
4.3.1 Intuitive Definition of a Limit . . . 39
4.3.2 Preview of Calc: Average/Instantaneous Rate of Change . . . 41

5 Week 5 **43**
5.1 Day 1 . . . 43
5.1.1 Calculation of Limits . . . 43
5.2 Day 2 . . . 45
5.2.1 Infinite Limits . . . 45
5.2.2 Continuity . . . 47
5.3 Day 3 . . . 47
5.3.1 Limit Mastery . . . 47
5.3.2 Two Facts, a Couple Tricks . . . 48

6 Week 6 **51**
6.1 Day 1 . . . 51
6.1.1 Continuity is Key . . . 51
6.1.2 Derivatives? Derivatives. . . . 52
6.2 Day 2 . . . 53
6.2.1 Examples of Derivatives and Related Computations . . . 53
6.2.2 Derivative as a Function . . . 55
6.2.3 Derivatives and Increasing/Decreasing . . . 55
6.3 Day 3 . . . 56

7 Week 7 **59**
7.1 Day 1 . . . 59
7.2 Day 2 . . . 62
7.2.1 Two More Derivative Rules . . . 62
7.3 Day 3 . . . 65
7.3.1 Listen to the wind blow, watch the sun rise . . . 65
7.3.2 Derivatives as Rates of Change: Marginal Functions . . . 66

8 Week 8 **68**
8.1 Day 1 . . . 68
8.1.1 Sections Covered . . . 68
8.1.2 Example Problems for Chapter 2 . . . 69
8.1.3 Example Problems for Chapter 3 . . . 71
8.2 Day 2 . . . 72
8.3 Day 3 . . . 72
8.3.1 Elasticity of Demand . . . 72

9 Week 9 **74**
9.1 Day 1 . . . 74
9.1.1 Elasticity of Demand, Continued . . . 74
9.1.2 Linear Approximations . . . 75
9.2 Day 2 . . . 76
9.2.1 Maxima and Minima . . . 77
9.3 Day 3 . . . 80

10 Week 10 **82**
10.1 Day 1 . . . 82
10.2 Day 2 . . . 84
10.3 Day 3 . . . 86

11 Week 11 **89**
11.1 Day 1 . . . 89
11.2 Day 2 . . . 91
11.2.1 Limits at Infinity . . . 91
11.2.2 End Behavior of Rational Functions . . . 92
11.3 Day 3 . . . 93

12 Week 12 **95**
12.1 Day 1 . . . 95
12.2 Day 2 . . . 97
12.3 Day 3 . . . 97
12.3.1 Antiderivatives . . . 98

13 Week 13 **100**
13.1 Day 1 . . . 100
13.1.1 Antiderivatives . . . 101
13.2 Day 2 . . . 104
13.3 Day 3 . . . 104

14 Week 14 **105**
14.1 Day 1 . . . 105
14.2 Day 2 . . . 107
14.2.1 Substitution . . . 107
14.3 Day 3 . . . 110
14.3.1 u-Substitution, cont. . . . 110
14.3.2 Review of Class Topics . . . 112

1 Week 1

Outline
Days: Wednesday, January 21; Friday, January 23
Topics: *Exponents, functions, basic set theory*

Album of the Week: *Ella and Louis Again* by *Ella Fitzgerald and Louis Armstrong (1957)*

1.1 Day 1

Goal: Review exponent rules and use them to manipulate/simplify algebraic expressions.

1.1.1 Intro to Exponents

In their simplest form, **exponents** are used as an abbreviation for repeated multiplication. For instance, if we want to multiply 2 by itself 5 times, we can write

$$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2.$$

A more concise way of expressing this would be

$$2^5$$

where the **base**, 2, is the number we are multiplying, and the **exponent**, 5, is the number of times we multiply. To represent an arbitrary exponent we use a variable, e.g. 2^x.

Example 1.1. Calculate the value of 3^b for $b = -2, -1, 0, 1, 2, 3, 4$. Keep in mind that $3 \cdot 3^b = 3^{b+1}$ and $\frac{3^b}{3} = 3^{b-1}$.

Solution 1.1.

b	3^b
4	$3^4 = 3 \cdot 3 \cdot 3 \cdot 3 = 81$
3	$3^3 = 3 \cdot 3 \cdot 3 = 27$
2	$3^2 = 3 \cdot 3 = 9$
1	3
0	$3^0 = 1$
-1	$3^{-1} = \frac{1}{3}$
-2	$3^{-2} = \frac{1}{9}$

This first example illustrates two basic rules.

Exponent Rules, Pt. 1 For any constants a, b, where $a \neq 0$:

1. $a^0 = 1$,
2. $a^{-b} = \frac{1}{a^b}$.

Note here that 0^0 is sometimes defined to be 1, but we will try to avoid this case.

Example 1.2. Construct a table of values showing 4^b for $b = -2, -1, 0, 1, 2, 3$.
Solution 1.2.

b	4^b
3	$4 \cdot 4 \cdot 4 = 64$
2	$4 \cdot 4 = 16$
1	4
0	1
-1	$\frac{1}{4}$
-2	$\frac{1}{4 \cdot 4} = \frac{1}{16}$

1.1.2 Exponents, Continued

We need to be able to handle expressions which contain the multiplication of terms involving exponents. For instance, in the previous section we saw $3^b \cdot 3 = 3^{b+1}$ and $\frac{3^b}{3} = 3^b \cdot 3^{-1} = 3^{b-1}$. For more intuition, consider the expression

$$x^3 \cdot x^4.$$

where x is a variable. We know $x^3 = x \cdot x \cdot x$ and $x^4 = x \cdot x \cdot x \cdot x$, so

$$x^3 \cdot x^4 = (x \cdot x \cdot x) \cdot (x \cdot x \cdot x \cdot x) = x^7 = x^{3+4}.$$

This tells us to **add exponents when multiplying terms with the same base**.

What happens if we raise an exponent to another power, e.g. $(3^3)^2$? Well, we can write

$$(3^3)^2 = (3^3) \cdot (3^3) = (3 \cdot 3 \cdot 3) \cdot (3 \cdot 3 \cdot 3) = 3^6 = 3^{3 \cdot 2}.$$

This tells us to **multiply exponents when raising one exponent to another power**.

Next, what if we raise multiple numbers to a power, e.g. $(xy)^3$? Following the definition and then rearranging, we see

$$(xy)^3 = (xy) \cdot (xy) \cdot (xy) = (x \cdot x \cdot x) \cdot (y \cdot y \cdot y) = x^3 y^3.$$

This tells us to **apply exponents to each factor in multiplication**.

Finally, we need one definition for purposes of notation before we give a second list of exponent rules.
Definition 1.1. An $n^{\textbf{th}}$ **root** of x, denoted $\sqrt[n]{x}$, is a number which satisfies $(\sqrt[n]{x})^n = x$.

Fact: $\sqrt[n]{x} = x^{\frac{1}{n}}$. This is true because the above rule tells us

$$(x^{\frac{1}{n}})^n = x^{\frac{1}{n} \cdot n} = x^1 = x.$$

Exponent Rules, Pt. 2 For all x, y, m, and n such that the given expression is defined,

1. $x^m \cdot x^n = x^{m+n}$
2. $\frac{x^m}{x^n} = x^{m-n}$

3. $(x^m)^n = x^{mn}$
4. $(xy)^n = x^n y^n$
5. $\sqrt[n]{x^m} = x^{\frac{m}{n}}$.

We can double check our understanding in the following examples.

Example 1.3. Determine the numerical value of each expression.

(a) $\frac{4^0}{(5^0)(6^1)}$

(b) $(-1)^4(2^2)^3$

(c) $\frac{(\sqrt[3]{8})}{2}$

Solution 1.3.

(a) $4^0 = 5^0 = 1$, $6^1 = 6$, so
$$\frac{4^0}{(5^0)(6^1)} = \frac{1}{6}.$$

(b) We can expand $(-1)^4 = (-1)(-1)(-1)(-1) = 1 \cdot 1 = 1$, and we can use Exponent Rules Pt. 2 number 3 to see $(2^2)^3 = 2^6 = 64$. Combining,
$$(-1)^4(2^2)^3 = 1 \cdot 64 = 64.$$

(c) Note that $8 = 2^3$, so from the definition of cube root
$$\frac{\sqrt[3]{8}}{2} = \frac{(2^3)^{\frac{1}{3}}}{2} = \frac{2}{2} = 1.$$

Example 1.4.

(a) True or False: For any real numbers x, y, and a, $x^a + y^a = (x+y)^a$. Explain your reasoning. (Hint: Pick a few different values of x, y, see if equality is true.)

(b) True or False: For any real numbers x, y, z, and a, $(xyz)^a = x^a y^a z^a$. Explain your reasoning.

Solution 1.4.

(a) False! Try $x = y = 1$, $a = 3$; on the left side we see
$$1^3 + 1^3 = 2.$$
On the right side we instead have
$$(1+1)^3 = 2^3 = 8$$
which is certainly not equal to 2. (Look at Freshman's Dream. Algebra for snobs!)

(b) True! This follows by extending Exponent Rules Pt. 2, number 4. If x and y are both real, then $w = xy$ is too, so by the aforementioned rule

$$(xyz)^a = (wz)^a = w^a z^a.$$

Then replacing $w = xy$ as before and applying the rule once again

$$w^a z^a = (xy)^a z^a = x^a y^a z^a.$$

1.2 Day 2

Goal: Give a broad introduction to functions, including their properties and representations.

1.2.1 Introduction to Functions

Definition 1.2.

1. A **function** consists of a set of items A called inputs, a set of items B called outputs, and a rule which assigns exactly one output to every input.
2. The set of all inputs A is called the domain and the set of all outputs *attained* by those inputs is the range.
3. The set B may also be referred to as the **codomain**. The difference between codomain and range is that not every element of the codomain need be *attained* by some input. (The range is a subset of the codomain. Many times we will ignore the codomain and simply work with the range.)

Notation: We will *usually* indicate a function by the letter f (the letters g and h are also frequently used), the input or independent variable will be represented by x, and the output of x under f will be written as $f(x)$ or y.

Example 1.5. The most common (and perhaps most useful) way to represent a function is by a **formula** or **equation**. For instance, for $x = -2, -1, 1, 2, 3$, define

$$f(x) = \frac{1}{x}.$$

Express the input-output pairs in a table of values.

Solution 1.5.

x	$f(x)$
3	$\frac{1}{3}$
2	$\frac{1}{2}$
1	1
-1	1
-2	$-\frac{1}{2}$

They can also be depicted in a graph, where the input is labeled on the horizontal axis (the x-axis) and the output is labeled on the vertical axis (the y-axis).

Remarks:

1. When the domain of a function f is not specified, the set of all inputs x such that $f(x)$ "makes sense" or is defined is called the **(natural) domain of** f.
2. A single input x in the domain of a function f can only be mapped to one output! If we encounter a representation for a mapping (e.g. a formula, graph, table) which shows an input x being sent to multiple distinct outputs, we say the function is **not well-defined**.

Example 1.6. True or False: We define a map f which takes as inputs the first names of students in a database and outputs the first letter (e.g. $f(\text{Cole}) = \text{C}$, $f(\text{Amanda}) = \text{A}$). Since $f(\text{Cole}) = f(\text{Catherine})$, f is not well-defined.

Solution 1.6.

False! A function is not well-defined if it sends a *single input* to *multiple outputs*. This function sends *multiple inputs* to a *single output*, which merely means the function is not **injective**. More on this later.

Solution 1.7.

No, the depicted function is not well-defined since it fails the vertical line test. In other words, it sends some individual inputs to multiple outputs.

1.2.2 Interval Notation and Sets in $\mathbb{R}$

We want to be able to refer to different collections or sets of real numbers in a standard way. The real numbers (sometimes denoted by $\mathbb{R}$) are any numbers which are rational (like 0, 1/2, -3/7, 10573829/19473289 for example) or irrational (like π or e to name two). If a and b are real numbers and a is less than b ($a < b$) we have the following notation:

- $(a, b) = \{x : a < x < b\}$ = All real numbers strictly between a and b
- $(a, b] = \{x : a < x \leq b\}$ = All real numbers strictly bigger than a, less than or equal to b
- $[a, b] = \{x : a \leq x \leq b\}$ = All real numbers greater than or equal to a, less than or equal to b
- $[a, b) = \{x : a \leq x < b\}$ = All real numbers greater than or equal to a, strictly less than b
- $(a, \infty) = \{x : a < x\}$ = All real numbers strictly greater than a
- $[a, \infty) = \{x : a \leq x\}$ = All real numbers greater than or equal to a
- $(-\infty, b) = \{x : x < b\}$ = All real numbers strictly less than b
- $(-\infty, b] = \{x : x \leq b\}$ = All real numbers less than or equal to b.
- $\{a\}$ = the single real number a.

One other useful tool when describing sets is the union operation, denoted by $\cup$. When we have two sets, for example $[0, 1]$ and $[5, \infty)$, we sometimes want to 'add the sets together,' i.e. look at the new set containing all elements in either $[0, 1]$ or $[5, \infty)$. This is written as:

$$\{x : 0 \leq x \leq 1 \text{ or } 5 \leq x < \infty\} = [0, 1] \cup [5, \infty).$$

Recall the following concept from Day 2.

Natural Domain: Sometimes we are given a function in equation form, like $f(x) = \sqrt{x}$, with no domain specified. In this case, the standard convention is to define the domain as the set of all real numbers x such that $f(x)$ is a real number (this is sometimes called the **natural domain**, but we'll just refer to it as the domain).

Example 1.7. We want to know what the (natural) domains are of the functions 1. $f(x) = \sqrt{x}$, 2. $f(x) = x^3$, and 3. $f(x) = \frac{1}{x-1}$.

Solution 1.8.

1. For what values of x is $\sqrt{x}$ *not* a real number? If $x < 0$, e.g. $x = -3$, then $\sqrt{-3} = 3i$ is imaginary, and therefore not real. As long as $x \geq 0$, $\sqrt{x}$ is real, so the domain of $f(x) = \sqrt{x}$ is
$$[0, \infty) = \{x : 0 \leq x\}.$$
2. What is the domain of $f(x) = x^3$? Explain.
The domain of f is all real numbers, since any number may be cubed with no issues. Denote this by $\mathbb{R}$ or $(-\infty, \infty)$.
3. Dividing by 0 is not a well-defined operation. In turn, the domain of $f(x) = \frac{1}{x-1}$ must exclude all numbers x such that $x - 1 = 0$; the only such number is $x = 1$. There is no other way for this function to be undefined, so the domain of $f(x) = \frac{1}{x-1}$ is
$$\{x : x < 0 \text{ or } 0 < x\} = (-\infty, 0) \cup (0, \infty).$$

Long story short, when asked to give the (natural) domain of a function $f(x)$, we should describe the set of inputs which are permissible. The *inputs which are not permissible* are ones that ...

- cause us to take the square root (or any even root) of a negative number,
- divide by 0,
- take the natural logarithm of a number which is less than or equal to 0 (more on this once logarithms are introduced).

Definition 1.3. Given a function $y = f(x)$, the **y-intercept** is the point $(0, f(0))$ on the graph assuming 0 is in the domain of f. The **zeros** (or **roots**) of $f(x)$ are the values of x such that $f(x) = 0$.

Example 1.8.

1. Find the y-intercept of $f(x) = x^2 - 5x + 6$.
2. Find the zeros of $f(x) = x^2 - 5x + 6$.

Solution 1.9.

1. According to the definition, the y-intercept is just the point $(0, f(0))$. For this case, $f(0) = 0^2 - 5(0) + 6 = 6$, so the y-intercept is $(0, 6)$. (Visually we can see this is where the graph crosses the y-axis.)
2. Zeros are x's such that $f(x) = 0$. Here, $f(x) = x^2 - 5x + 6 = 0$ is the same (by factoring) as $(x - 2)(x - 3) = 0$. The left side of this equation is 0 exactly when $x = 2$ or $x = 3$, meaning $x = 2, 3$ are the zeros of this function. (Visually we see this as where the graph meets the x-axis.)

1.3 Day 3

Goal: Continue gaining information about functions. Build new functions through various operations (e.g. adding, subtracting, composing, etc.), then identify when a function is increasing/decreasing/concave down/concave up.

Given two functions $f(x)$ and $g(x)$, we can add, subtract, multiply, divide, or compose them. The new resulting function will (in almost all cases) have different properties from both f and g.

Example 1.9. Let $f(x) = x^2 + x - 2$, $g(x) = 2x + 3$. We will calculate
(1) $f(x) + g(x)$, (2) $f(x) - g(x)$, (3) $f(x)g(x)$, (4) $\frac{f(x)}{g(x)}$, and (5) $(f \circ g)(x) = f(g(x))$.

Solution 1.10.

1. Combine 'like terms,' meaning we add the coefficients of terms with the same power of x:
$$f(x) + g(x) = (x^2 + x - 2) + (2x + 3) = x^2 + (x + 2x) + (-2 + 3) = \boxed{x^2 + 3x + 1}.$$

2. Again we combine like terms, but the key with subtraction is to distribute the negative:
$$f(x) - g(x) = (x^2 + x - 2) - (2x + 3) = (x^2 + x - 2) - 2x - 3 = x^2 + (x - 2x) + (-2 - 3)$$
$$= \boxed{x^2 - x - 5}.$$

3. Multiplication of polynomials is all about distribution:
$$f(x)g(x) = (x^2 + x - 2)(2x + 3)$$
$$= \boxed{2x^3 + 5x^2 - x - 6}.$$

4. We can try to factor the numerator and denominator in order to cancel terms, but in this case there's no simplification which can be done:
$$\frac{f(x)}{g(x)} = \frac{x^2 + x - 2}{2x + 3} = \boxed{\frac{(x + 2)(x - 1)}{2x + 3}}.$$

5. Composition of functions may take some getting used to, but $f(g(x))$ is computationally intuitive. We just replace any x which appears in $f(x)$ with the formula for $g(x)$:
$$f(x) = x^2 + x - 2,\ g(x) = 2x + 3 \implies f(g(x)) = (g(x))^2 + (g(x)) - 2,$$
$$f(g(x)) = (4x^2 + 12x + 9) + (2x + 3) - 2 = \boxed{4x^2 + 14x + 10}.$$

Note: in general, it is not the case that $f(g(x)) = g(f(x))$! Order matters when it comes to composition of functions.

1.3.1 Function Shapes

Definition 1.4. A function $f(x)$ defined on an interval $[a, b]$ is

- **increasing** if for any x, y in $[a, b]$, $x < y$, we have $f(x) \leq f(y)$; in words, as the input increases, so does the output.

- **decreasing** if for any x, y in $[a, b]$, $x < y$, we have $f(x) \geq f(y)$; as the input increases, the output decreases.
- **concave down** if the graph of f 'bends down' on the interval $[a, b]$, i.e. if secant lines lie below the graph. Imagine a frown :(
- **concave up** if the graph of f 'bends up' on the interval $[a, b]$, i.e. if secant lines lie above the graph. Imagine a smiley face :)

Remarks:

1. Notice the inequalities in the definitions of increasing and decreasing are *not* strict. This means constant functions are technically both increasing and decreasing!
2. When we want to specify $f(x)$ is increasing (respectively decreasing) and nonconstant, we sometimes say $f(x)$ is **strictly increasing** (respectively **strictly decreasing**).
3. In a month or two, we will see some convenient ways of expressing the above definitions in terms of derivatives.

Example 1.10. Determine (or at least roughly estimate) where each of the following functions is increasing, decreasing, concave up, and concave down:

(a) $f(x) = 2x^2 - 1$

(b) $f(x) = x^3 - 7x + 6$

(c) $f(x) = -4$ (Hint: consider the definitions!)

Solution 1.11.

(a) Visually, this is an upward facing parabola with vertex at $x = 0$. This means f is decreasing on $(-\infty, 0]$, increasing on $[0, \infty)$. Additionally, f is concave up everywhere since secant lines always remain above the graph.

(b) We can factor $x^3-7x+6 = (x-1)(x-2)(x+3)$ so we have zeros at $x = -3, 1, 2$. Since $f(-4) = -30 < 0$, $f(0) = 6 > 0$, $f(3/2) = -1.125 < 0$ we can sketch the graph and see f is increasing **approximately** on $(-\infty, -1.5] \cup [1.5, \infty)$, decreasing on $[-1.5, 1.5]$, concave down on $(-\infty, 0)$, and concave up on $(0, \infty)$.

(c) The constant function is both increasing *and* decreasing, as well as both concave up *and* concave down. This is due to the non-strict inequalities involved in the definitions of these terms, but if you said it was neither concave up nor concave down that's fine.

2 Week 2

Outline
Days: Monday (January 26), Wednesday (January 28), Friday (January 30)
Topics/Keywords: *Equations of lines, slope, polynomials, rational functions, zeros.*
Goals:

- Become comfortable with point-slope and slope-intercept forms of lines.
- Broaden our knowledge of basic function types to include polynomials and rational functions.
- Use functions to model cost, revenue, and profit.

Album of the Week: *A Swingin' Affair* by *Frank Sinatra (1957)*

2.1 Day 1

Goal: Learn about what kind of equation has a graph which is a line. In particular, use point-slope form and slope-intercept form to describe the function.

2.1.1 Linear Functions

Definition 2.1. A function is **linear** if it is represented by an equation of the form $f(x) = mx + b$ for some constants m, b.

Remarks:

1. If $m > 0$, then the linear function is increasing on $(-\infty, \infty)$; if $m < 0$, then the linear function is decreasing on $(-\infty, \infty)$; if $m = 0$ then the linear function is a horizontal line (constant) at height b.
2. Lines have a **constant rate of change** or **slope**. This is shown by the fact that if I find the difference between $f(x_1)$ and $f(x_2)$ and scale by the difference between x_1 and x_2...

$$\frac{f(x_1) - f(x_2)}{x_1 - x_2} = \frac{(mx_1 + b) - (mx_2 + b)}{x_1 - x_2} = \frac{m(x_1 - x_2)}{x_1 - x_2} = m.$$

Suppose we want to construct the (unique) line $f(x)$ passing through two points (x, y) and (x_1, y_1), $x \neq x_1$, in the plane. By the above remarks, this line must have slope

$$m = \frac{y - y_1}{x - x_1}.$$

But then we can just rearrange and see

$$y - y_1 = m(x - x_1) \implies f(x) = y = mx + \underbrace{(y_1 - mx_1)}_{\text{intercept}}.$$

This leads to the following definition.

Definition 2.2. Given a point (x_1, y_1), the line with slope m passing through (x_1, y_1) is given by the **point-slope form**

$$y - y_1 = m(x - x_1).$$

The same line has **slope-intercept form**

$$y = mx + b$$

where $b = y_1 - mx_1$ is the y-intercept.

Example 2.1.

(a) Compute the slope of the line passing through the points $(-1, 10)$, $(1, 4)$.

(b) Determine the equation of the line passing through the point $(-3, 3)$ with slope $-\frac{3}{2}$.

Solution 2.1.

(a) $m = \frac{y-y_1}{x-x_1} = \frac{10-4}{-1-2} = -\frac{6}{3} = -2$.

(b) Use point-slope form: $y - y_1 = m(x - x_1)$ where m is the slope and (x_1, y_1) is any point on the line. Therefore

$$y - 3 = -\frac{3}{2}(x - (-3)) \implies y = -\frac{3}{2}x - \frac{9}{2} + 3$$
$$\therefore y = -\frac{3}{2}x - \frac{3}{2}$$

2.2 Day 2

Goal: Discuss three more basic classes of functions, namely polynomials, power, and rational.

Definition 2.3. A **polynomial** is a function of the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

where $a_n, a_{n-1}, \ldots, a_1, a_0$ are real constants (numbers), $a_n \neq 0$, and n is a nonnegative integer called the **degree** of f.

Example 2.2. Determine if each of the following functions are polynomials; state the degree where applicable.

(a) $f(x) = -2x^3 + 4x^5 + x - 1$

(b) $f(x) = x^2 - \sqrt{3x} + 6$

(c) $f(x) = -x^{\frac{4}{2}} + x + 4$

(d) $f(x) = 2x^7 + x^6 - 4x^5 + x^{-4} + 3$.

Solution 2.2.

(a) Yes, this is a polynomial. Only one thing to really check: everywhere we see an x, is it raised to a positive, whole-number power? This is clearly the case, since the powers in this function are 5, 3, and 1. The constant -1 term at the end is also allowed, since a_0 in the definition is any real constant.

(b) This is not a polynomial since $\sqrt{3x} = \sqrt{3}x^{\frac{1}{2}}$ has a fractional power, which is not allowed. Cannot state the degree because it's not a polynomial.

(c) This is a polynomial of degree 2 because we can simplify

$$f(x) = -x^{\frac{4}{2}} + x + 4 = -x^2 + x + 4.$$

(d) No, this is not a polynomial. This is because the power -4 on the term x^{-4} is not a *positive* whole number. All the other terms would be okay!

Definition 2.4. A **power function** is a function of the form

$$f(x) = ax^b$$

where a, b are real numbers (NOT necessarily integers or nonnegative).

Example 2.3. Determine which of the following are power functions. Explain your reasoning.

(a) $f(x) = x^5$

(b) $f(x) = ex^{\pi}$

(c) $f(x) = 3^x$

(d) $f(x) = 3 + x^{-3.2}$

(e) $f(x) = 3x^{-3.2}$

Solution 2.3.

(a) Yes, $f(x) = x^5$ is a power function because it takes the form $f(x) = ax^b$ where $a = 1$, $b = 5$, matching the above definition.

(b) Yes, this is a power function because it takes the form $f(x) = ax^b$ for $a = e$ (Euler's number), $b = \pi$.

(c) No, this is not a power function because it has a *constant base* and *variable exponent* instead of the other way around.

(d) No, this is not a power function. While $g(x) = x^{-3.2}$ would constitute a power function, adding a constant is not allowed per the definition.

(e) Yes, this is a power function because it takes the form $f(x) = ax^b$ where $a = 3$, $b = -3.2$. In contrast with (d), *scaling* by a real number is okay while *adding/translating* is not.

2.2.1 End Behavior of Power Functions and Polynomials

If we let the input x become arbitrarily large, positive, i.e. $x \to \infty$, or let x become arbitrarily large, negative, meaning $x \to -\infty$, what happens to the output of a power function $f(x) = ax^n$, where n is a positive integer? This is called **end behavior**.

The idea of end behavior can seem confusing when n is an arbitrary exponent and a is an arbitrary coefficient, but a few examples will illuminate the pattern.

Example 2.4. Determine the end behavior of the following power functions by describing what happens to the output as $x \to -\infty$ and $x \to \infty$. (Note: don't be put off by the use of $g(x)$, $h(x)$, or any other label to represent a function!)

(a) $g(x) = 2x^2$

(b) $h(x) = 3x^3$

(c) $f(x) = -5x^4$

(d) $P(x) = -x$

Solution 2.4.

(a) One way to describe what happens to $g(x)$ is by writing out a table of input-output pairs. We'll make one for $x \to -\infty$ and one for $x \to \infty$.

x	$g(x)$
-1	2
-10	200
-100	20000
-1000	2000000

x	$g(x)$
1	2
10	200
100	20000
1000	2000000

From the first table, as $x \to -\infty$ it is clear that $g(x)$ becomes arbitrarily large, positive, i.e. $g(x) \to \infty$. From the second table the same pattern emerges: as $x \to \infty$, $g(x)$ becomes arbitrarily large, positive, i.e. $g(x) \to \infty$.

(b) Let's use the same method as in (a), since tables are effective.

x	$h(x)$
-1	-3
-10	-3000
-100	-3000000
-1000	-3000000000

x	$h(x)$
1	3
10	3000
100	3000000
1000	3000000000

From the first table, as $x \to -\infty$ we see $h(x)$ becomes arbitrarily large, negative, i.e. $h(x) \to -\infty$. This is because, for example, $3(-10)^3 = 3(-1)^3(10)^3 = 3(-1)(1000) = -3000$. Crucially, $(-1)^{\text{odd power}} = -1$.

On the other hand, as $x \to \infty$ we see $h(x)$ becomes arbitrarily large, positive, i.e. $h(x) \to \infty$.

(c) Use the formal setup or structure of a table if that helps keep you organized, but we can intuitively reason this out (at least save the keystrokes for making the table). If x is large and positive, e.g. $x = 1000$, we see

$$f(1000) = -5(1000)^4 = -5 \times 10^{12}$$

which is a very large, negative number. The negative sign will persist and the magnitude will continue to grow as $x \to \infty$, so $f(x) \to -\infty$ as $x \to \infty$.

Similarly, try plugging in a large negative x, e.g. $x = -1000$. We see

$$f(-1000) = -5(-1000)^4 = -5(-1)^4(1000)^4 = -5 \times 10^{12} = f(1000).$$

Changing the sign of the input doesn't change the output (due to the even power 4), so the limit as $x \to -\infty$ is the same as the limit as $x \to \infty$. Therefore $f(x) \to -\infty$ as $x \to -\infty$.

(d) As $x \to \infty$, $P(x) \to -\infty$; as $x \to -\infty$, $P(x) \to \infty$. This is because $P(x)$ behaves exactly as x does, but multiplied by a negative.

From the above example, we can extrapolate and have the following fact about the limiting behaviour of any power function with positive integer exponent.

Fact: Suppose $f(x) = ax^n$ is a power function where a is a nonzero real number and n is a positive integer. We have the following cases.

1. Suppose n is *even*. Then ...
   (a) ... if $a > 0$, $f(x) \to \infty$ as $x \to \infty$ or $x \to -\infty$.
   (b) ... if $a < 0$, $f(x) \to -\infty$ as $x \to \infty$ or $x \to -\infty$.
2. Suppose n is *odd*. Then ...
   (a) ... if $a > 0$, $f(x) \to \infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$.
   (b) ... if $a < 0$, $f(x) \to -\infty$ as $x \to \infty$ and $f(x) \to \infty$ as $x \to -\infty$.

Remarks:

1. If the **Fact** stated here is too much to memorize, it is never a bad idea to make a table of values! That's a foolproof method for any example.
2. The cases where $a < 0$ are the same as when $a > 0$, the behavior of $f(x)$ just differs by a negative sign.

2.2.2 Polynomials

Recall the definition of a polynomial.

Definition 2.5. A function $f(x)$ is a **polynomial** if it takes the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

for real numbers $a_n, \ldots, a_0$ and integer $n \geq 0$. If $a_n \neq 0$, we say the **degree** of the polynomial $f(x)$ is n.

Example 2.5. Determine whether each of the following functions is a polynomial. If it is, state the degree; if it isn't, explain why. (Note: the input or independent variable doesn't always have to be denoted by x, don't let other labels throw you off!)

(a) $f(x) = 3x + 2$

(b) $h(t) = -e^{2t}$

(c) $g(t) = -2t^e$

(d) $r(x) = x^5 + x^4 + x^3 + x^2 + x + 1$

(e) $p(x) = 5x^7$

Solution 2.5.

(a) This is a polynomial of degree 1, since the largest power of x that appears is 1.

(b) The function $h(t) = -e^{2t}$ is not a polynomial. A polynomial is comprised of terms of the form ct^k, while this is an exponential. (There's a massive difference between having t as a base with a constant exponent k and having a constant base e with t in the exponent.)

(c) This is not a polynomial because the power on t which appears is not a whole number. ($e = 2.718...$)

(d) This is a polynomial of degree 5 since it has the appropriate form and the largest exponent is 5.

(e) This is a polynomial of degree 7. Note this implies that *all power functions with nonnegative integer exponents are polynomials.*

It turns out that the end behavior of polynomials can be determined using what we know about the end behavior of power functions.

Fact: Given a polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, the end behavior of $f(x)$ is the same as the end behavior of $a_n x^n$.

Remarks:

1. No matter how many terms a polynomial has, we can ignore everything except the term with highest exponent when determining end behavior.
2. The term $a_n x^n$ is often referred to as the **leading term** or **highest-order term** of the polynomial.

Example 2.6. Determine the end behavior of each polynomial listed below. That is, determine the behavior of the function as $x \to \infty$ or $x \to -\infty$.

(a) $f(x) = -x^3 - 17x^7 + 1 - 37x^{103827}$

(b) $f(x) = 23x^{124748922} + x^{11111} - 17973x^{15} + 13$

Solution 2.6.

(a) The leading term of this polynomial is $-37x^{103827}$ (be sure to retain the entire coefficient, including the negative sign). Since 103827 is an odd number and $-37 < 0$, using our earlier Fact implies that $f(x) \to \infty$ as $x \to -\infty$ and $f(x) \to -\infty$ as $x \to \infty$. (Be sure you are able to identify 103827 as an odd number; use a table of values to confirm the limiting behavior if not fully comfortable with the Fact(s).)

(b) The end behavior of the whole polynomial is the same as the end behavior of the leading term $p(x) = 23x^{1234789922}$, which is a power function with *even integer exponent* and *positive coefficient*. By our previous work, this means $f(x) \to \infty$ as $x \to \infty$ or $x \to -\infty$. (Make sure you are able to identify that 1234789922 is even!)

2.2.3 Zeros and Factoring Polynomials

Definition 2.6. A function $f : \mathbb{R} \to \mathbb{R}$ has a **zero** or **root** at $x = x_0$ if $f(x_0) = 0$. Visually, the graph of $f(x)$ intersects the x-axis at $(x_0, 0)$.

Example 2.7. Determine the root(s) of the functions listed below, if they exist.

(a) $h(x) = -8x + 16$

(b) $g(x) = 5$

(c) $f(x) = 7x - 24$

Solution 2.7.

(a) We set $h(x) = 0$ and solve for the value(s) of x which satisfy the equation:

$$h(x) = 0 \iff -8x + 16 = 0 \iff 8x = 16$$
$$\therefore x = 2.$$

(b) The function $g(x) = 5$ is constant and identically equal to 5, meaning it cannot have a root! Algebraically, if we set $g(x) = 0$ and try to solve for x...

$$g(x) = 0 \iff 5 = 0$$

which is impossible, a contradiction. This means g has no roots.

(c) To find roots, set $f(x) = 0$ and solve for x:

$$7x - 24 = 0 \iff 7x = 24 \iff x = \frac{24}{7}.$$

In the previous example we found roots of linear functions, i.e. degree 1 polynomials. Can we do the same for quadratic polynomials, i.e. polynomials of degree 2? There are two methods in general.

Quadratic Formula: Given a degree two polynomial $f(x) = ax^2 + bx + c$, the roots of f are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Remarks:

1. An important feature of the quadratic formula is the term $b^2 - 4ac$ under the square root. This is called the **discriminant**, and determines what type of roots the polynomial has:

(a) If $b^2 - 4ac = 0$, $x = \frac{-b \pm \sqrt{0}}{2a} = -\frac{b}{2a}$, meaning there is exactly one root.

(b) If $b^2 - 4ac > 0$, then $x = \frac{-b+\sqrt{b^2-4ac}}{2a}$ and $x = \frac{-b-\sqrt{b^2-4ac}}{2a}$ are two distinct roots.

(c) If $b^2 - 4ac < 0$, then $\sqrt{b^2 - 4ac}$ is an imaginary number. This means that $f(x)$ has no real roots.

2. For convenience, it is useful to have this formula memorized. I know that's a bit of a boomer mentality, but some things just aren't worth taking the time to look up.

Sometimes we don't need to resort to using the quadratic formula, and instead we can reduce the quadratic polynomial to a product of linear ones.

Factoring: Given $f(x) = ax^2 + bx + c$, *try* to rewrite

$$f(x) = (a_1x + c_1)(a_2x + c_2)$$

which would imply (by setting equal to 0) that the roots of f are $x = -\frac{c_1}{a_1}$ and $x = -\frac{c_2}{a_2}$. Note this is not always easy to do by hand! Identifying quadratics which factor "nicely" tends to come with practice if you're not already familiar.

Example 2.8. Find the roots of:

(a) $f(x) = x^2 + 6x + 8$

(b) $f(x) = x^2 + 4x - 10$

(c) $f(x) = 2x^2 - 8x + 8$

(d) $f(x) = x^3 - 4x^2 - 21x$

Solution 2.8.

(a) This polynomial can be factored as $x^2 + 6x + 8 = (x + 2)(x + 4)$. We can double check by computing the multiplication of the linear terms:

$$(x + 2)(x + 4) = x^2 + 4x + 2x + 2 \cdot 4 = x^2 + 6x + 8.$$

Importantly, 2 and 4 *multiply* to give 8 (the constant term), while 2 and 4 *add* to give 6 (the coefficient on the linear term). The roots of $f(x)$ are found by setting equal to zero and solving for x, meaning

$$f(x) = 0 \iff x^2 + 6x + 8 = 0 \iff (x + 2)(x + 4) = 0.$$

A product of terms equals zero only if at least one of the terms equals zero, so here $(x + 2)(x + 4) = 0 \implies x + 2 = 0$ or $x + 4 = 0$. Therefore $x = -2$, $x = -4$ are the roots.

(b) To factor, we want two numbers which multiply to -10 and add to 4; the only pairs of numbers which multiply to -10 are $(\pm 1, \mp 10), (\pm 2, \mp 5)$. No combination here will add to 4, so we resort to the quadratic formula:

$$x = \frac{-4 \pm \sqrt{4^2 - 4(1)(-10)}}{2} = \frac{-4 \pm \sqrt{16 + 40}}{2} = \frac{-4 \pm \sqrt{56}}{2}.$$

The value of $\sqrt{56}$ is not an integer, so we can just leave our roots written as $x = \frac{-4+\sqrt{56}}{2}$ and $x = \frac{-4-\sqrt{56}}{2}$.

(c) When possible, we want the coefficient on the leading term to be 1. In this case, each term has a common factor of 2

$$f(x) = 2x^2 - 8x + 8 = 2(x^2 - 4x + 8).$$

This shows $f(x) = 0 \iff x^2-4x+4 = 0$ after dividing by 2. But $x^2-4x+4 = (x-2)(x-2) = (x-2)^2$, so the only root is $x = 2$.

(d) Always check for common factors! Each term here has an x in it which we can take out:

$$x^3 - 4x^2 - 21x = x(x^2 - 4x - 21).$$

Now we factor the quadratic:

$$x\underbrace{(x^2 - 4x - 21)} = x\underbrace{(x - 7)(x + 3)}.$$

Setting equal to 0 (as we do when finding roots) shows $x(x-7)(x+3) = 0 \implies x = 0, x-7 = 0, x+3 = 0$. Thus $x = -3, 0, 7$ are the roots of the given polynomial.

Example 2.9. Find the roots of the following polynomials.

(a) $h(t) = 2t^2 - t$

(b) $p(x) = x^2 - 16$

(c) $f(x) = x^2 + 15x + 54$

Solution 2.9.

(a) $h(t) = 0 \iff 2t^2 - t = 0 \iff t(2t - 1) = 0$. Setting each factor equal to 0 shows $x = 0$ or $2t - 1 = 0 \implies t = \frac{1}{2}$. The roots are then $t = 0, \frac{1}{2}$.

(b) There are two somewhat distinct but equivalent ways to go about this. First, we can factor:

$$x^2 - 16 = 0 \iff (x + 4)(x - 4) = 0; \quad \therefore x = -4, 4.$$

Alternatively:

$$x^2 - 16 = 0 \implies x^2 = 16 \implies x = \pm\sqrt{16} = \pm 4.$$

(c) Factor: $x^2 + 15x + 54 = (x + 6)(x + 9) = 0 \implies x = -9, -6$.

Example 2.10. True or False: To find the roots of $f(x) = x^2 + 9$, we can write

$$x^2 + 9 = 0 \implies \sqrt{x^2 + 9} = \sqrt{0} \implies x + 3 = 0 \implies x = -3.$$

Solution 2.10.

False! We saw last week that $(x + y)^n \neq x^n + y^n$ in general. Here we have the particular case where $n = \frac{1}{2}$ and $y = 9$, so we **cannot** split the square root across terms! Instead, to find roots:

$$\sqrt{x^2 + 9} = 0 \iff (\sqrt{x^2 + 9})^2 = 0^2$$
$$\iff x^2 + 9 = 0 \iff x^2 = -9$$
$$\therefore x = \pm\sqrt{-9} = \pm 3i.$$

The function has two *purely imaginary roots*, i.e. it has no real roots.

2.3 Day 3

Goal: Briefly wrap up discussion of functions. Introduce methods of modeling cost, revenue, and profit.

2.3.1 Rational Functions

Definition 2.7. A **rational function** is a function of the form

$$f(x) = \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0},$$

where the numerator and denominator are polynomials.

Example 2.11. Find the domain of the rational function

$$f(x) = \frac{x^2 + 5x + 4}{x^2 + 3x + 2}.$$

Solution 2.11. To do this, think back to our discussion of *natural* domain: where can this function be undefined? The only thing which can go wrong is if we divide by 0, i.e. the denominator is equal to 0. We solve for x satisfying

$$x^2 + 3x + 2 = 0 \iff (x+2)(x+1) = 0 \iff x = -2, -1.$$

Thus the domain of f is $D = (-\infty, -2) \cup (-2, -1) \cup (-1, \infty)$. NOTE: although we can simplify $f(x) = \frac{(x+4)(x+1)}{(x+2)(x+1)} = \frac{x+4}{x+2}$, ***as written*** the point $x = -1$ is still not in the domain of f! Try plotting in Desmos and evaluating $f(-1)$.

2.3.2 Piecewise Functions

Definition 2.8. A function $f(x)$ is **piecewise** if there are at least two functions f_1, f_2 and domains D_1, D_2 such that

$$f(x) = \begin{cases} f_1(x), & x \text{ in } D_1 \\ f_2(x), & x \text{ in } D_2. \end{cases}$$

Example 2.12. Define the piecewise function

$$f(x) = \begin{cases} f_1(x) = 3x + 4, & x \leq -1 \\ f_2(x) = x, & x > -1 \end{cases}.$$

(1) Is f a well-defined function? (2) Is f continuous? Explain why or why not.

Solution 2.12.

1. Yes, f is a well-defined function. For a piecewise function this amounts to checking that f_1 and f_2 are functions and that $f_1(x) = f_2(x)$ at any x contained in both domains. Here, f_1 and f_2 are both lines which we know are functions; moreover, their domains do not overlap, so the second condition holds automatically.

2. No, f is not continuous. We can see this by looking at the graph which has a jump discontinuity at $x = -1$. We'll see in the next couple weeks that the definition of continuity at -1 is

$$\lim_{x \to -1} f(x) = f(-1),$$

which does not hold here because $\lim_{x \to -1} f(x)$ does not exist (left and right limits are not equal).

2.3.3 Models of Cost, Revenue, and Profit

Suppose we are operating a business; if we produce x units of our product or service, the **cost function** $C(x)$ represents the amount that we paid in order to produce x units. It is common for there to be two components of this cost: **fixed** and **variable**. As their names suggest, fixed costs are constant while variable costs *depend on the input, i.e. the number of units produced, x*. When the cost function or cost model is linear, it takes the form

$$\boxed{C(x) = mx + b} \tag{1}$$

where m is the **cost per unit** or **marginal cost** (in other words, it determines the variable cost) and b is the **fixed cost** (colloquially known as operating cost or overhead). Note that even if we produce 0 units, meaning $x = 0$, the cost function is still nonzero (assuming $b \neq 0$) since $C(0) = b$.

The **revenue function** $R(x)$ represents the payment which our business receives as a result of selling x units of product/service. Assuming we are paid a constant amount *per* unit (corresponding to a constant rate of change or slope), we have a linear model

$$R(x) = px \tag{2}$$

where p is the **selling price per unit**. If we want to know how much money we're netting when costs are taken into account, we calculate the **profit function** $P(x)$:

$$P(x) = R(x) - C(x). \tag{3}$$

Keep in mind that the profit function is the revenue minus the costs *even if revenue and cost are nonlinear*.

2.3.4 Break-Even Analysis

It is important (obviously) for a company to know how many units of a product they need to produce such that $P(x) > 0$, since this means money is being gained, as opposed to where $P(x) < 0$. Thus knowing when $P(x) = 0$ is valuable because it separates the above domains (keep reading to make more sense of this).

Definition 2.9. The **break-even quantity** is the number of units x which must be produced in order to have $R(x) = C(x)$, i.e. the profit is zero: $P(x) = 0$.

Remark: Geometrically, the break-even quantity x represents the input value where the lines $R(x)$ and $C(x)$ intersect.

Example 2.13. Let's say we're operating a lemonade stand. We sell cups of lemonade for \$125 apiece (listen, we only sell to the rich and foolish); it cost us \$250 to set up the stand (very fancy stand, naturally) and costs \$5 in supplies for each cup we sell. Write down the revenue function, cost function, profit function, and break-even quantity for this enterprise.

Solution 2.13.

Revenue: $\boxed{R(x) = 125x}$ since we are charging \$125 for each cup.

Cost: Our variable cost is \$5, since this is the amount we have to pay to supply each cup, while our overhead cost is \$250, so $\boxed{C(x) = 5x + 250}$.

Profit: Profit is revenue minus cost, so

$$\boxed{P(x) = R(x) - C(x) = 125x - (5x + 250) = 120x - 250}.$$

Break-Even Quantity: Set $P(x) = 0$ and solve for x

$$P(x) = 0 \iff 120x - 250 = 0 \iff 120x = 250 \iff \boxed{x = \frac{250}{12} = \frac{25}{12} \approx 2.083\ldots}.$$

This means that we only need to sell roughly 2 cups of lemonade before we break even, and anything more than that will put us 'in the green'.

2.3.5 Supply and Demand Models

Definition 2.10.

(i) The **demand function** $p_d(x)$ gives you the (selling) price when x units are wanted by the consumer. (**Think: if the consumer wants x units, we'll set the price to be $p_d(x)$.**)

(ii) The **supply function** $p_s(x)$ gives you the (selling) price the supplier needs in order to bring x units to market. (**Think: in order for us to be willing to produce x units, the selling price should be $p_s(x)$.**)

Remarks:

- First, this all feels backwards! Typically we think of supply and demand as a function of the price, i.e. $x = f(p)$ in our above notation, but either approach is valid in this rough linear model.
- If price is high we expect demand to be low, and if price is low we expect demand to be high. Said in our order: **if demand is low the price should be set high, while if demand is high the price should be low.** This means demand $p_d(x)$ has a negative slope.
- If price is high we expect supply to be high, while if price is low we expect supply to be low. Said in our order: **if supply is low the price is low, while if supply is high the price is high**. This means supply $p_s(x)$ has a positive slope.
- Sometimes people don't use the subscript d or s to indicate demand price versus supply price and instead just use $p(x)$ for both. Context clues!

Definition 2.11. The **equilibrium point** (x, p) represents the number of units x and price p such that supply and demand are equal, i.e. $p_d(x) = p = p_s(x)$. More specifically, x is the **equilibrium quantity** and p is the **equilibrium price**.

Example 2.14. Suppose our lemonade stand has a supply function given by

$$p_s(x) = 10x + 3$$

and a demand function given by

$$p_d(x) = -3x + 20.$$

What is the equilibrium quantity of our business? What is the equilibrium price?

Solution 2.14.

The equilibrium point is where the two lines intersect, meaning the functions are equal. That is, we can solve for x by setting

$$10x + 4 = -3x + 30 \implies x = \frac{26}{13} = 2.$$

This means the lines intersect when the number of units is $x = 2$, so we can find the corresponding price by evaluating $p(2)$ (either function will yield the same value by the previous calculation):

$$p(2) = 10(2) + 4 = 20 + 4 = \boxed{24}.$$

Here's a more involved example concerning supply and demand. Fun anecdote: when I originally wrote these notes, I was only Amanda's fiancé! Time flies in Math 1071.

Example 2.15. Suppose Amanda and her husband Cole own and operate a small bookstore. Their current list price is roughly \$30 per book, a price at which they are willing to supply 80 books. If the price per book was raised to \$40, they would be willing to supply 110. They're currently selling 70 books per day, but they expect to still sell 55 books every day if the price was indeed raised to \$40. (1) Determine linear functions modeling supply and demand; (2) Find the equilibrium quantity of books and price per book.

Solution 2.15.

1. We are given two points relevant to demand, namely $(70, 30)$ (currently selling 70 per day at \$30 apiece) and $(55, 40)$ (the expected number sold at \$40 apiece is 55). This means the slope of the demand line is
$$m = \frac{40-30}{55-70} = -\frac{10}{15} = -\frac{2}{3}.$$
Plugging into point-slope form shows demand equation is
$$p - p_1 = m(x - x_1) \to p - 30 = -\frac{2}{3}(x - 70) \to \boxed{p(x) = -\frac{2}{3}x + \frac{230}{3}}.$$
Similarly, we are given two points relevant to supply: $(80, 30)$ (they currently supply 80 books per day and charge \$30) and $(110, 40)$ (they'd be willing to supply 110 at \$ 40 apiece). By the same process
$$m = \frac{40-30}{110-80} = \frac{1}{3},$$
$$p - p_1 = m(x - x_1) \to p - 30 = \frac{1}{3}(x - 80) \to p(x) = \frac{1}{3}x + \frac{-80+90}{3} \to \boxed{p(x) = \frac{1}{3}x + \frac{10}{3}.}$$
2. To find equilibrium quantity and price, we first set demand and supply equal, solve for x:
$$-\frac{2}{3}x + \frac{230}{3} = \frac{1}{3}x + \frac{10}{3} \implies x = \frac{220}{3} = 73.\overline{3}.$$
This tells us the equilibrium quantity is about $\frac{220}{3}$ books. As a consequence, the equilibrium price is
$$p\left(\frac{220}{3}\right) = \frac{1}{3}\left(\frac{220}{3}\right) + \frac{10}{3} = \frac{220+30}{9} = \frac{250}{9} \approx 27.78$$
which means the equilibrium price is \$27.78 per book.

2.3.6 Quadratic Models

Definition 2.12. A quadratic polynomial is in **standard form** if it is written
$$f(x) = ax^2 + bx + c \tag{4}$$
where $a \neq 0$: if $a > 0$ the parabola opens upward, and if $a < 0$ the parabola opens downward. The **vertex** or peak/valley of the parabola occurs at the x-coordinate $x = -\frac{b}{2a}$.

Example 2.16. (Bonus) Show that the vertex of $f(x) = ax^2 + bx + c$ really does occur at $x = -\frac{b}{2a}$.

Solution 2.16. Hint: complete the square. Since scaling the function won't impact where the vertex occurs, we can consider $f(x) = x^2 + \frac{b}{a}x + \frac{c}{a}$. Completing the square shows $f(x) = (x + \frac{b}{2a})^2 + \frac{c}{a} - \frac{b^2}{4a^2}$, which attains its minimum value when $x + \frac{b}{2a} = 0$, i.e. $x = -\frac{b}{2a}$ as expected/claimed.

Example 2.17. Suppose now our lemonade stand has a supply function given by

$$p(x) = 100 + \frac{3}{2}x$$

and the cost function for making x units is

$$C(x) = 25 - 5x + 2x^2.$$

1. Find the revenue function $R(x)$
2. Find the profit function $P(x)$
3. Calculate the number of units we should sell to maximize profit
4. Calculate the maximum profit
5. Find the number of units we should make to minimize cost.

Solution 2.17.

1. Recall from previous lectures that the revenue is given in terms of the sell price p as $R(x) = px$. In this case we are given $p = p(x) = 100 + \frac{3}{2}x$, therefore

$$R(x) = \left(100 + \frac{3}{2}x\right)x = 100x + \frac{3}{2}x^2.$$

2. Profit = Revenue - Cost, so

$$P(x) = 100x + \frac{3}{2}x^2 - \left(25 - 5x + 2x^2\right) = -25 + 105x - \frac{1}{2}x^2.$$

3. From the previous part we can see profit is given by a downward facing parabola, meaning it attains a unique maximum at the vertex. In this case $a = -\frac{1}{2}$, $b = 105$, $c = -25$ so the vertex occurs at

$$x = -(105/2(-1/2)) = 105.$$

   This means we should sell 105 cups of lemonade in order to maximize our profit.

4. The maximum profit is $P(105)$, namely the value of the profit function at the vertex:

$$P(105) = -25 + 105(105) - \frac{1}{2}(105)^2 = 5487.5$$

   This means if we sell 105 cups of lemonade, we'll net \$ 5487.50.

5. Our cost function is the upward facing parabola $C(x) = 25 - 5x + 2x^2$, which means it attains a minimum value at

$$x = -\frac{b}{2a} = -\frac{(-5)}{2(2)} = \frac{5}{4}.$$

   So, we can sell roughly 1 cup of lemonade to minimize cost in this scenario.

3 Week 3

Outline
Days: Monday (February 2), Wednesday (February 4), Friday (February 6)
Topics: *Modeling cost, revenue, and profit. Compounded interest. Exponential and logarithmic functions.*

Goals:

- Interpret models of cost, revenue, and profit. Utilize break-even analysis.
- Understand and compute interest which is compounded annually, at more frequent discrete intervals, or continuously,
- Use inverse functions, specifically $\log_b(\cdot)$ or $\ln(\cdot)$.

Album of the Week: *Revolver* by *The Beatles (1966)*

3.1 Day 1

Goal: Continue to explore how functions are utilized as mathematical models of cost, revenue, and profit.

Example 3.1. Suppose that when 100 units of a certain item are supplied, they are sold at a price of \$1.25 per unit. When 80 more units of a certain item are supplied, the price per unit increased by \$1. Assuming a linear model, find the supply equation $p_s(x)$ where x is the number of units sold.

Solution 3.1.

This is where we put our knowledge of lines into practice. We are told that $p_s(x)$ is a line satisfying $p_s(100) = 1.25$ and $p_s(180) = 2.25$. As a result,

$$m = \frac{p_2 - p_1}{x_2 - x_1} = \frac{2.25 - 1.25}{180 - 100} = \frac{1}{80}.$$

Plugging into point-slope form:

$$p_s(x) - p_1 = m(x - x_1) \iff p_s(x) - 1.25 = \frac{1}{80}(x - 100)$$

$$\therefore \boxed{p_s(x) = \frac{1}{80}x}.$$

3.1.1 Quadratic Models, Continued

Example 3.2. A company finds that the profit they make in a day when selling x units is $P(x) = -25x^2 + 2875x - 75000$.

How many items should be sold to maximize profit? What is the maximum profit which can be achieved, assuming the monetary unit is dollars?

Solution 3.2.

The maximum or peak of a (downward-facing) parabola occurs at the vertex. Here, the vertex occurs at

$$x = -\frac{b}{2a} = -\frac{2875}{2(-25)} = 57.5 \text{ units.}$$

This means that 57.5 units of product should be made in order to maximize profit; at this level of production, the profit is

$$P(57.5) = -25(57.5)^2 + 2875(57.5) - 75000 = \$7656.25$$

3.2 Day 2

Goal: Introduce ourselves to exponential functions. We should be able to recognize the difference between $f(x) = x^b$ and $f(x) = b^x$ as well as understand the significance of the number e in compounded interest.

3.2.1 Exponential Functions

We discussed exponents back in Week 1, but here is a reminder.

Definition 3.1. An **exponential function** is a function of the form

$$f(x) = b^x, \ b > 0.$$

Example 3.3. Sketch the graph of and/or complete the table of values for (a) $f(x) = 2^x$, and (b) $f(x) = \left(\frac{1}{3}\right)^x$.

Solution 3.3.

x	$f(x) = 2^x$
-2	$\frac{1}{4}$
-1	$\frac{1}{2}$
0	1
1	2
2	4

x	$f(x) = \left(\frac{1}{3}\right)^x$
-2	9
-1	3
0	1
1	$\frac{1}{3}$
2	$\frac{1}{9}$

Exponential functions are applicable to real life through compound interest, among other topics.

3.2.2 Practical Application: Compounded Interest

Suppose we invest a quantity P of money in a savings account with annual interest rate r (e.g. $r = 0.1$; for these purposes, r is a number in $(0, 1)$, so if we're given $r = 3.8\%$ we should convert it to say $r = \frac{3.8}{100} = 0.038$) that is compounded annually. This means that we take rP and add it to our original investment P, so our new account value at the end of 1 year is

$$A(t = 1) = P + rP = P(1 + r).$$

What if we let the money continue to sit for a second year? Then we take $rA(1)$ and add it to the amount we had at the end of year 1, $A(1)$:

$$A(t = 2) = A(1) + r(A(1)) = P(1 + r) + r(P(1 + r)) = P(1 + r)(1 + r) = P(1 + r)^2.$$

If we continue this pattern, the amount of money in the account after t years is

$$\boxed{A(t) = P(1 + r)^t}.$$

This is the equation for interest compounded annually, or once per year. Naturally, we want to know what happens if we compound multiple times per year. For a simple case to start, let's say the money gets compounded twice per year rather than once; we should get half of the interest at the 6 month mark, then the other half at the end of the year. At the 6 month mark, we should then add $\left(\frac{r}{2}\right) P$ to our original investment:

$$A\left(\frac{1}{2}\right) = P + \left(\frac{r}{2}\right) P = P\left(1 + \frac{r}{2}\right).$$

Then after one full year, we should add $\left(\frac{r}{2}\right) A\left(\frac{1}{2}\right)$ to $A\left(\frac{1}{2}\right)$:

$$\begin{aligned} A(1) &= A\left(\frac{1}{2}\right) + \left(\frac{r}{2}\right) A\left(\frac{1}{2}\right) = P\left(1 + \frac{r}{2}\right) + \left(\frac{r}{2}\right) P\left(1 + \frac{r}{2}\right) \\ &= P\left(1 + \frac{r}{2}\right)\left(1 + \frac{r}{2}\right) = P\left(1 + \frac{r}{2}\right)^2. \end{aligned}$$

Repeating this process shows that if our money is compounded n times per year with interest rate r, then the total amount in the account after t years is

$$\boxed{A(t) = P\left(1 + \frac{r}{n}\right)^{nt}}.$$

This formula will ultimately show us the importance of Euler's number e, but for now let's do an example.

Example 3.4. We're investing \$500 and have to choose which bank to put our money in, Bank A or Bank B. At Bank A, our money will be compounded monthly at an interest rate of 4.2%; at Bank B, we are offered a \$100 bonus for joining (which is deposited at the start), and the money is compounded every 3 months at an interest rate of 2.75%. If we're planning to leave the money for 10 years, which option should we go with?

Solution 3.4.

If we go with Bank A: $P = 500$, $r = 0.042$, $n = 12$, and $t = 10$, meaning

$$A(10) = (500)\left(1 + \frac{0.042}{12}\right)^{12(10)} = 760.42$$

If we go with Bank B: $P = 600$, $r = 0.0275$, $n = 4$ (since there are 4 periods of 3 months in a year), and $t = 10$, meaning

$$A(10) = (600)\left(1 + \frac{0.0275}{4}\right)^{4(10)} \approx 789.18$$

This tells us that **Bank B offers the better option** for our 10 year investment.

3.2.3 Continuously Compounded Interest

What if we let the number of times our money gets compounded, n, tend to infinity? This leads to what is known as **continuously compounded interest** (oh look, that's the name of the subsection). To compute this limit, let $m = \frac{n}{r}$; then our previous formula for $A(t)$ can be written

$$A(t) = P\left(1 + \frac{1}{m}\right)^{mrt} = P\left[\left(1 + \frac{1}{m}\right)^{m}\right]^{rt}.$$

If we can see where $\left(1 + \frac{1}{m}\right)^m$ goes as $m \to \infty$, we will obtain an expression for $A(t)$ with continuously compounded interest. This limit is a bit too difficult for us to use algebraic methods, but we can plug in large values of m and see if it tends to a value:

$$m = 100: \qquad \left(1 + \frac{1}{100}\right)^{(100)} = 2.7048\ldots$$

$$m = 1000: \qquad \left(1 + \frac{1}{1000}\right)^{(1000)} = 2.7169\ldots$$

$$m = 10000: \qquad \left(1 + \frac{1}{10000}\right)^{(10000)} = 2.7181\ldots$$

We can keep going, but the conclusion is that as $m \to \infty$, $\left(1 + \frac{1}{m}\right)^m$ approaches some number which is roughly 2.718. This number gets it's own definition.

Definition 3.2. **Euler's number** (named for Swiss mathematician Leonhard Euler), denoted by e, is defined as

$$e = \lim_{m\to\infty}\left(1 + \frac{1}{m}\right)^{m}.$$

With this in mind, we have the following formula for interest compounded continuously:

$$\boxed{A(t) = Pe^{rt}}.$$

Example 3.5. Flash back to Example 2.2 and now assume a mysterious Bank C has joined the race for our \$500 investment, offering an account with continuously compounded interest at a rate of 4.6%. Is Bank C the best choice?

Solution 3.5.
We know $P = 500$, $r = 0.046$, and $t = 10$, so

$$A(10) = 500e^{0.046(10)} = 792.04.$$

Bank C is just a slightly better option than Bank B (by around \$3), and better than Bank A by around \$32. One more banking example for today.

Example 3.6. Now we're investing \$100 in three different banks: Bank 1 offers **annually compounded interest at a rate of 3%**; Bank 2 offers **monthly compounded interest at a rate of 3%**; Bank 3 offers **continuously compounded interest at a rate of 3%**. When we return to collect our money in 25 years, how much do we have in total across all three?

Solution 3.6. We can calculate the total money in each bank (round to nearest hundredth place), then add them up. In Bank 1, $P = 100$, $r = 0.03$, $n = 1$, and $t = 25$:

$$\boxed{A(25) = (100)\,(1 + 0.03)^{25} = 209.38}$$

In Bank 2 the only difference is that $n = 12$:

$$A(25) = 100\left(1 + \frac{0.03}{12}\right)^{25(12)} = \boxed{211.5}$$

In Bank 3, we use the formula $A(t) = Pe^{rt}$:

$$\boxed{A(t) = (100)e^{(0.03)(25)} = 211.70}$$

Thus the total amount of money in the accounts is $\boxed{209.38 + 211.5 + 211.7 = 632.58}$.

3.3 Day 3

Goal: Introduce ourselves to logarithmic functions, understand their relationship with exponential functions. Be able to calculate a logarithm with different bases.

3.3.1 An Overview of Inverse Functions

To skip technical details, go directly to Definition 3.5!

Q: Given a function $f : A \to B$, when is there a function $g : B \to A$ such that $g(f(a)) = a$ for every a in A and $f(g(b)) = b$ for every b in B?

Definition 3.3. Two functions $f : A \to B$ and $g : B \to A$ are **inverses** if the above properties holds. Commonly written $g = f^{-1}$, $f = g^{-1}$, we say the two functions are **invertible**.

Example 3.7. Let $f(x) = -x + 3$. If possible, find a function g such that $g(f(x)) = x$ and $f(g(x)) = x$ for all x. If not possible, explain why. Let $f(x) = 4x^2$ and answer the same question.

Solution 3.7. For notation, let $y = f(x) = -x + 3$. To find such a function g, **solve for x in terms of y.**

$$y = -x + 3 \implies x = 3 - y.$$

This means we can define $g(x) = 3 - x$ (replacing y with x). Now check:

$$g(f(x)) = g(-x + 3) = 3 - (-x + 3) = x;$$
$$f(g(x)) = f(3 - x) = -(3 - x) + 3 = x.$$

The desired properties are satisfied!

Instead set $y = f(x) = 4x^2$. Following the same principle as before,

$$y = 4x^2 \implies 2|x| = \sqrt{y} \implies x = \pm\frac{1}{2}\sqrt{y}.$$

But we cannot write $g(x) = \pm\frac{1}{2}\sqrt{x}$, as this is not a well-defined function! Thus $f(x) = 4x^2$ is not invertible.

What was different about $4x^2$ that made it **non-invertible**? For each output y, there are two distinct inputs x such that $4x^2 = y$, namely $x = -\frac{1}{2}\sqrt{y}$ and $x = \frac{1}{2}\sqrt{y}$ from our calculations. On the other hand there is one and only one input x such that $-x + 3 = y$.

Definition 3.4. A function $f : A \to B$ is called **injective** if for each b in B there is at most one input a such that $f(a) = b$. Alternatively, if $f(a_1) = f(a_2)$ then $a_1 = a_2$.

Remark:

1. In order for a function f to have an inverse it must be injective. However, while this condition is *necessary*, it is *not sufficient*.
2. Technically, if $f : A \to B$ does not cover all of B, then f^{-1} can only be defined on $f(A)$, the range of f.

3.3.2 Logarithms as Inverses

During our study of exponential functions $f(x) = b^x$, $b > 0$, we saw that **exponentials are one-to-one and hit every value in** $(0, \infty)$. This means there must be an inverse map $f^{-1} : (0, \infty) \to \mathbb{R}$. The inverse of $f(x) = b^x$ is called the **logarithm with base** b, denoted $\log_b$. Let's be precise and give a definition.

Definition 3.5. Let $b > 0$ and set $f(x) = b^x$. The **logarithm with base** b is the unique function

$$\log_b : (0, \infty) \to \mathbb{R}$$

which assigns to each number x in $(0, \infty)$ the output $y = \log_b(x)$, satisfying

$$b^y = x.$$

In words, $\log_b(b^x) = x$ for every real x and $b^{\log_b(x)} = x$ for every x in $(0, \infty)$.

Example 3.8.

(a) If $3^x = 81$, what is x?

(b) Solve for x such that $7^{2x-3} = 35$.

Solution 3.8.

(a) There are two ways to do this. First, if we remember our powers of 3 then we see $3^1 = 3, 3^2 = 9, 3^3 = 27, 3^4 = 81$, so it must be the case that $x = 4$. Alternatively, since $\log_3$ is the inverse of 3^x we can apply it to each side of the equation to solve for x:

$$3^x = 81 \implies \log_3(3^x) = \log_3(81) \implies x = \log_3(81).$$

Both of these approaches are valid, and indeed $\log_3(81) = 4$.

(b) This is a bit more tricky, and remembering powers of 7 won't help us. We're forced to use the logarithm approach, applying $\log_7$ to both sides:

$$7^{2x-3} = 35 \implies \log_7(7^{2x-3}) = \log_7(35) \implies 2x - 3 = \log_7(35) \implies x = \frac{\log_7(35) + 3}{2}$$
$$\therefore\ x = 2.4135\ldots$$

Definition 3.6. The inverse of $f(x) = e^x$ is called the **natural logarithm**, or the logarithm with base e, which we denote by $\ln(\cdot)$. This is the function which satisfies $\ln(e^x) = x$ for all real x and $e^{\ln(x)} = x$ for all $x > 0$.

Example 3.9. Suppose we know $4\ln(2x+3) - 17 = 1$. What value(s) of x solve this equation?

Solution 3.9. We know the exponential function e^x is the inverse of the natural log, so once we get the logarithm on the left alone (i.e. without a coefficient of 4 or a subtracted 17) we can **exponentiate** each side and solve for x:

$$4\ln(2x+3) - 17 = 1 \implies \ln(2x+3) = \frac{9}{2} \implies e^{\ln(2x+3)} = e^{\frac{9}{2}}$$
$$\implies 2x + 3 = e^{\frac{9}{2}} \implies x = \frac{e^{\frac{9}{2}} - 3}{2} = 43.5085\ldots$$

One last example for you to complete on your own.

Example 3.10. True or False: If $2e^{-x} = 5$, then $x = -\frac{\ln(5)}{\ln(2)}$. Explain your reasoning.

Solution 3.10. We can solve directly

$$2e^{-x} = 5 \implies e^{-x} = \frac{5}{2} \underbrace{\implies}_{\ln()} -x = \ln\left(\frac{5}{2}\right)$$
$$\therefore x = -\ln\left(\frac{5}{2}\right).$$

But is this the same as $x = -\frac{\ln(5)}{\ln(2)}$? No! Check numerically with a calculator.

Properties of Logarithms

1. Product Property: $\log_b(ac) = \log_b(a) + \log_b(c)$
2. Coefficient Property: $a\log_b(c) = \log_b(c^a)$
3. Quotient Property: $\log_b\left(\frac{a}{c}\right) = \log_b(a) - \log_b(c)$

Example 3.11. Solve for values of x which satisfy the following equation.

$$\ln\left(x + \frac{2}{x}\right) + \ln(x) = \ln(6)$$

Solution 3.11.

We need to collect all x's together on the left side within a single logarithm so that we can then apply the exponential function to both sides. By the Product Property:

$$\ln\left(x+\frac{2}{x}\right)+\ln(x)=\ln\left(\left(x+\frac{2}{x}\right)x\right)=\ln(x^2+2) \text{ for all } x\neq 0.$$

It is fine to throw away any $x \leq 0$ since the logarithm terms on the left are undefined there. The original equation now says

$$\ln(x^2+2)=\ln(6).$$

At this point we need to undo the logarithm on both sides; this is done by applying its inverse, the exponential

$$e^{\ln(x^2+2)}=e^{\ln(6)}.$$

Since they are inverse functions, we are left with

$$x^2+2=6 \iff x^2=4.$$

Taking the square root of both sides, we obtain $x=\pm 2$. One more subtlety to account for: $x=-2<0$ cannot be plugged into the original equation because $\ln(-2)$ is undefined! Therefore our only solution is $\boxed{x=2}$.

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4 Week 4

Outline
Days: Monday (February 9), Wednesday (February 11), Friday (February 13)
Topics: *Review for Exam 1, take Exam 1, evaluate limits.*

Goals:

- Discuss logarithms, then review for the first exam :(
- Take the first exam and ace it :)
- Give a preview of calculus using limits.

Album of the Week: *Axis: Bold as Love* by *the Jimi Hendrix Experience (1967)*

4.1 Day 1

Goal: Wrap up required material for the first exam which hasn't been covered yet. Review material for first exam.

4.1.1 Exam 1 Topics

Here is a list of textbook sections which we've covered so far. Please note this list is not meant to be exhaustive, only an outline.

- **Functions and Graphs**
  - Exponents
  - Domain, codomain, range
  - Intercepts, zeros
- **Classes of Functions**
  - Polynomials
  - Rational functions
  - Exponential
  - Logarithms
- **Modeling**
  - Compound interest
  - Revenue, cost, profit
  - Break-even analysis
  - Supply, demand
  - Equilibrium point

4.1.2 Review Problems

A collection of problems which can be used for studying, either with notes or as a practice exam.

Problem 1: Simplify the expression to include only positive and negative powers (no division or radicals should be left):

$$\frac{x^{\frac{5}{2}}\sqrt[5]{y^4z^5}x^2}{\sqrt[4]{x}\,y^{-2}z^{\frac{1}{2}}}.$$

(a) $x^{-\frac{1}{3}}y^{\frac{14}{5}}z$

(b) $x^3y^{-\frac{3}{2}}z^2$

(c) $x^{\frac{17}{4}}y^{\frac{14}{5}}z^{\frac{1}{2}}$

(d) $x^{\frac{17}{4}}y^3z^{-\frac{1}{2}}$.

Solution 4.1.

$$\frac{x^{\frac{5}{2}}\sqrt[5]{y^4z^5}x^2}{\sqrt[4]{x}\,y^{-2}z^{\frac{1}{2}}} = \left(x^{\frac{5}{2}}y^{\frac{4}{5}}zx^2\right)\left(x^{-\frac{1}{4}}\,y^2z^{-\frac{1}{2}}\right) = x^{\frac{17}{4}}y^{\frac{14}{5}}z^{\frac{1}{2}} = \text{(c)}$$

Problem 2: Let $f(x) = -x^2 - 4x + 4$, $g(x) = 4x^2$.

(a) Evaluate $f(3)$, $f(g(3))$, and $f(3g(x))$

(b) What type of function is f? (Two acceptable adjectives.)

(c) State the y-intercept of f. Write your answer as a coordinate point (x, y).

(d) Find the roots (zeros) of f.

Solution 4.2.

(a) First, $f(3) = -(3)^2 - 4(3) + 4 = -9 - 12 + 4 = -17$. Next,

$$f(g(3)) = f(4(3)^2) = f(28) = -1436.$$

Finally,

$$f(3g(x)) = f(12x^2) = -(12x^2)^2 - 4(12x^2) + 4 = -144x^4 - 48x^2 + 4.$$

(b) The function $f(x)$ is a quadratic or degree-2 polynomial.

(c) The y-intercept of f has y-coordinate $y = f(0) = 4$. The full coordinate point is then $(0, 4)$.

(d) Set $f(x) = 0$, factor and solve:

$$-x^2 - 4x + 4 = 0 \iff x^2 + 4x - 4 = 0.$$

Well, so much for factoring; this one doesn't have integer roots, so we have to use the quadratic formula instead.

$$x = \frac{-4 \pm \sqrt{16 - 4(-4)}}{2} = \frac{-4 \pm \sqrt{32}}{2}.$$

Problem 3: Let $f(x) = (x^2 - 2)^2$. Compute $f(\sqrt{2x+3})$.

Solution 4.3.

Input $\sqrt{2x+3}$ for x in the equation for $f(x)$:

$$f(\sqrt{2x+3}) = ((\sqrt{2x+3})^2 - 2)^2 = (2x+3-2)^2 = (2x+1)^2.$$

This is just a quadratic function, $f(\sqrt{2x+3}) = 4x^2 + 4x + 1$.

Problem 4: We're investing \$1000 in either Bank A or Bank B. At Bank A, our money will be compounded continuously at an interest rate of 3.5%; at Bank B we will receive a \$150 bonus at the start but the money will be compounded at a 2.5% rate. If we plan to invest for 25 years, which bank should we choose and how much more money will we gain there?

Solution 4.4.

(a) Invest at Bank A, as it yields roughly \$250 more than Bank B

(b) Invest at Bank B, as it yields roughly \$250 more than Bank A

(c) Both banks yield the same amount of money.

(d) None of the above.

For justification, at Bank A (rounding to nearest cent):

$$A(25) = 1000e^{(0.035)(25)} = 2398.88$$

At Bank B:

$$A(25) = 1150e^{(0.025)(25)} = 2148.48$$

Thus despite the initial bonus at Bank B, the higher interest rate at Bank A wins out over time. (Correct response is (a), Bank A yields $\sim$ \$250 more.)

Problem 5: Amanda visits Martin House Books where they charge \$30 for each of the first 4 books purchased, then \$20 for each book purchased *after the first 4*. What is the function representing the price $p(x)$ of x books?

Solution 4.5.

(a) $p(x) = \begin{cases} 30x, & 0 \le x \le 4, \\ 20x, & x > 4 \end{cases}$

(b) $p(x) = \begin{cases} 30x, & 0 \le x \le 4, \\ 20x + 70, & x > 4 \end{cases}$

(c) $p(x) = \begin{cases} 20x, & 0 \le x \le 4, \\ 30(x-4) + 80, & x > 4 \end{cases}$

(d) $p(x) = \begin{cases} 30x, & 0 \le x \le 4, \\ 20x + 40, & x > 4 \end{cases}$.

The correct option is (d). She pays \$30 for each book up until 4, so for $0 \le x \le 4$ the price is $p(x) = 30x$. If she then buys a fifth book, we add \$20 to get to $120 + 20 = 140$; if she buys a sixth book we add \$20 again to get to $120 + 20(2) = 160$, etc. This means for $x > 4$ we have $p(x) = 120 + 20(x - 4) = 20x + 40$. Thus $p(x) = \begin{cases} 30x, & 0 \le x \le 4, \\ 20x + 40, & x > 4 \end{cases}$.

Problem 6: Assume a linear cost model in the following scenario.

(a) Cole sells used golf balls. He has a fixed cost of \$50, and he spends \$75 to package 900 balls on average. What is the cost function for this business?

(b) The demand function at Cole's company is $p(x) = -5x + 100$. What is his revenue function?

(c) How many golf balls does he need to sell to break even?

Solution 4.6.

(a) $C(x) = mx + b = \underbrace{\frac{75}{900}}_{\text{cost per unit produced}} x + \underbrace{50}_{\text{fixed cost}} = \boxed{\frac{1}{12}x + 50}$.

(b) Revenue = price times quantity, so

$$R(x) = p(x) \cdot x = (-5x + 100)x = \boxed{-5x^2 + 100x}.$$

(c) Break-even quantity is the value of x such that $P(x) = 0$ (note this is profit, not price!). We see

$$P(x) = 0 \iff R(x) - C(x) = 0 \iff (-5x^2 + 100x) - (\frac{1}{12}x + 50) = 0$$

$$\iff -5x^2 + \frac{1199}{12}x - 50 = 0.$$

We can use the quadratic formula to solve for the zeros of this quadratic, and they are $\boxed{x = 0.514, 19.47}$.

Problem 7: Solve for x. Give an exact value, not a decimal.

$$-3\ln(6 + 5x) + 2 = 11.$$

Solution 4.7.

$$-3\ln(6 + 5x) + 2 = 11 \implies \ln(6 + 5x) = -3 \implies e^{\ln(6+5x)} = e^{-3}$$

$$\implies 6 + 5x = e^{-3} \implies \boxed{x = \frac{e^{-3} - 6}{5}}.$$

Problem 8: Find the domain of

$$f(x) = \frac{-2x + 5}{\sqrt{16 - 2x}}.$$

Solution 4.8.

We can't divide by 0, so we find

$$\sqrt{16-2x}=0 \iff x=8.$$

We also can't take the square root of a negative, so we see

$$16-2x<0 \iff 16<2x \iff 8<x.$$

This gives the domain as

$$\boxed{D=(-\infty,8)}.$$

4.2 Day 2

Goal: Take an exam, survive the day.

4.3 Day 3

Goal: Gain a preliminary understanding of limits, what they tell us about functions, and how they form the foundation of calculus.

4.3.1 Intuitive Definition of a Limit

Let's say we're given a function $f(x)$ which may or may not be well-defined at $x=a$. We still want to know where the graph moves when x gets close to a, or in other words we want to look at the values of $f(x)$ for $x\approx a$, but $x\neq a$. We can make this notion precise:

Definition 4.1. Suppose $f(x)$ is defined on an open interval containing a, but maybe not at a itself. As x approaches a (but is not equal to a), if the values $f(x)$ all approach a number L we say **the limit as x approaches a of $f(x)$ is L**. This is denoted

$$\lim_{x\to a} f(x)=L.$$

Remarks/Takeaways:

- When taking the limit as $x\to a$ of $f(x)$, we don't care about $f(a)$ itself! It may not be well-defined.
- Since we only care about $f(x)$ for inputs x which are close to a but not equal, we can **estimate the limit as $x\to a$ by plugging in values like $x=a\pm 0.1$, $x=a\pm 0.01$, $x=a\pm 0.001$**, etc.
- It is entirely possible that for $x>a$, $f(x)$ approaches one value L_1 while for $x<a$, $f(x)\to L_2$ where $L_1\neq L_2$. In this instance we say (1) $\lim_{x\to a^+} f(x)=L_1$ (**the left hand limit is L_1**), (2) $\lim_{x\to a^-} f(x)=L_2$ (**the right hand limit is L_2**), and (3) $\lim_{x\to a} f(x)$ does not exist. We'll see this in an example.

Example 4.1. Determine $\lim_{x\to 2} f(x)$ for each of the following functions. If the limit does not exist, write DNE.

1. (1) $f(x)=x^2-1$,

2. (2) $f(x) = \begin{cases} x^2 - 1, & x < 0 \\ 0, & x = 0 \\ 2x + 5, & x > 0 \end{cases}$,

3. (3) $f(x) = \frac{1}{x^2}$.

Solution 4.9.

1. We know this is a parabola, which is a 'nice' function in that the graph has no jumps/holes/asymptotes, so we would expect that $\lim_{x\to 0} f(x) = f(0) = -1$ (this notion is called **continuity**, which we'll define properly in a moment). We can verify this graphically or through the suggested table of values:

x	$x^2 - 1$
-0.1	-0.99
-0.01	-0.9999
-0.001	-0.999999
0.001	-0.999999
0.01	-0.9999
0.1	-0.99

Table 1: Table of points $(x, f(x))$ for x near 0. As x gets close to 0 from either side, $f(x) \to -1$.

2. In this case we have a piecewise function with pieces defined on each side of 0 (it doesn't matter that $f(0) = 0$ because we don't care about the function value at the limit point). From the graph, we can see $\lim_{x\to 0^-} f(x) \neq \lim_{x\to 0^+} f(x)$, hence the limit does not exist. This is reflected in the table of values, which show $\lim_{x\to 0^-} f(x) = -1$, while $\lim_{x\to 0^+} f(x) = 5$.

x	$f(x)$
-0.1	-0.99
-0.01	-0.9999
-0.001	-0.999999
0.001	5.002
0.01	5.02
0.1	5.2

Table 2: Table of points $(x, f(x))$ for x near 0. As x gets close to 0 from the left, $f(x) \to -1$; as x gets close to 0 from the right, $f(x) \to 5$.

3. From the table of values we see that as $x \to 0$, $f(x) = \frac{1}{x^2}$ becomes arbitrarily large and positive (it's a vertical asymptote with both sides tending upward). In this case we say $\lim_{x\to 0} f(x) = \infty$.

x	$1/x^2$
-0.1	100
-0.01	10000
-0.001	1000000
0.001	1000000
0.01	10000
0.1	100

Table 3: Table of points $(x, f(x))$ for x near 0. We see $f(x)$ can be made as large as we want by choosing x close enough to 0.

Definition 4.2. A function $f(x)$ is **continuous at** $x = a$ if $f(a)$ is well-defined and $\lim_{x\to a} f(x) = f(a)$.

4.3.2 Preview of Calc: Average/Instantaneous Rate of Change

Given a function $f(x)$, we say the **average rate of change of f over** $[a, b]$ is the slope of the line passing through $(a, f(a))$ and $(b, f(b))$. This line is known as a **secant line**.

Example 4.2.

1. Let $f(x) = 2x + 5$. Find the equation of the secant line passing through $(1, f(1))$ and $(3, f(3))$. Determine the average rate of change of f over $[1, 3]$.
2. Let $f(x) = 2x^2 + 5$. Find the equation of the secant line passing through $(1, f(1))$ and $(3, f(3))$. Determine the average rate of change of f over $[1, 3]$.

Solution 4.10.

1. This is just the equation of a line passing through two points, something we've seen before. To find the slope,

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{f(3) - f(1)}{3 - 1} = \frac{11 - 7}{2} = 2.$$

   Then from point-slope form, the secant line y is given by

$$y - 7 = 2(x - 1) \iff y = 2x + 5$$

   which is the same as $f(x)$ itself. The average rate of change of $f(x)$ is just given by the slope of the secant line, which is the same as the line itself, 2.

2. The secant line will have slope

$$m = \frac{f(3) - f(1)}{3 - 1} = \frac{23 - 7}{2} = 8.$$

   Using point-slope form, the equation of the secant line is

$$y - 7 = 8(x - 1) \implies \boxed{y = 8x - 1}.$$

In the above example, we found the secant line through $(x, f(x))$ and $(x + h, f(x + h))$ for some values of x and $h > 0$. If f is a function representing the position of an object, then the average rate of change of f is the average velocity of the object from time x to time $x + h$. But what if we don't want an average velocity, but rather the velocity at a specific moment x in time? **This is where limits come in, as we have to let** $h \to 0$.

Definition 4.3. The **instantaneous rate of change of f at $x = a$**, also known as the **derivative of f at** a is

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h},$$

provided the limit exists.

Example 4.3. Suppose $p(t) = -2t^2 + t - 4$ represents the displacement of a person from a stop sign on their street at time t while they go for their daily walk. At what time(s) do they stop walking (i.e. when is their velocity 0)?

Solution 4.11. We know the velocity of the person is given by the derivative of their position function:

$$p'(t) = \lim_{h\to 0} \frac{p(t+h)-p(t)}{h} = \lim_{h\to 0} \frac{-2(t^2+2th+h^2)+(t+h)-4-(-2t^2+t-4)}{h}$$

$$= \lim_{h\to 0} \frac{-4th+h^2+h}{h} = \lim_{h\to 0} -4t+h+1 = -4t+1.$$

The person is stopped or has velocity zero when $p'(t) = -4t+1 = 0$, therefore $t = \frac{1}{4}$. (**Notice: The vertex of the parabola** $p(x) = -2t^2+t-4$ **occurs at** $t = -\frac{b}{2a} = -\frac{1}{2(-2)} = \frac{1}{4}$**!** It's not a coincidence that this coincides with $p'(t) = 0$...)

5 Week 5

Outline
Days: Monday (February 16), Wednesday (February 18), Friday (February 20)
Topics: *Limits, asymptotes, (dis)continuity.*

Goals:

- Calculate limits through algebraic, numeric, or visual methods.
- Define continuity of a function at a point.
- Discuss types of discontinuities: holes, jumps, asymptotes.

Album of the Week: *Rumours* by *Fleetwood Mac (1977)*

5.1 Day 1

Goal: Continue gaining familiarity with limits (visually, algebraically, and how they relate to rates of change).

5.1.1 Calculation of Limits

Example 5.1.

1. Let $f(x) = \frac{x^2+6x+9}{x^2+x-6}$. Calculate $\lim_{x\to -3} f(x)$ and $\lim_{x\to 2} f(x)$.
2. In terms of graph structures, what can we say occurs at these two x values?

Solution 5.1.

1. The limit as $x \to -3$ is equal to 0, while the limit as $x \to 2$ does not exist because the left and right hand limits do not agree ($\lim_{x\to 2^-} f(x) = -\infty$, $\lim_{x\to 2^+} f(x) = +\infty$).
2. Notice $f(x) = \frac{(x+3)^2}{(x+3)(x-2)}$ has a hole at $x = -3$ and a vertical asymptote at $x = 2$.

Limit Laws 1
Let a, c be real numbers (constants). Then

1. $\lim_{x\to a} x = a$,
2. $\lim_{x\to a} c = c$.

Limit Laws 2
Let $f(x)$ and $g(x)$ be functions and assume $\lim_{x\to a} f(x) = L_1$, $\lim_{x\to a} g(x) = L_2$. Then

1. (Sum/Difference) $\lim_{x\to a}(f(x) \pm g(x)) = L_1 \pm L_2$

2. (Product) $\lim_{x\to a}(f(x)g(x)) = L_1L_2$
3. (Powers) $\lim_{x\to a}(f(x))^n = L_1^n$ for positive integers n
4. (Roots) $\lim_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{L_1}$ if n is odd; also true for n even as long as $L_1 \geq 0$.

Example 5.2. Let $f(x) = (x-1)^2\sqrt{3x^2+2}$. Using the above limit laws, calculate...

(a) $\lim_{x\to 0} f(x)$

(b) $\lim_{x\to 2} f(x)$

Solution 5.2.

(a)

$$\lim_{x\to 0}(x-1)^2\sqrt{3x^2+2} = (\lim_{x\to 0}(x-1)^2)(\lim_{x\to 0}\sqrt{3x^2+2})$$

$$= ((\lim_{x\to 0} x) - 1)^2\sqrt{3(\lim_{x\to 0} x^2)+2} = 1\cdot\sqrt{2} = \boxed{\sqrt{2}}.$$

(b) Same steps as (a), just plug in $x = 2$ at the end: $\lim_{x\to 2} f(x) = 1\cdot\sqrt{14} = \boxed{\sqrt{14}}$.

We saw already that limits are what help us construct calculus, so here's an example to work with that idea more.

Example 5.3. Let $f(x) = ax^2 + bx + c$ where $a < 0$.

(i) Calculate $f(x+h)$

(ii) Find $\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

(iii) Find $\lim_{x\to -\frac{b}{2a}} f(x)$

(iv) Find the coordinates (x, y) of the vertex of the parabola.

Solution 5.3.

(i)

$$f(x+h) = a(x+h)^2 + b(x+h) + c = a(x^2+2xh+h^2) + bx + bh + c$$

$$= \boxed{ax^2 + 2axh + ah^2 + bx + bh + c}.$$

(ii) First, we see

$$\frac{f(x+h)-f(x)}{h} = \frac{ax^2+2axh+ah^2+bx+bh+c-(ax^2+bx+c)}{h} = 2ax + b + ah.$$

As a result,

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = 2a(\lim_{h\to 0} h) + b + a(\lim_{h\to 0} h) = \boxed{2ax+b}.$$

(iii) Since f is a quadratic polynomial, it is continuous, hence

$$\lim_{x\to -\frac{b}{2a}} f(x) = f\left(-\frac{b}{2a}\right) = a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c$$

$$= \frac{b^2}{4a} - \frac{b^2}{2a} + c = -\frac{b^2}{4a} + c.$$

(iv) We saw in a previous set of lecture notes that the vertex of a parabola occurs at $x = -\frac{b}{2a}$, so that means the coordinates of the vertex are (based on (iii)) $\left(-\frac{b}{2a}, -\frac{b^2}{4a} + c\right)$.

5.2 Day 2

Goal: Calculate several different kinds of limits using polynomials, rational functions, and functions with vertical asymptotes. Define continuity at a point.

5.2.1 Infinite Limits

We've seen many examples up to this point where a function $f(x)$ increases or decreases without bound as x approaches some value a. For instance, we saw $f(x) = \frac{1}{x^2}$ becomes arbitrarily large, positive as x approaches 0 *from either side*. We can classify three types of limits in this vein:

Definition 5.1. Let $f(x)$ be a function defined on an open interval containing $x = a$.

1. (**Infinite limits from the left**) Suppose as $x \to a$ but $x < a$ that the values of $f(x)$ *increase without bound*. Then we say **the limit as x approaches a from the left of $f(x)$ is positive infinity** and express this by
$$\lim_{x \to a^-} f(x) = \infty.$$
If as $x \to a$ but $x < a$ we see the values of $f(x)$ *decrease without bound*, then we say **the limit as x approaches a from the left of $f(x)$ is negative infinity** and express this by
$$\lim_{x \to a^-} f(x) = -\infty.$$

2. (**Infinite limits from the right**) Suppose as $x \to a$ but $x > a$ that the values of $f(x)$ *increase without bound*. Then we say **the limit as x approaches a from the right of $f(x)$ is positive infinity** and express this by
$$\lim_{x \to a^+} f(x) = \infty.$$
If as $x \to a$ but $x > a$ we see the values of $f(x)$ *decrease without bound*, then we say **the limit as x approaches a from the right of $f(x)$ is negative infinity** and express this by
$$\lim_{x \to a^+} f(x) = -\infty.$$

3. (**2-sided infinite limits**) Suppose as $x \to a$ that the values of $f(x)$ increase without bound. Then we say **the limit as x approaches a of $f(x)$ is positive infinity** and express this by
$$\lim_{x \to a} f(x) = \infty.$$
If as $x \to a$ we see the values of $f(x)$ *decrease without bound*, then we say **the limit as x approaches a of $f(x)$ is negative infinity** and express this by
$$\lim_{x \to a} f(x) = -\infty.$$

Important observations:

- If $f(x)$ has an infinite limit from both sides at $x = a$, meaning $\lim_{x \to a^-} f(x) = \pm\infty$ and $\lim_{x \to a^+} f(x) = \pm\infty$, then $f(x)$ **has a vertical asymptote at** a.

- According to our definition of a limit, $\lim_{x\to a} f(x)$ exists if the values of $f(x)$ approach **a real number** L **for** x **near** a. In the cases $\lim_{x\to a} f(x) = \infty$ or $\lim_{x\to a} f(x) = -\infty$, ∞ **and** $-\infty$ **are not real numbers** so the limit technically does not exist. However, we will still say the limit equals $\pm\infty$ rather than saying the limit DNE.

Example 5.4. Compute the following limits:

1. $\lim_{x\to 1^-} \frac{1}{x-1}$
2. $\lim_{x\to 1^+} \frac{1}{x-1}$
3. $\lim_{x\to 1} \frac{1}{x-1}$
4. $\lim_{x\to 2^-} \frac{-1}{(x-2)^2(x+1)}$
5. $\lim_{x\to 2^+} \frac{-1}{(x-2)^2(x+1)}$
6. $\lim_{x\to 2} \frac{-1}{(x-2)^2(x+1)}$

Solution 5.4.

1. Note if we try to plug in the limiting x value we end up dividing by 0! For infinite limits/limits of rational functions at points where the function is not defined we should use a table of values. Since we want $\lim_{x\to 1^-}$ we should plug in values of x which are close to but stictly less than one.

$$x = \frac{9}{10} \longrightarrow f\left(\frac{9}{10}\right) = \frac{1}{-1/10} = -10$$

$$x = \frac{99}{100} \longrightarrow f\left(\frac{99}{100}\right) = \frac{1}{-1/100} = -100$$

$$x = \frac{999}{1000} \longrightarrow f\left(\frac{999}{1000}\right) = \frac{1}{-1/1000} = -1000.$$

   If we continue in this fashion, we see (heuristically) that for $x \approx 1$ but $x < 1$, $f(x) \approx \frac{1}{\text{very small negative number}} =$ very large negative number. In this case we conclude $\lim_{x\to 1^-} \frac{1}{x-1} = -\infty$.

2. We follow a similar procedure as in the previous part except now $x > 1$ (since we want $\lim_{x\to 1^+}$).

$$x = \frac{11}{10} \longrightarrow f\left(\frac{11}{10}\right) = \frac{1}{1/10} = 10$$

$$x = \frac{101}{100} \longrightarrow f\left(\frac{101}{100}\right) = \frac{1}{1/100} = 100$$

$$x = \frac{1001}{1000} \longrightarrow f\left(\frac{1001}{1000}\right) = \frac{1}{1/1000} = 1000.$$

   If we continue in this fashion, we see that for $x \approx 1$ but $x > 1$, $f(x) \approx \frac{1}{\text{very small positive number}} =$ very large positive number. The only difference from (1) is that $x - 1 > 0$ for $x > 1$! In this case we conclude $\lim_{x\to 1^+} \frac{1}{x-1} = \infty$.

3. From the previous two parts we know $\lim_{x\to 1^-} f(x) = -\infty \neq \infty = \lim_{x\to 1^+} f(x)$. Since left and right side limits don't agree, $\lim_{x\to 1} f(x)$ does not exist.

   Parts (4)-(6) follow a similar procedure...but are the results analogous? What are the differences between the functions? **Squaring matters!**

4. $\lim_{x\to 2^-} \frac{-1}{(x-2)^2(x+1)} = -\infty$
5. $\lim_{x\to +2} \frac{-1}{(x-2)^2(x+1)} = -\infty$
6. $\lim_{x\to 2} \frac{-1}{(x-2)^2(x+1)} = -\infty$

5.2.2 Continuity

Informally, we say a function $f(x)$ is **continuous at** $x = a$ if we can *trace the graph* of f over a *without lifting our pencil off the page.* Let's explore this through an example before giving the formal definition.

Example 5.5. In each case, decide based on the informal definition above if $f(x)$ is continuous at $x = 1$.

1. $f(x) = x - 1$
2. $f(x) = \frac{x^2-2x+1}{x^2-1}$
3. $f(x) = \frac{1}{x-1}$

Solution 5.5.

1. This is the equation of a line with slope 1 and y-intercept -1. We've drawn the graphs of lines before and seen that we never have to pick up the writing utensil, so $f(x)$ is continuous at $x = 1$.
2. This time, $f(x)$ is a rational function which we can write as $f(x) = \frac{(x-1)^2}{(x+1)(x-1)}$. From our discussion of rational functions, we see f has a hole at $x = 1$ since we divide by 0 but could cancel this division if allowed. This means we have to lift our pencil when drawing the graph because there's no output at $x = 1$. This means $f(x)$ is discontinuous at $x = 1$.
3. We saw in the previous section about infinite limits that this function has a vertical asymptote at $x = 1$. In particular, f is not defined at $x = 1$ and we have to lift our pencil when drawing the graph. This means $f(x)$ is discontinuous at $x = 1$.

When we are able to 'draw the graph without lifting our pencil at $x = a$,' we are physically showing that $\lim_{x\to a} f(x)$ exists, $f(a)$ is well-defined, and moreover $\lim_{x\to a} f(x) = f(a)$. These conditions make up the formal definition of continuity.

Definition 5.2. A function $f(x)$ is **continuous at** $x = a$ if

1. $f(a)$ is well-defined
2. $\lim_{x\to a} f(x)$ exists
3. $\lim_{x\to a} f(x) = f(a)$.

5.3 Day 3

Goal: Become masters of limits. Be able to identify when a function is continuous at a point, classify different types of discontinuities.

5.3.1 Limit Mastery

To make sure we're on the same page with limits so far, let's look at a couple of standard examples.

Example 5.6. Let

$$f(x) = \begin{cases} -x^2, & x < -2 \\ 4, & x = -2 \\ \frac{x}{\frac{1}{2}-x}, & -2 < x \leq 1 \\ -3, & 1 < x < 2 \text{ and } x > 2 \\ 5, & x = 2 \end{cases}.$$

1. Identify a point a where $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x)$ but $f(a)$ is not equal to the right or left limit.
2. Compute $\lim_{x\to -2} f(x)$.
3. Identify a point a where the function has an infinite limit. What are the right and left limits there?
4. What is $\lim_{x\to 1} f(x)$?
5. What is $\lim_{x\to 2} f(x)$?

Solution 5.6.
Great idea to sketch this function so that all the information becomes visual.

1. The given conditions mean the function must approach the same limit from either side, but the value of $f(a)$ is not equal to that limit. This kind of point is exhibited at $x = 2$ since $\lim_{x\to 2} f(x) = -3$ but $f(2) = 5$.
2. Although $f(-2) = 4$, we don't care about the function value when calculating limits! Instead we can look at the left and right sided limits separately before comparing their values. We see

$$\lim_{x\to -2^-} f(x) = \lim_{x\to -2^-} -x^2 = -4,$$

$$\lim_{x\to -2^+} f(x) = \lim_{x\to -2^+} \frac{x}{\frac{1}{2}-x} = \frac{-2}{\frac{1}{2}+2} = -\frac{4}{5}.$$

   Since the left limit and right limit are not equal, $\lim_{x\to -2} f(x)$ DNE.
3. This should be an easy one, the only rational function piece is $\frac{x}{\frac{1}{2}-x}$ which is defined over $-2 < x \leq 1$ and has an asymptote at $x = \frac{1}{2}$. Approaching from the left, $\frac{1}{2} - x > 0$ and arbitrarily small, therefore $\frac{x}{\frac{1}{2}-x} \to +\infty$ as $x \to \frac{1}{2}^-$. By similar reasoning, $f(x) \to -\infty$ as $x \to \frac{1}{2}^+$.
4. Much like part (2) we compute left and right limits, then compare.

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \frac{x}{\frac{1}{2}-x} = -2,$$

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} -3 = -3.$$

   Since the limits from either side do not agree, the limit doesn't exist.
5. The function jumps up and attains the value 5 at $x = 2$, but limits don't care about the value of the function at $x = a$! We see the function is identically equal to -3 on either side of $x = 2$, so $\lim_{x\to 2} f(x) = 2$.

5.3.2 Two Facts, a Couple Tricks

Facts: Assume $p(x)$ and $q(x)$ are polynomials.

1. $\lim_{x\to a} p(x) = p(a)$ for any real number a.
2. Provided $q(a) \neq 0$, $\lim_{x\to a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}$.

Recall **a function $f(x)$ is continuous at $x = a$ if $f(a)$ is well-defined, $\lim_{x\to a} f(x)$ exists, and $f(a) = \lim_{x\to a} f(x)$**. The above facts then state **polynomials are continuous everywhere and rational functions are continuous whenever their denominator is nonzero**.

Despite all the limit laws and related ideas we've already seen there are still more difficulties we need to deal with, like when **our function has the form $\frac{f(x)}{g(x)}$ and $\lim_{x\to a} f(x) = \lim_{x\to 0} g(x) = 0$**. Luckily, there are some standard tricks that we can apply.

Steps:

- Check that the given function has the indeterminate form stated above (otherwise the limit likely can be computed using other limit laws).
- Try to simplify $\frac{f(x)}{g(x)}$ in one of three ways: **If f and g are polynomials, cancel common factors**; **If f or g involves a difference including a square root, try multiplying f and g by the *conjugate***; **If f/g is a complex fraction, try simplifying it**.
- Once the previous steps are complete, try to apply limit laws.

Example 5.7.

1. Evaluate $\lim_{x\to -1} \frac{x^2-1}{x^2+3x+2}$
2. Evaluate $\lim_{x\to 5} \frac{\sqrt{x-1}-2}{x-5}$
3. Evaluate $\lim_{x\to 2} \frac{\frac{2}{x}-1}{3(x-1)}$

Solution 5.7.

1. We have a rational function whose numerator and denominator evaluate to zero at the limit point $x = -1$:

$$x^2 - 1\Big|_{x=-1} = (-1)^2 - 1 = 1 - 1 = 0; \qquad x^2 + 3x + 2\Big|_{x=-1} = (-1)^2 - 3 + 2 = 0.$$

   We should try to factor and cancel. The numerator is a difference of squares, $x^2 - 1 = (x + 1)(x - 1)$, while the denominator factors as

$$x^2 + 3x + 2 = (x + 2)(x + 1).$$

   This means we can return to the original limit and simplify:

$$\lim_{x\to -1} \frac{x^2 - 1}{x^2 + 3x + 2} = \lim_{x\to -1} \frac{(x + 1)(x - 1)}{(x + 2)(x + 1)} = \lim_{x\to -1} \frac{x - 1}{x + 2} = \frac{-2}{1} = \boxed{-2}.$$

2. If we evaluate the numerator and denominator at the limit point $x = 5$, we find we're in an indeterminate form:

$$\sqrt{x - 1} - 2\Big|_{x=5} = \sqrt{4} - 2 = 0; \qquad x - 5\Big|_{x=5} = 5 - 5 = 0.$$

   Since the function involves a square root, we should immediately think of multiplying by the conjugate, $-\sqrt{x - 1} - 2$:

$$\frac{\sqrt{x-1}-2}{x-5} = \frac{(\sqrt{x-1}-2)(-\sqrt{x-1}-2)}{(x-5)(-\sqrt{x-1}-2)} = \frac{-(x-1)-2\sqrt{x-1}+2\sqrt{x-1}+4}{(x-5)(-\sqrt{x-1}-2)}$$
$$= \frac{-x+5}{(x-5)(-\sqrt{x-1}-2)} = \frac{-(x-5)}{(x-5)(-\sqrt{x-1}-2)} = \frac{-1}{-\sqrt{x-1}-2}.$$

Now we can take the limit, since plugging in the limit point no longer results in division by 0 or an indeterminate form:

$$\lim_{x\to 5}\frac{\sqrt{x-1}-2}{x-5} = \lim_{x\to 5}\frac{-1}{-\sqrt{x-1}-2} = \boxed{\frac{1}{4}}.$$

3. This is **not an indeterminate form!** We always need to check that we end up with $\frac{0}{0}$ if we try to evaluate at the limit point, but here we don't:

$$\text{Numerator} : \left.\frac{2}{x}-1\right|_{x=2} = 1-1=0$$

$$\text{Denominator} : \left.3(x-1)\right|_{x=2} = 3(1)=3.$$

This is why it pays to **always (always!) check to see if it's in an indeterminate form**. If it's not, and we're not dividing by 0, just plug in:

$$\lim_{x\to 2}\frac{\frac{2}{x}-1}{3(x-1)} = \boxed{0}.$$

6 Week 6

Outline
Days: Monday (February 23), Wednesday (February 25), Friday (February 27)
Topics: *Continuity/Types of discontinuities, limit definition of derivative, derivative rules.*

Goals:

- Identify when a function is (dis)continuous.
- Understand and be able to use the limit definition of a derivative.
- Use the limit definition of a derivative to compute $f'(a)$, find equation of tangent line.
- Develop foundational derivative rules for ease of computation.

Album of the Week: *A Night at the Opera* by *Queen (1975)*

6.1 Day 1

Goal: Wrap up our coverage of continuity. Emphasize identification of different types of discontinuity from equation and/or graph.

6.1.1 Continuity is Key

Recall the following definitions:

Definition 6.1. A function $f(x)$ defined on an open interval containing $x = a$ is **continuous at** a if

1. $f(a)$ is well-defined
2. $\lim_{x\to a} f(x)$ exists, and
3. $\lim_{x\to a} f(x) = f(a)$.

Definition 6.2. A function $f(x)$ is **discontinuous at** $x = a$ if it is not continuous at $x = a$. There are three types of discontinuities:

- (**Hole**)
  $\lim_{x\to a} f(x)$ exists but is not equal to $f(a)$.
- (**Jump**) $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ both exist and are finite, but not equal to each other.
- (**Infinite/Asymptote**) $\lim_{x\to a^-} f(x) = \pm\infty$ or $\lim_{x\to a^+} f(x) = \pm\infty$.

The following example will give broad coverage of some typical questions regarding continuity.

Example 6.1. Suppose we define

$$f(x) = \begin{cases} x, & x < 2, \\ 5 - 2x, & 2 \leq x < 3 \\ 0, & x = 3 \\ \frac{x-4}{x^2-9x+20}, & 3 < x \end{cases}.$$

1. Evaluate $\lim_{x \to 2^-} f(x)$.
2. Calculate $\lim_{x \to 2} f(x)$ or explain why it does not exist.
3. $\lim_{x \to 3^+} f(x)$
4. Identify all points of discontinuity of $f(x)$ and classify their types. Explain your reasoning.

Solution:

1. Crucially, this is a one-sided limit which we can compute with the appropriate function from the piecewise definition:
$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} x = 2$$
2. The input $x = 2$ is at a juncture between two pieces of the piecewise function, so we compute the left and right limits separately:
$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} x = 2;$$
$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} 5 - 2x = 1.$$
Since the left and right limits are not equal, the limit does not exist.
3. Even though $f(3) = 0$, limits do not care about the function value at the limit point! Instead, we see
$$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} \frac{x-4}{x^2 - 9x + 20} = \frac{3-4}{9 - 27 + 20} = -\frac{1}{2}.$$
4. There are several discontinuities, all listed below.
   - @ $x = 2$: jump discontinuity from parts 1 and 2.
   - @ $x = 3$: jump discontinuity.
   - @ $x = 4$: hole! This one is not obvious from how the function is written, important to check for the roots of $x^2 - 9x + 20 = (x - 4)(x - 5)$.
   - @ $x = 5$: vertical asymptote.

6.1.2 Derivatives? Derivatives.

We recently encountered the concept of a secant line of a function over an interval $[a, b]$ and noted that the **slope of the secant line represents the average rate of change of f over the given interval**. This is a crucial starting point as we now move into the realm of instantaneous rate of change.

Let a be a real number and suppose we want to find the instantaneous rate of change of f at a. It's reasonable to say that **the instantaneous rate of change at f can be approximated well by secant lines over intervals of the form $[a - h, a]$ or $[a, a + h]$** where h is a very small, positive number. (**Think of it this way: we want to know the velocity of a car at a specific point in time $t = t_0$; the average speed of the car over an extremely small time span around $t = t_0$ will effectively give us the velocity at $t = t_0$.**) But how do we turn these approximations into an exact value? **Take the limit as $h \to 0$!** Visually, this shifts the secant line of $f(x)$ over $[a, a + h]$ to the tangent line to $f(x)$ at $x = a$.

Fact: The slope of the secant line to the function $f(x)$ over the interval $[a, a + h]$ is $m = \frac{f(a+h)-f(a)}{h}$. This is just rise over run.

Definition 6.3. The **derivative of $f(x)$ at $x = a$** is defined as

$$\boxed{f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}}$$

provided this limit exists. If the limit does not exist we say f **is not differentiable at** $x = a$. Equivalently, the derivative of $f(x)$ at $x = a$ is the slope of the tangent line to the graph of $f(x)$ at $x = a$, provided a well-defined tangent line exists.

Remark: It's not uncommon to see $a+h$ replaced simply by x in the above definition, in which case $h \to 0$ is equivalent to $a + h = x \to a$. This means we could also define

$$\boxed{f'(a) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a}}.$$

Example 6.2. Let (1) $f(x) = 3x^2 - x$, (2) $f(x) = |x|$. Compute $f'(0)$ or show why $f(x)$ is not differentiable at 0.

Solution 6.1.

1. $$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{3h^2 - h - 0}{h} = \lim_{h\to 0} 3h - 1 = -1.$$

2. Recall that $|x| = \begin{cases} -x, & x \leq 0 \\ x, & x \geq 0 \end{cases}$. This means that in order to compute $f'(0)$ we'll have to use the left and right limits:

$$\lim_{h\to 0^-} \frac{f(h) - f(0)}{h} = \lim_{h\to 0^-} \frac{-h}{h} = -1;$$
$$\lim_{h\to 0^+} \frac{f(h) - f(0)}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1.$$

Since the left and right limits don't agree, the limit does not exist. Thus $f(x) = |x|$ is not differentiable at 0.

6.2 Day 2

Goal: Understand how to use the limit definition of the derivative at a point. Extrapolate this definition to view derivatives as functions.

6.2.1 Examples of Derivatives and Related Computations

Last time we saw **the derivative of $f(x)$ at a** is defined by

$$\boxed{f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}}$$

and we said f **is differentiable at a if the above limit exists**.

Buzz words that often get used in relation to derivatives which can be confusing or get forgotten (but shouldn't be!):

- "instantaneous rate of change at $x = a$" is synonymous with $f'(a)$
- "the slope of the tangent line to the graph of $f(x)$ at a" is synonymous with $f'(a)$
- "velocity at time $t = t_0$" is synonymous with $f'(t_0)$ where $f(t)$ is the position function.

Example 6.3. Let $f(x) = -2x^2 + 3x + 1$. Let's compute...

(a) The derivative of f at $x = 1$.

(b) The instantaneous rate of change of f at $x = -1$.

(c) The **equation of the tangent line** to f at $x = 0$.

Solution 6.2.

(a) We use the limit definition:

$$f'(1) = \lim_{h\to 0} \frac{f(1+h) - f(1)}{h} = \lim_{h\to 0} \frac{(-2(1+h)^2 + 3(1+h) + 1) - 2}{h}$$
$$= \lim_{h\to 0} \frac{-2(h^2 + 2h + 1) + 3h + 3 + 1 - 2}{h} = \lim_{h\to 0} \frac{-2h^2 - 4h - 2 + 3h + 2}{h}$$
$$= \lim_{h\to 0} \frac{-2h^2 - h}{h} = \boxed{-1}.$$

(b) Remember, instantaneous rate of change is just fancy talk for derivative. The computation is similar to (1):

$$f'(-1) = \lim_{h\to 0} \frac{f(-1+h) - f(-1)}{h} \boxed{= 7}.$$

(c) This one is a bit more involved since it asks for the equation of a line. Recall **the definition of the derivative at a is the slope of the tangent line at a**, so we should set $m = f'(0)$, $x_1 = 0$, $y_1 = f(0)$ in point-slope form.

$$f'(0) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{-2h^2 + 3h + 1 - 1}{h} = 3.$$

This means the slope of our tangent line is $m = 3$. The equation of the tangent line at $x = 0$ is then

$$y - 1 = 3(x - 0) \to \boxed{y = 3x + 1}.$$

Example 6.4. Let $f(x) = \begin{cases} 2x, & x \leq 1 \\ -x - 3, & x > 1 \end{cases}$. Compute $f'(0)$, $f'(1)$, and $f'(2)$.

1. Since we want the derivative at $x = 0 < 1$, we only use the piece of the function given by $2x$:

$$\lim_{h\to 0} \frac{2h}{h} \boxed{= 2}.$$

Notice that **the derivative of a line is just its slope**.

2. This is tricky; we have to take left and right side limits separately since the input $x = 2$ is at a transition of the piecewise definition. We see

$$\lim_{h\to 0^-} \frac{f(1+h)-f(1)}{h} = \lim_{h\to 0^-} \frac{2(1+h)-2}{h} = 2,$$

since when $h < 0$, $1 + h < 1$, so $f(1 + h) = 2(1 + h)$. On the other hand, when $h > 0$, $1 + h > 1$, so $f(1 + h) = -(1 + h) - 3 = -h - 4$...

$$\lim_{h\to 0^+} \frac{f(1+h)-f(1)}{h} = \lim_{h\to 0^+} \frac{-h-6}{h} = -\infty.$$

Uh oh...this means f is not differentiable at 1. **The reason why? f is not continuous at 1, and a function cannot be differentiable where it's not continuous!**

3. By the same reasoning as in (1) (or a similar computation), $\boxed{\text{f'(2)=-1}}$.

6.2.2 Derivative as a Function

So far we've only dealt with $f'(a)$ where a is a fixed number. However, **we can just treat $f'(x)$ as a function of x in its own right** and plug in different values of x.

Definition 6.4. The **derivative function of f**, known simply as **the derivative of f**, is the function $f'(x)$ whose domain consists of all x such that the following limit exists:

$$\boxed{f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}}.$$

We'll be computing derivatives and looking at their properties for the next month or two, so let's keep getting experience through examples.

Example 6.5. Let $f(x) = \sqrt{x+1}$. Compute $f'(x)$. (Hint: multiplying by the conjugate never stops being useful...)

Solution 6.3. Just apply the limit definition:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{\sqrt{x+h+1}-\sqrt{x+1}}{h}$$

$$= \lim_{h\to 0} \frac{\sqrt{x+h+1}-\sqrt{x+1}}{h} \cdot \frac{\sqrt{x+h+1}+\sqrt{x+1}}{\sqrt{x+h+1}+\sqrt{x+1}} = \lim_{h\to 0} \frac{h}{h(\sqrt{x+h+1}+\sqrt{x+1})}$$

$$\boxed{= \frac{1}{2\sqrt{x+1}}}.$$

6.2.3 Derivatives and Increasing/Decreasing

Example 6.6. This problem will help us solidify the connection between a function being increasing/decreasing and the sign of its derivative. Let $f(x) = -2x^2 + 3x$.

(i) Calculate $f'(x)$.

(ii) Determine the intervals on which $f(x)$ is strictly increasing and strictly decreasing. (Hint: remember the vertex of $ax^2 + bx + c$ occurs as $x = -b/2a$).

(iii) Determine the intervals on which $f'(x) > 0$ and $f'(x) < 0$.

(iv) Is there a connection between the answers to (2) and (3)? Explain.

Solution 6.4.

(i) We apply the limit definition (which we now have burned into our memories):

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\big(-2(x+h)^2 + 3(x+h)\big) - \big(-2x^2 + 3x\big)}{h}$$
$$= \lim_{h\to 0} \frac{-2(x^2 + 2xh + h^2) + 3x + 3h + 2x^2 - 3x}{h} = \lim_{h\to 0} \frac{-4xh - 2h^2 + 3h}{h} = \lim_{h\to 0} -4x - 2h + 3$$
$$= \boxed{-4x + 3}.$$

(ii) The vertex of $f(x)$ occurs at $x = -\frac{b}{2a} = \frac{-3}{2(-2)} = \frac{3}{4}$. Since it's a downward facing parabola, this means $f(x)$ **is strictly increasing on** $(-\infty, 3/4)$ and $f(x)$ **is strictly decreasing on** $(3/4, \infty)$.

(iii) We can solve for where $f'(x) > 0$:

$$f'(x) = -4x + 3 > 0 \iff 3 > 4x \iff x < \frac{3}{4},$$

i.e. $f'(x) > 0$ **on the interval** $(-\infty, 3/4)$. By a similar calculation,

$$f'(x) = -4x + 3 < 0 \iff 3 < 4x \iff \frac{3}{4} < x,$$

so $f'(x) > 0$ **on the interval** $(3/4, \infty)$.

(iv) Yes, there's a connection. **The function is strictly increasing exactly where** $f'(x) > 0$, and similarly **the function is strictly decreasing exactly where** $f'(x) < 0$. This makes sense because having $f'(x) > 0$ (resp. < 0) means the slope of the tangent line is positive (resp.negative), and a line with a positive (resp. negative) slope is an increasing (resp. decreasing) function.

Definition 6.5. A differentiable function is **strictly increasing** when $f'(x) > 0$ and **strictly decreasing** when $f'(x) < 0$.

6.3 Day 3

Goal: Understand more basic computations of derivatives using the limit definition. See higher order derivatives, first derivative 'rules'.

Example 6.7. Let $f(x) = \sqrt{x}$. Use your knowledge of indeterminate forms to compute $f'(x)$ via the limit definition. What is $\lim_{x\to 0^+} f'(x)$? Does this visually make sense?

Solution 6.5.

Applying the definition we see

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}.$$

This is **an indeterminate form involving a square root** since the numerator and denominator both go to 0 as $h \to 0$. Our next step is to **multiply by the conjugate**: we can use either $\sqrt{x+h}+\sqrt{x}$ **or** $-\sqrt{x+h}-\sqrt{x}$!

$$\lim_{h\to 0}\frac{\sqrt{x+h}-\sqrt{x}}{h}=\lim_{h\to 0}\frac{(\sqrt{x+h}-\sqrt{x})(\sqrt{x+h}+\sqrt{x})}{h(\sqrt{x+h}+\sqrt{x})}$$

$$=\lim_{h\to 0}\frac{h}{h(\sqrt{x+h}+\sqrt{x})}=\lim_{h\to 0}\frac{1}{(\sqrt{x+h}+\sqrt{x})}=\boxed{\frac{1}{2\sqrt{x}}}.$$

For the second part of the question, this tells us $\boxed{\lim_{x\to 0^+} f'(x)=+\infty}$. This makes sense because **the graph of $\sqrt{x}$ is very close to being a vertical line near $x=0$, i.e. the rate of change is positive infinity**.

Notation: So far we've denoted the derivative of $f(x)$ by $f'(x)$ or simply f'. Another common notation is as follows:

$$f'=\frac{df}{dx}.$$

The strange looking fraction $\frac{d}{dx}$ is the *derivative operator*; we think of $\frac{d}{dx}$ as being a tool which we use to take the derivative of a function. Assuming a function f is 'nice' we can take the derivative of f', that is the derivative of the derivative. We denote this new function by f'' and define it by the only reasonable limit

$$f''(x)=\lim_{h\to 0}\frac{f'(x+h)-f'(x)}{h}.$$

We can continue taking derivatives as long as the resulting function is nice (meaning the limit exists)...

Definition 6.6. The **nth derivative of** $f(x)$, $f^{(n)}(x)$ is defined recursively by the limit

$$f^{(n)}(x)=\lim_{h\to 0}\frac{f^{(n-1)}(x+h)-f^{(n-1)}(x)}{h}.$$

We **denote the nth derivative by** $f^{(n)}(x)$ **or** $\frac{d^n f}{dx^n}$.

Example 6.8. Let $f(x)=3x+5$. Compute the first and second derivatives of f.

Solution 6.6.

We can find $f'(x)$ just based on what we've learned about lines. We know **lines have constant rate of change equal to their slope**, so because the derivative gives the rate of change at any given input x, $\boxed{f'(x)=3}$. This can also be determined using the limit definition:

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}\frac{3(x+h)+5-(3x+5)}{h}=\lim_{h\to 0}\frac{3h}{h}=3.$$

To compute the second derivative, f'', we can again intuitively reason: f'' is the derivative of $f'(x)=3$ which is constant; a constant function never changes, i.e. has a rate of change of 0, so $\boxed{f''(x)=0}$. Or, back to the limit definition (using the fact that $f'(x+h)=f'(x)=3$)

$$f''(x)=\lim_{h\to 0}\frac{f'(x+h)-f'(x)}{h}=\lim_{h\to 0}\frac{3-3}{h}=\lim_{h\to 0}0=0.$$

From what we've seen above or by using the limit definition, we can find the following rules for derivative computations.

Foundational Derivative Rules:

- (Constant Rule) If $f(x) = c$ is a constant, then $f'(x) = 0$ for all x.
- (Scaling) Let c be any constant. If $f(x)$ is differentiable then $\frac{d}{dx}(cf(x)) = cf'(x)$.
- (Additivity) If $f(x)$ and $g(x)$ are differentiable, then $\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$.
- (Power Rule) Let $f(x) = x^n$. Then $f'(x) = nx^{n-1}$.

Example 6.9. Calculate $f'(x)$ for $f(x) = 3x^2$ (i) using the limit definition, and (ii) using the derivative rules.

Solution 6.7.

(i) According to the limit definition we have

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{3(x+h)^2 - 3x^2}{h}$$
$$= 3\left(\lim_{h\to 0} \frac{x^2 + h^2 + 2xh - x^2}{h}\right) = 3\left(\lim_{h\to 0} h + 2x\right) = 6x.$$

This shows $f'(x) = 6x$.

(ii) Using the power rule, the exponent of 2 comes down as a coefficient and then the new exponent is $2 - 1 = 1$:

$$f'(x) = 2 \cdot 3x^1 = 6x.$$

The two answers agree as expected.

7 Week 7

Outline
Days: Monday (March 2), Wednesday (March 4), Friday (March 6)
Topics: *Derivatives, chain rule, applications to business functions.*

Goals:

- Compute (many) derivatives using derivative rules.
- Understand how to gain information about function shape and other properties via derivatives.
- Introduce the notion of marginal functions and their uses.

Album of the Week: ***Control*** **by** ***Janet Jackson (1986)***

7.1 Day 1

Goal: See all the basic derivative rules, begin understanding how to use them in computations.

Here is the list of all basic derivative rules covered in class up to and including this date.

Theorem: Let f, g be differentiable functions, c a constant.

1. (**Constant Rule**) The derivative of a constant is zero:
$$\frac{d}{dx}(c) = 0.$$
2. (**Power Rule**) For any real number n,
$$\frac{d}{dx}(x^n) = nx^{n-1}.$$
3. (**Sum/Difference Rule**) The derivative of a sum/difference of two functions is equal to the sum/difference of the derivatives:
$$\frac{d}{dx}(f(x) \pm g(x)) = \frac{df}{dx} \pm \frac{dg}{dx} = f'(x) \pm g'(x).$$
4. (**Constant Multiple Rule**) The derivative of a constant multiple of a function f is the constant times the derivative of f:
$$\frac{d}{dx}(cf(x)) = c\frac{df}{dx} = cf'(x).$$
5. (**Product Rule**)
$$\frac{d}{dx}(f(x)g(x)) = \left(\frac{df}{dx}\right)g(x) + f(x)\left(\frac{dg}{dx}\right) = f'(x)g(x) + f(x)g'(x).$$

6. (**Quotient Rule**)

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\left(\frac{df}{dx}\right)g(x) - \left(\frac{dg}{dx}\right)f(x)}{(g(x))^2}$$

wherever $g(x) \neq 0$.

Seeing the statement of these rules won't guarantee you know how to use them! Seeing someone else (an instructor or a friend or ChatGPT) use these rules won't help guarantee you know how to use them! The only way to become well-versed in taking derivatives is to struggle through many (many) examples on your own.

These examples increase in difficulty as they go. The first couple will only utilize the Power Rule, the Sum/Difference Rule, but then the rest will be incorporated. In some places I've tried to indicate which rule is being used at each step in the calculation by writing the rule number underneath the equals signs.

Example 7.1. Calculate $f'(x)$.

1. $f(x) = x^4$
2. $f(x) = x^{-\frac{7}{16}}$

Solution 7.1.

1. $f'(x) = 4x^{4-1} = 4x^3$.
2. $f'(x) = -\frac{7}{16}x^{-\frac{7}{16}-1} = -\frac{7}{16}x^{-\frac{23}{16}}$.

Example 7.2. Compute the derivative of each function.

1. $f(x) = x^{65537} - x^{-\frac{1}{2}}$
2. $g(x) = x^2 + x + 1$

Solution 7.2.

1. $$f'(x) \underbrace{=}_{\mathbf{3}} \frac{d}{dx}(x^{65537}) - \frac{d}{dx}(x^{-\frac{1}{2}}) \underbrace{=}_{\mathbf{2}} 65537x^{65536} + \frac{1}{2}x^{-\frac{3}{2}}.$$

2. $$g'(x) \underbrace{=}_{\mathbf{3}} \frac{d}{dx}(x^2) + \frac{d}{dx}(x) + \frac{d}{dx}(1) \underbrace{=}_{\mathbf{1,2}} 2x + 1.$$

Example 7.3. Compute the derivatives of the following functions.

1. $h(x) = -\frac{1}{4}x^2 + 5x - 7$
2. $f(x) = -x^{\frac{2}{3}} + 51x^3 - x$

Solution 7.3.

1.
$$h'(x) \underbrace{=}_{\mathbf{3,4}} -\frac{1}{4}\frac{d}{dx}(x^2) + 5\frac{d}{dx}(x) - \frac{d}{dx}(7) \underbrace{=}_{\mathbf{1,2}} -\frac{1}{4}(2x) + 5$$
$$= -\frac{1}{2}x + 5.$$

2.
$$f'(x) \underbrace{=}_{\mathbf{3,4}} -\frac{d}{dx}(x^{\frac{2}{3}}) + 51\frac{d}{dx}(x^3) - \frac{d}{dx}(x)$$
$$\underbrace{=}_{\mathbf{2}} -\frac{2}{3}x^{\frac{2}{3}-1} + 51(3x^2) - 1 = -\frac{2}{3}x^{\frac{2}{3}-1} + 153x^2 - 1$$

Example 7.4. We know from the power rule and constant multiple rule that if $f(x) = 2x^3$, $f'(x) = 6x^2$. Using the product rule, show the derivative of $g(x) = (2x)(x^2)$ is also $6x^2$.

Solution 7.4.
$$g'(x) \underbrace{=}_{\mathbf{5}} \left(\frac{d}{dx}(2x)\right)(x^2) + (2x)\left(\frac{d}{dx}(x^2)\right) \underbrace{=}_{\mathbf{2}} (2)(x^2) + (2x)(2x)$$
$$= 2x^2 + 4x^2 = 6x^2.$$

Example 7.5. Let $f(x) = \frac{\sqrt{x}+3x}{x^2+9}$.

1. Find $f'(x)$
2. Find the equation of the tangent line at $x = 1$.

Solution 7.5.

1. The function is a quotient, so we use the quotient rule (shockingly).
$$f'(x) \underbrace{=}_{6} \frac{\left(\frac{d}{dx}(\sqrt{x}+3x)\right)(x^2+9) - \left(\frac{d}{dx}(x^2+9)\right)(\sqrt{x}+3x)}{(x^2+9)^2}$$
$$\underbrace{=}_{1,2,3,4} \frac{\left(\frac{1}{2\sqrt{x}}+3\right)(x^2+9) - (2x)(\sqrt{x}+3x)}{(x^2+9)^2}.$$

2. To find the equation of the tangent line at $x = 1$ we need two pieces of information: the slope of the tangent line at $x = 1$, i.e. $f'(1)$, and the coordinates of a point which the line passes through. Since the line is tangent to f at $x = 1$ it passes through $(1, f(1)) = (x_1, y_1)$. Thus
$$f'(1) = \frac{\left(\frac{1}{2\sqrt{1}}+3\right)(1^2+9) - (2(1))(\sqrt{1}+3(1))}{(1^2+9)^2} = \frac{\left(\frac{1}{2}+3\right)(10) - (2)(4)}{100}$$
$$= \frac{\frac{7}{2}\cdot 10 - 8}{100} = \frac{\frac{70-16}{2}}{100} = \frac{54}{200} = \frac{27}{100}.$$
We also know $f(1) = \frac{\sqrt{1}+3(1)}{1^2+9} = \frac{4}{10} = \frac{2}{5}$. Plugging into point-slope form,

$$y - y_1 = m(x - x_1) \iff y - f(1) = f'(1)(x - 1)$$

$$\iff \boxed{y - \frac{2}{5} = \frac{27}{100}(x - 1)}.$$

As a sanity check we can plot the function and this tangent line in Desmos to see if it looks correct.

7.2 Day 2

Goal: Learn about the rules for differentiating e^x and $\log_b(x)$, then spend time putting the derivative rules into practice with more examples.

7.2.1 Two More Derivative Rules

So far we've collected six different derivative rules, but we need to add a couple more in order to differentiate functions involving e^x and $\log_b(x)$.

Theorem:

1. $$\frac{d}{dx}(b^x) = (\ln(b))b^x.$$

2. $$\frac{d}{dx}(\log_b(x)) = \frac{1}{x\ln(b)}, \quad x \neq 0.$$

Corollary: Since $\ln(e) = 1$, the above Theorem implies

$$\frac{d}{dx}(\ln(x)) = \frac{1}{x}, \quad x \neq 0$$

and

$$\frac{d}{dx}(e^x) = e^x.$$

Example 7.6. Calculate $f'(x)$ in each of the following cases.

1. $f(x) = 3\log_2(x) - 4e^x + \frac{1}{2}x^6$
2. $f(x) = \frac{\ln(x)}{e^x}$

Solution 7.6.

1. $f'(x) = \frac{3}{x\ln(2)} - 4e^x + 3x^5$
2. $f'(x) = \frac{\left(\frac{1}{x}\right)e^x - e^x\ln(x)}{e^{2x}}$.

Here is our updated list of derivative rules:

Theorem: Let f, g be differentiable functions, c a constant.

1. (**Constant Rule**) The derivative of a constant is zero:
$$\frac{d}{dx}(c) = 0.$$

2. (**Power Rule**) For any real number n,
$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

3. (**Sum/Difference Rule**) The derivative of a sum/difference of two functions is equal to the sum/difference of the derivatives:
$$\frac{d}{dx}(f(x) \pm g(x)) = \frac{df}{dx} \pm \frac{dg}{dx} = f'(x) \pm g'(x).$$

4. (**Constant Multiple Rule**) The derivative of a constant multiple of a function f is the constant times the derivative of f:
$$\frac{d}{dx}(cf(x)) = c\frac{df}{dx} = cf'(x).$$

5. (**Product Rule**)
$$\frac{d}{dx}(f(x)g(x)) = \left(\frac{df}{dx}\right) g(x) + f(x) \left(\frac{dg}{dx}\right) = f'(x)g(x) + f(x)g'(x).$$

6. (**Quotient Rule**)
$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\left(\frac{df}{dx}\right) g(x) - \left(\frac{dg}{dx}\right) f(x)}{(g(x))^2}$$
wherever $g(x) \neq 0$.

7. (**Exponential Rule**)
$$\frac{d}{dx}(b^x) = (\ln(b))b^x.$$

8. (**Logarithm Rule**)
$$\frac{d}{dx}(\log_b(x)) = \frac{1}{x\ln(b)}, \quad x \neq 0.$$

Problems to Work Through

1. Differentiate $f(x) = -2x^4 + 3x^3 - 5x^2 + 8x - 13$.

2. Find $g'(x)$ where $g(x) = -2e^x(2x - 5)$.

3. Compute the derivative of $f(x) = 1000 + \frac{5-x}{e^x+3}$.

4. Find the equation of the tangent line to $g(x) = \frac{x^3}{3} - x + 9$ at $x = 1$.

5. Compute the derivative of $f(x) = \frac{x \ln(x)}{3x^2+1}$. Is the domain of $f'(x)$ the same as the domain of $f(x)$?

6. Differentiate $g(x) = e^{7 \ln(x^2)}$. (Hint: Logarithm rules or chain rule.)

7. Compute the derivative of $f(x) = \frac{\log_5(x)+x^{-\frac{1}{5}}}{-x-5}$.

8. Let $f(x) = \frac{\sqrt{2}}{2}x^2 - \sqrt{2}x + 10$. What is the minimum value of $f(x)$? What is the value of the derivative where the minimum occurs?

9. Evaluate $g'(0)$ where

$$g(x) = \begin{cases} -\frac{1}{2}, & x \leq -3, \\ \frac{1}{x+1}, & -3 \leq x \leq -2 \\ \frac{1}{2}x, & -2 \leq x \leq 1 \\ \frac{1}{2}, & 1 \leq x \end{cases}.$$

10. Bonus: Differentiate

$$f(x) = \frac{x \ln(x^2)}{(2x+3)e^x}.$$

7.3 Day 3

Goal: Understand how to use the **chain rule** to compute the derivative of a composition of functions.

7.3.1 Listen to the wind blow, watch the sun rise

If anyone sees this and understands the musical reference in the subsection title, let me know.

Chain Rule: Suppose f and g are differentiable functions and f is well-defined on the range of g. Then

$$\boxed{\frac{d}{dx}(f(g(x))) = f'(g(x))g'(x)}.$$

Remark: Verbally we read the right side as saying **the derivative of the outside function, composed with the inside function, times the derivative of the inside function**. The trickiest part at first is often to identify the 'inside' function.

Example 7.7. Let $f(x) = (x+2)^2$. Compute $f'(x)$ first by foiling out, then by using the chain rule.

Solution 7.7.

If we foil we see $f(x) = x^2 + 4x + 4$, therefore $f'(x) = 2x + 4$. Using the chain rule, the 'outside function' is the squaring function, while the 'inside function' is $x + 2$. In other words, $f(x) = (x^2) \circ (x + 2)$. We then see via chain rule

$$f'(x) = 2(x+2)(1) = 2(x+2) = 2x + 4$$

which gives the same result.

Example 7.8. Compute the first derivative of the following functions using the chain rule.

1. $f(x) = 2e^{x^2+2}$
2. $f(x) = \frac{1}{\ln(x)+x}$
3. Show that the derivative of $f(x) = x^r$ is $f'(x) = rx^{r-1}$ by using the chain rule to differentiate $g(x) = e^{r\ln(x)}$.

Solution 7.8.

1. The 'outside function' is e^x while the 'inside function' is $x^2 + 2$. The chain rule then gives

$$f'(x) = 2e^{x^2+2} \cdot \frac{d}{dx}(x^2 + 2) = 4xe^{x^2+2}.$$

2. This requires a bit of rearrangement. If we write $f(x) = (\ln(x) + x)^{-1}$, then the 'outside function' is x^{-1} while the 'inside function' is $\ln(x) + x$. As a result

$$\begin{aligned} f'(x) = -(\ln(x) + x)^{-2} \cdot \frac{d}{dx}(\ln(x) + x) &= -(\ln(x) + x)^{-2} \left(\frac{1}{x} + 1\right) \\ &= -\frac{\frac{1}{x} + 1}{(\ln(x) + x)^2}. \end{aligned}$$

3. Note $g(x) = f(x)$, it's just written in a different form using exponent and log rules. The 'outside function' is e^x while the 'inside function' is $r\ln(x)$, so

$$g'(x) = \underbrace{e^{r\ln(x)}}_{=x^r} \cdot \frac{d}{dx}(r\ln(x)) = x^r \left(\frac{r}{x}\right) = rx^{r-1}.$$

7.3.2 Derivatives as Rates of Change: Marginal Functions

In business math, the term **marginal** means we use the derivative of some function.

Definition 7.1.

1. If $C(x)$ is the cost of producing x items, the **marginal cost** is $\boxed{MC(x) = C'(x)}$ (sorry for the awful notation $MC(x)$).
2. If $R(x)$ is the revenue from making x items, the **marginal revenue** is $\boxed{MR(x) = R'(x)}$.
3. If $P(x) = R(x) - C(x)$ is the profit from making x items, the **marginal profit** is

$$\boxed{MP(x) = P'(x) = R'(x) - C'(x) = MR(x) - MC(x)}.$$

Recall from the limit definition of the derivative that the marginal cost (similar for revenue and profit) can be written

$$MC(x) = \lim_{h\to 0} \frac{C(x+h) - C(x)}{h}.$$

If we only need to approximate $MC(x)$, then it suffices to choose a particular small value of h and evaluate $\frac{C(x+h)-C(x)}{h}$. Since our input x is a number of items, it's reasonable to choose $h = 1$. For this approximation we see the following fact (written for cost, revenue, and profit):

Fact:

1. $MC(x) = C'(x) \approx C(x+1) - C(x)$, therefore $C'(x)$ **is approximately the cost to produce the** $(x+1)^{\textbf{st}}$ **item, i.e. one more item after** x.
2. $MR(x) = R'(x) \approx R(x+1) - R(x)$, therefore $R'(x)$ **is approximately the revenue associated with the** $(x+1)^{\textbf{st}}$ **item, i.e. one more item after** x.
3. $MP(x) = P'(x) \approx P(x+1) - P(x)$, therefore $P'(x)$ **is approximately the profit from the** $(x+1)^{\textbf{st}}$ **item, i.e. one more item after** x.

It is important to be able to interpret and distinguish between what $C(x)$, $MC(x)$, $R(x)$, $MR(x)$, $P(x)$, and $MP(x)$ all mean. This is best done through examples.

Example 7.9. Suppose the price function for a product is $p(x) = 9 - 0.03x$ for $0 \leq x \leq 300$. (1) Find $R(x)$. (2) Estimate the revenue from the 101st product using $R'(x)$ (in the **Fact**). (3) Calculate the actual revenue from the 101st item; is the estimate decent?

Solution 7.9.

1. $R(x) = p(x) \cdot x = -0.03x^2 + 9x$
2. Note $R'(x) = -0.06x + 9$, so $R'(100) = -0.06(100) + 9 = -6 + 9 = 3$, so according to the **Fact** the revenue from the 101st item will be \$3.
3. The actual revenue is
$$R(101) - R(100) = 602.97 - 600 = \$2.97$$
so the estimate is only off by 3 cents.

8 Week 8

Outline
Days: Monday (March 9), Wednesday (March 11), Friday (March 13)
Topics: *Exam review, Exam 2, elasticity of demand.*

Goals:

- Review for the second exam.
- Ace the second exam.
- See elasticity of demand for the first time.

Album of the Week: *Master of Puppets* by *Metallica (1986)*

8.1 Day 1

Goal: Review topics which will be covered on Exam 2.

8.1.1 Sections Covered

Chapter 2 - Limits and Continuity

- (2.1) Understand how average rates of change correspond to slopes of secant lines; see that instantaneous rates of change correspond to slopes of tangent lines (limits of secant lines).
- (2.2) Be able to evaluate limits (both full and one sided) using a table of values or the graph of a function. Determine when a limit is infinite (vertical asymptote) or does not exist.
- (2.3) Recognize and be able to use basic limit laws, how to determine limits of polynomials and rational functions. Know how to evaluate a limit in indeterminate form by factoring and canceling, multiplying by the conjugate, or simplifying a complex fraction.
- (2.4) Know the definition of continuity of a function at a point. Be able to identify points of discontinuity and their types.

Chapter 3 - Derivatives

- (3.1) Understand the definition of the derivative at a point as the slope of the tangent line or the instantaneous rate of change. Be able to calculate $f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ (the limit definition of the derivative). Be able to find the equation of a tangent line.
- (3.2) Define the derivative $f'(x)$ as a function and understand its connection to continuity (differentiable implies continuous; continuous may not be differentiable). Be able to sketch derivative based on a graph of $f(x)$.
- (3.3, 3.6, 3.9) Know how to use derivative rules to calculate $f'(x)$ for various functions $f(x)$.
- (3.4) Be able to interpret the derivative of a function as a rate of change. In particular, be familiar with interpretations of marginal functions (cost, revenue, profit).

8.1.2 Example Problems for Chapter 2

Example 8.1. Let $f(x) = -2x^2 + 1$.

(a) Find the equation of the secant line to $f(x)$ over the interval $[0, 3]$.

(b) Calculate the instantaneous rate of change of $f(x)$ at $x = 2$.

Solution 8.1.

Example 8.2. Sketch the graph of a function $f(x)$ which satisfies

(i) $\lim_{x\to -2} f(x) = +\infty$

(ii) $\lim_{x\to -1^-} f(x) = 2$, $\lim_{x\to -1^+} f(x) = -2$, and $f(-1) = 1$

(iii) $f(0) = 0$

(iv) $\lim_{x\to 1} f(x) = -1$, $f(1) = 3$

(v) $\lim_{x\to 2^-} f(x) = -\infty$, $\lim_{x\to 2^+} f(x) = +\infty$

Solution 8.2.

Example 8.3. Find a value of a so that the function

$$g(t) = \begin{cases} at^2 - 1, & t \leq 1 \\ -t + 2, & t > 1 \end{cases}$$

is continuous at $t = 1$, or show why such a value of a does not exist.

Solution 8.3.

Example 8.4. Let $h(x) = \frac{x+1}{(x^2-1)(x-3)}$.

(a) Evaluate $\lim_{x\to 3^-} h(x)$.

(b) Evaluate $\lim_{x\to -1} h(x)$.

Solution 8.4.

8.1.3 Example Problems for Chapter 3

Example 8.5. Let $f(x) = 3x^3 - x$. State but do not evaluate the limit definition of $f'(1)$.

Solution 8.5.

Example 8.6. True or False: If the graph of a function has a sharp corner or cusp at $x = 2$, then $f'(2) = 0$.

Solution 8.6.

Example 8.7. Compute the derivative $f'(x)$ for each of the following functions $f(x)$.

(i) $f(x) = xe^{-2x}$

(ii) $f(x) = 3x^{-2} - 2\ln(3x)$

(iii) $f(x) = \frac{x}{1+e^x} + 5\sqrt{x}$

Solution 8.7.

8.2 Day 2

Goal: Take an exam, try to maintain a healthy blood pressure and outlook on life!

8.3 Day 3

Goal: Understand what is meant by 'elasticity of demand' and interpret associated examples. Mention what a linear approximation is and how it can be used.

8.3.1 Elasticity of Demand

Elasticity of demand is a measure of how sensitive the demand function of a product is to changes in price. Recall we originally defined the demand function as $p(x)$, the price when x units are wanted by the customers...now we're going to switch things around and say the demand is $x = f(p)$, i.e. the quantity the customer wants when the price is set to be p. We want to quantify how the number of units x consumers will buy changes as the price p changes, but there are some factors to consider:

1. Some products are necessities while others are definitely not. For instance, gas for vehicles isn't something people can typically just stop buying, while a subscription to Max (or Hulu or Netflix or Prime or Peacock or Disney Plus...or all of them) is easily expendable. To that end, increasing the price of gas by $ 1 per gallon will not induce a decrease in demand like a $ 1 per month increase in the price of Max.

2. If we increase the price of gas from $ 3 to $ 4 per gallon, an enormous change; if we change the price of a laptop from $ 2000 to $ 2001, it's a minuscule difference. We'll need to **normalize our comparisons by considering the percent change in price**.

With these examples in mind and noticing that x will always decrease as price increases, we want to compare $\frac{\Delta x}{x}$ (percent change in units purchased) to $\frac{\Delta p}{p}$ (percent change in price)...but with a negative sign out front to make things positive. This leads us to

$$-\frac{\frac{\Delta x}{x}}{\frac{\Delta p}{p}} = -\frac{p}{x}\frac{\Delta x}{\Delta p}.$$

If we make the deltas into true differential d's, we obtain the following definition.

Definition 8.1. If a product has demand function $x = f(p)$, then the **elasticity of demand** associated with the product is

$$E(p) = -\frac{p}{x}\frac{dx}{dp}.$$

We want to know if a 5% change in price will induce more, less, or equal to 5% change in quantity demanded. This leads to another definition.

Definition 8.2.

1. If the percent change in demand is less than the percent change in price, leading to $E(p) < 1$, we say the demand is **inelastic**.

2. If the percent change in demand is equal to the percent change in price is **unit elastic**.

3. If the percent change in demand is greater than the percent change in price, leading to $E(p) > 1$, then we say the demand is **elastic**.

Example 8.8. Suppose the demand function for a can of Pepsi is $p = 5 - \frac{3}{2}x$.

1. Find the function $E(p)$ for the elasticity of demand.

2. When $p = 1.5$, determine if the demand is elastic, inelastic, or unit elastic.

Solution:

1. First order of business is to rearrange so we have $x = f(p)$:

$$\frac{3}{2}x = 5 - p \implies x = \frac{10}{3} - \frac{2}{3}p.$$

   From the definition above we then find

$$E(p) = -\frac{p}{x}\frac{dx}{dp} = -\frac{p}{\frac{10}{3} - \frac{2}{3}p}\left(-\frac{2}{3}\right) = \frac{2p}{10 - 2p}.$$

2.

$$E(1.5) = 0.42 \implies \text{Inelastic.}$$

9 Week 9

Outline
Days: Monday (March 16), Wednesday (March 18), Friday (March 20)
Topics: *Linear approximations, maxima/minima/optimization, critical points, Fermat's theorem.*
ENJOY SPRING BREAK!

Goals:

- Visually and numerically understand how the tangent line at $x = a$ is the best linear approximation of $f(x)$ near $x = a$.
- Know via the Extreme Value Theorem under what conditions a function is guaranteed to attain its maximum and minimum values.
- Use Fermat's Theorem and the idea of critical points to find maxima/minima.
- Differentiate between *local* extrema and *global* extrema.

Album of the Week: *CrazySexyCool* by *TLC*

9.1 Day 1

Goal: Review elasticity of demand. Be able to use tangent lines to approximate function values which are otherwise difficult to compute.

9.1.1 Elasticity of Demand, Continued

We begin by recalling the definition of elasticity of demand.

Definition 9.1. If a product has demand function $x = f(p)$, then the **elasticity of demand** associated with the product is

$$E(p) = -\frac{p}{x}\frac{dx}{dp}.$$

We want to know if a 5% change in price will induce more, less, or equal to 5% change in quantity demanded. This leads to another definition.

Definition 9.2.
1. If the percent change in demand is less than the percent change in price, leading to $E(p) < 1$, we say the demand is **inelastic**.
2. If the percent change in demand is equal to the percent change in price is **unit elastic**.
3. If the percent change in demand is greater than the percent change in price, leading to $E(p) > 1$, then we say the demand is **elastic**.

This can be applied to revenue maximization through the following reasoning:

- If $E(p) < 1$, i.e. demand is inelastic, then we can increase price without a significant decrease in demand. This will lead to an increase in revenue!

- If $E(p) > 1$, i.e. demand is elastic, then we cannot increase price without causing a significant decrease in demand. In other words, increasing price when demand is elastic will lead to a decrease in revenue!
- If $E(p) = 1$, then we are situated directly between the previous two cases. This means **revenue is maximized when** $E(p) = 1$.

Example 9.1. A product has demand function given by $x = 100 + \frac{1}{2p^2}$ for $0 < p$.

(a) Determine the elasticity of demand $E(p)$ associated with this product.

(b) Find the price p at which revenue will be maximized.

Solution 9.1.

(a) To find elasticity of demand, we calculate

$$E(p) = -\frac{p}{x}\frac{dx}{dp} = -\frac{p}{\left(100 + \frac{1}{2p^2}\right)}\frac{d}{dp}\left(100 + \frac{1}{2p^2}\right)$$

$$= -\frac{p}{\left(100 + \frac{1}{2p^2}\right)}\left(-\frac{1}{p^3}\right) = \frac{1}{100p^2 + \frac{1}{2}}$$

Thus $E(p) = \frac{1}{100p^2+\frac{1}{2}}$.

(b) Revenue is maximized when $E(p) = 1$; in this case that implies

$$\frac{1}{100p^2 + \frac{1}{2}} = 1 \implies 100p^2 + \frac{1}{2} = 1$$

$$\implies 100p^2 = \frac{1}{2}$$

$$\therefore p = \frac{1}{\pm 10\sqrt{2}}$$

Since prices cannot be negative, we conclude revenue is maximized when $p = \frac{1}{10\sqrt{2}} \approx 0.07$ dollars per unit.

9.1.2 Linear Approximations

When we find the tangent line to a function $f(x)$ at $x = a$, it looks like

$$y = f'(a)(x - a) + f(a).$$

Zooming in very close to the graph of $f(x)$ near $x = a$, the **tangent line approximately follows the graph**. This is the whole idea behind linear approximations! It means that for $x_0 \approx a$, we may write

$$f(x_0) \approx f'(a)(x_0 - a) + f(a)$$

The point is that evaluating nonlinear functions is computationally difficult while linear functions are simple.

Definition 9.3. The **linear approximation** or **tangent line approximation** or **linearization** of $f(x)$ near $x = a$ is

$$L(x) = f'(a)(x - a) + f(a).$$

Example 9.2. Let $f(x) = \frac{1}{x^2+5}$. (a) Find the linearization of $f(x)$ near $x = 2$, and (b) use it to approximate $\frac{1}{(2.1)^2+5}$.

Solution 9.2.

(a) To find the equation of the tangent line we need $f'(x)$. We compute the derivative

$$f'(x) = \frac{d}{dx}\left(\frac{1}{x^2+5}\right) = -\frac{2x}{(x^2+5)^2}.$$

This implies

$$f'(2) = -\frac{4}{81}$$

and so

$$\boxed{L(x) = -\frac{4}{81}(x-2) + \frac{1}{9}}.$$

(b) To estimate $\frac{1}{(2.1)^2+5} = f(2.1)$, use the fact that $L(2.1) \approx f(2.1)$ and instead calculate $L(2.1)$.

$$L(2.1) = -\frac{4}{81}\left(\frac{1}{10}\right) + \frac{1}{9} = \frac{-4+90}{810} = \frac{86}{810}$$

$$= \boxed{\frac{86}{810} = 0.106172839506}$$

Compare with the actual value: $f(2.1) = 0.106269925611$. The estimate is very good and $L(x)$ may be somewhat less computationally intensive or expensive to calculate.

Example 9.3. Suppose the profit function for a product is given by $P(x) = \sqrt{4x-4}$. Use the linearization of P to estimate the change in profit between selling $x = 26$ units and $x = 29$ units.

Solution 9.3.

We compute the linearization $L(x)$ associated with P at $x = 26$, then calculate $L(92) - L(90)$.

$$L(x) = P'(26)(x-26) + P(26) = \frac{4}{2\sqrt{4(26)-4}}(x-26) + \sqrt{4(26)-4}$$

$$\implies L(x) = 2(1/10)(x-26) + 10 = \frac{1}{5}(x-26) + 10.$$

This then tells us $P(29) \approx L(29) = \frac{1}{5}(3) + 10 = \frac{53}{5} = 10.6$, and therefore

$$P(29) - P(26) \approx 10.6 - 10 = \boxed{0.6}$$

We can also calculate the actual value: $P(29) - P(26) = 0.58300...$, so our approximation is decent.

9.2 Day 2

Goal: Define global and local extrema as well as critical points. State and use the Extreme Value Theorem, Fermat's Theorem.

9.2.1 Maxima and Minima

We've already seen some specific instances where finding the maximum or minimum of a function is relevant to a business (e.g. maximizing profit, minimizing cost). Let's get a more in-depth look at maxima and minima.

Definition 9.4. Let f be a function defined on an interval I which contains the real number c. We say f has an **absolute maximum over I at c** if $f(c) \geq f(x)$ for every x in I. Similarly, we say f has an **absolute minimum over I at c** if $f(c) \leq f(x)$.

Remark: Note if we say '$f(x)$ has an absolute maximum over I at $x = c$,' **the absolute maximum value of f is $f(c)$ and NOT c**. The number c is the input where the maximum value takes place.

An interesting question to ask is whether every function f defined on an interval I necessarily attains its maximum/minimum values. With a couple key assumptions concerning the interval I and the function f, we get a nice answer.

Extreme Value Theorem: Suppose f is a **continuous function** defined on an interval I which is **closed and bounded**, i.e. $I = [a, b]$ for some real numbers a, b. Then there exist real numbers $a \leq c_1, c_2 \leq b$ such that $f(c_1)$ is the absolute maximum value of f and $f(c_2)$ is the absolute minimum of f over I.

Remark: The key assumptions are that f is continuous on all of I and I is closed and bounded. If either of these conditions fail, f need not necessarily attain its maximum and minimum values.

Example 9.4. Consider $f(x) = \frac{1}{x}$ and $I = [0, 1]$.

(i) What is the maximum value of f on I?

(ii) Explain why (i) is not a contradiction of EVT. That is, explain which assumption of the theorem fails.

Solution 9.4.

(i) There is no maximum value of $f(x)$ on I. If we suppose $x = c$ is a real number where f is maximized, we get an immediate contradiction by noticing that $f(c/2) = 2f(c) > f(c)$. Importantly, $f(0)$ is undefined and not equal to ∞.

(ii) The conclusion to (i) is not a contradiction of EVT because f is not continuous on all of $I = [0, 1]$. In particular, f is discontinuous at $x = 0$ (vertical asymptote).

Absolute extrema represent the value of a function at its highest peaks and lowest valleys **over the whole real line**, but what about smaller peaks and valleys?

Definition 9.5. A function f has a **local maximum at** $x = c$ if there exists some interval I (within the domain of f) containing c such that $f(c) \geq f(x)$ for all x in I. We say f **has a local minimum at** $x = c$ if there exists some interval I (within the domain of f) containing c such that $f(c) \leq f(x)$ for all x in I.

Remark: Note that in the above definition the point $x = c$ is only a *local* maximum (respectively minimum) because we don't have information outside the interval I! For instance, the function $f(x) = \frac{x^3}{3} - x + 1$ has a local maximum at $x = -1$ but it is certainly not an absolute maximum.

These definitions are all fine and good, but how can we use the equation of a function $f(x)$ to find where it has (local) maximums or minimums? We can see from intuition that local maximums and minimums occurs where the function has a horizontal tangent line; this leads us to the following definition.

Definition 9.6. The input $x = c$ is called a **critical point** of $f(x)$ if $f'(c) = 0$ or $f'(c)$ is undefined.

Our previous discussion about the fact that local maximums and minimums occur where the function has horizontal tangent lines gives us the following theorem of Fermat (not to be confused with Fermat's Last Theorem):

Fermat's Theorem: If $f(x)$ has a local maximum or minimum at $x = c$ and f is differentiable at c, then $f'(c) = 0$.

Remarks:

- Something which is commonly misunderstood about Fermat's Theorem is that **if $f'(c) = 0$, this does not guarantee that f has a local max/min at c!** Consider the function $f(x) = x^3$; $f'(x) = 3x^2 = 0$ exactly when $x = 0$, but f does not have a max or a min there.
- Phrased a different way, Fermat's Theorem tells us that **if f is differentiable at $x = a$ and $f'(a) \neq 0$, then $f(a)$ cannot be a local max/min value of the function**.
- (**Most important comment for applications**) With the Extreme Value Theorem in mind, suppose we want to know the maximum/minimum values of a function f defined on a closed, bounded interval $[a, b]$. Then by Fermat's Theorem, we only have to check points where (1) $f'(c) = 0$, (2) $f'(c)$ is undefined, and (3) the endpoints $x = a$ and $x = b$.

Let's do some examples relevant to the material discussed here.

Example 9.5.

(a) Let $f(x) = \frac{x^3}{3} - x + 1$. What are the critical points of f?

(b) Suppose a company produces a product with cost function $C(x)$ which is differentiable everywhere. If we are told that producing $x = 100$ units results in a local minimum of cost, what do we know about $MC(100)$ (the value of the marginal cost function at $x = 100$)?

(c) Find the maximum and minimum values of $f(x) = x^2 - 4x + 3$ over the interval $[1, 3]$ and state where they occur.

Solution 9.5.

(a) To find critical points we find all values of x such that $f'(x) = 0$ or $f'(x)$ is undefined. Since the given function is a polynomial, $f'(x)$ is defined for all x, so we only need to find where $f'(x) = 0$.

$$f'(x) = x^2 - 1 = 0 \iff (x - 1)(x + 1) = 0 \iff x = \pm 1.$$

Thus the critical points of f are $x = -1$ and $x = 1$.

(b) Recall that the marginal cost function is just the derivative of the cost function, i.e. $MC(x) = C'(x)$. If we know $C(100)$ is a local minimum value, then Fermat's Theorem implies that (since C is differentiable) $MC(100) = 0$.

(c) We follow the process outlined in the Remarks following Fermat's Theorem (and outlined again below). First, we **find** $f'(x)$:

$$f'(x) = 2x - 4.$$

Next, **we find the critical points of** f. In this case there's only one:

$$f'(x) = 2x - 4 = 0 \iff x - 2 = 0 \iff x = 2.$$

Importantly, 2 is in the interval $[1, 4]$! If it wasn't we wouldn't care/we'd ignore it. Next, we **evaluate $f(x)$ at the critical points and the endpoints of the interval**:

$$f(1) = 1 - 4(1) + 3 = 0; \qquad f(2) = (2)^2 - 4(2) + 3 = -1; \qquad f(4) = (4)^2 - 4(4) + 3 = 3.$$

Lastly, **we compare the values determined in the previous step and determine which ones correspond to the global max/min**. In this case we found $f(2) = -2$ was the lowest value, hence is the global min, while $f(4) = 3$ was the highest value, hence the global max.

To reiterate the above process: Assume we want to know the maximum and minimum values of $f(x)$ over a closed, bounded interval $[a, b]$. Then...

1. Compute $f'(x)$.
2. Find all critical points of f which *lie in the interval* $[a, b]$. Remember this includes points where the derivative is undefined!
3. Evaluate $f(x)$ at the x-values determined in Step 3 as well as at a and b (the endpoints of the interval). Compare these values to determine absolute maximum and minimum.

What if we want to know the global maximum/minimum of f over an interval which is *not* closed? Then we take limits toward the endpoint(s) instead of evaluating there (since the function may not be defined at the endpoint(s)):

Assume we want to know the global maximum and minimum values of $f(x)$ over an interval which is open or half open. Then...

1. Compute $f'(x)$.
2. Find all critical points of f which lie in the interval. Remember this includes points where the derivative is undefined!
3. Evaluate $f(x)$ at the x-values determined in Step 3 as well as at a and b (the endpoints of the interval). If the interval is of the form (a, b), also evaluate $\lim_{x\to a^+} f(x)$ and $\lim_{x\to b^-} f(x)$; if the interval is of the form $(a, b]$, evaluate $\lim_{x\to a^+} f(x)$ and $f(b)$; if the interval is of the form $[a, b)$, evaluate $f(a)$ and $\lim_{x\to b^-} f(x)$. Compare all appropriate values.

Example 9.6. Let

$$f(x) = |x| = \begin{cases} -x, & x \le 0 \\ x, & x \ge 0 \end{cases}.$$

Find the maximum and minimum values of f over $I = [-1, 3)$.

Solution 9.6. By Fermat's Theorem, we look at $f'(x)$ and find critical points. From prior work with the absolute value function we know

$$f'(x) = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \\ \text{Undefined}, & x = 0 \end{cases}$$

because $|x|$ has a sharp corner at $x = 0$. This means the only critical point of $f(x)$ is at $x = 0$. We check the value of f at $x = -1$ (the left endpoint), $x = 0$ (the only critical point in the interval), and check $\lim_{x\to 3^-} f(x)$ (behavior of f toward the right endpoint which is **not** in the interval):

$$f(-1) = 1, \quad f(0) = 0, \quad \lim_{x\to 3^-} f(x) = 3.$$

The minimum value of f is 0 and is attained at $x = 0$. There is no maximum value of f which is attained over $[-1, 3)$ because the function is increasing as $x \to 3^-$, the endpoint which is not included. (Suggestion: Sketch the picture of $f(x)$ over the given interval and explain why there is no maximum.)

9.3 Day 3

Goal: Continue with examples relating to optimization. Includes applications to real-world problems with *constraints*, as well as minimization of mean squared error cost function.

Example 9.7. A company must produce a sturdy rectangular container with a square base and a volume of 128 cubic ft. The cost to produce the top and sides of this box is \$2 per square foot and the cost to produce the bottom is \$6 per square foot. Find the dimensions of the box that will minimize cost.

Solution 9.7.

Let x be the side length of the square base (and therefore top) of the box, let h be the height of the box. Then:

Area of bottom: x^2 square feet.
Area of top: x^2 square feet.
Area of each side: xh square feet.
Cost of bottom: $6x^2$
Cost of top: $2x^2$
Cost of *all* sides (combined): $4 \cdot (2xh) = 8xh$
Constraint:

$$\text{Vol} = 128 = lwh = x^2 h \implies h = \frac{128}{x^2}.$$

Putting this all together, the cost to make the box with length and width x, height $h = 128/x^2$ is

$$C(x) = \underbrace{6x^2}_{\text{bottom cost}} + \underbrace{2x^2}_{\text{top cost}} + \underbrace{8xh}_{\text{sides cost}} = 8x^2 + \frac{1024}{x}.$$

To find the absolute or global minimum of $C(x)$, we find the critical points of $C(x)$.

$$C'(x) = 16x - \frac{1024}{x^2} = 0 \implies x^3 = 64.$$

Thus the only critical point of the cost function is $x = 64^{\frac{1}{3}} = (4^3)^{\frac{1}{3}} = 4$ feet. As a result, the minimum cost must be

$$C(4) = 8(4^2) + \frac{1024}{4} = \boxed{\$384}.$$

We can use our knowledge of critical points and extrema to minimize a simplified version of the mean squared error cost function J. That is, we can find the line of best fit given a dataset!

Example 9.8. Suppose we are given a dataset with four data points: $(x_1, y_1) = (0, 1)$, $(x_2, y_2) = (1, 1)$, $(x_3, y_3) = (3, 6)$, and $(x_4, y_4) = (5, 11)$. We want to find the best model of the form $f(x) = mx$ to fit this data by minimizing the mean squared error cost function $J(m)$.

1. Write the equation of $J(m)$. (Hint: It's a function of m since our error depends on the slope of the model function.)
2. Find the value of the slope parameter m which minimizes $J(m)$.

Solution 9.8.

1. As the name suggests, the mean squared error cost function computes the average of the squared error of all predictions. The predicted value of our model is just the output of $f(x) = mx$ based on the x coordinate of the given data point, while the error is the difference between the predicted y-value

and the actual y-coordinate of the data point. For instance, the model predicts that given the input $x_1 = 0$, the output will be

$$f(x_1) = f(0) = m \cdot 0 = 0.$$

The actual output at x_1 is $y_1 = 1$. This means the squared error associated with the prediction is

$$(f(x_1) - y_1)^2 = (0 - 1)^2 = 1.$$

Since we have four data points, this means

$$\begin{aligned} J(m) &= \frac{1}{4}\left((f(x_1) - y_1)^2 + (f(x_2) - y_2)^2 + (f(x_3) - y_3)^2 + (f(x_4) - y_4)^2\right) \\ &= \frac{1}{4}\left((m \cdot 0 - 1)^2 + (m \cdot 1 - 1)^2 + (m \cdot 3 - 6)^2 + (m \cdot 5 - 11)^2\right) \\ &= \frac{1}{4}(1 + (m-1)^2 + (3m-6)^2 + (5m-11)^2) \\ &= \frac{1}{4}(1 + m^2 - 2m + 1 + 9m^2 - 36m + 36 + 25m^2 - 110m + 121) \\ &= \frac{1}{4}(35m^2 - 148m + 159). \end{aligned}$$

2. The function $J(m)$ is an upward-facing quadratic, so it has a unique minimum at its vertex. We can either write the m value where the vertex occurs directly or can find critical points by taking the derivative with respect to m (there will be only one critical point, corresponding to the unique minimum).

$$J'(m) = \frac{1}{4}(70m - 148) = 0 \implies m = \frac{148}{70} \approx 2.1143$$

This means that the minimum mean squared error of our model occurs when $m = 2.1143$, i.e. the line of best fit is given by the equation

$$f(x) = 2.1143x.$$

Sketch this line along with the data to see if it looks correct.

10 Week 10

Outline

Days: Monday (March 30), Wednesday (April 1), Friday (April 3)
Topics: *Local extrema, First Derivative Test, Second Derivative Test, graph shape.*
Goals:

- Use Fermat's Theorem to identify potential locations of maxima/minima.
- Follow the idea of the First Derivative Test to determine whether $f(x)$ has a local maximum, minimum, or neither at given $x = c$.
- Follow the idea of the Second Derivative Test to determine whether $f(x)$ has a local maximum, minimum, or neither at given $x = c$. Recall what concavity/convexity is/are.
- Appreciate the greatness of Nina Simone with the best vibes.

Bonus: Find all the ghosts haunting these notes (which were originally written for class during Halloween week)!

Album of the Week: *The Chronic* by *Dr. Dre (1992)*

10.1 Day 1

Goal: Work through a variety of examples for finding maxima and minima. Emphasize the fact that not all critical points correspond to even local maxima/minima!

Let's recall the following from last week:

Definition 10.1. An input $x = c$ is a **critical point** of $f(x)$ if $f'(c) = 0$ or $f'(c)$ is undefined (this second option meaning f is not differentiable at c).

Using this definition we stated an important result:

Fermat's Theorem: If $f(x)$ has a local maximum or minimum at $x = c$, then c must be a critical point of f.

We want to use Fermat's Theorem and a consistent method for finding critical points in order to find where (local) maxima and minima of a function occur. In other words, we're going into great detail in topics relating to the Extreme Value Theorem.

Example 10.1. Let $f(x) = x^3 - 2$ on $[-5, 2]$.

(a) Find all critical points of f in the given interval.

(b) Determine the maximum and minimum values of f on the given interval.

(c) Does $x = 0$ correspond to a (local) maximum, a (local) minimum, or neither? Explain.

Solution 10.1.

(a) To find critical points of a polynomial, which is differentiable everywhere, we set $f'(x) = 0$ and solve for x.

$$f'(x) = 3x^2 = 0 \iff x^2 = 0 \iff x = 0.$$

Thus the only critical point of the function is $x = 0$.

(b) To determine the maximum and minimum values of a function on a closed, bounded interval, we **test $f(x)$ at the critical points and the endpoints of the interval**. In this case we test and compare $f(-5), f(0)$, and $f(2)$.

$$f(-5) = (-5)^3 - 2 = -125 - 2 = -127, \qquad f(0) = 0^3 - 2 = -2, \qquad f(2) = 2^3 - 2 = 6.$$

This tells us that **the maximum value of $f(x)$ over $[-5, 2]$ is 6, and this maximum occurs at $x = 2$, while the minimum value of $f(x)$ over $[-5, 2]$ is -127 and this minimum occurs at $x = -5$.**

(c) The critical point $x = 0$ is neither a local max nor a local min! We cannot find an open interval I containing $x = 0$ such that $f(0) = 0$ is the max or min on I. This is because we can always move a tiny bit to the right of 0 and achieve larger values or move a tiny bit to the left of 0 and achieve lower values. **This shows that not all critical points correspond to (local) maximums or minimums!**

Example 10.2. A company makes a product with profit function $P(x)$ and we know that within the production range $10 \leq x \leq 1000$ (where x is the number of units produced), $P(555)$ is the largest value. What can we then say about $MP(555)$? (Recall the marginal profit function $MP(x) = P'(x)$.)

Solution 10.2. This is a direct application of Fermat's Theorem! If $P(x)$ is maximized at $x = 555$, then either $MP(555) = P'(555) = 0$ or $MP(555)$ is undefined. We don't know which option is correct because we don't know if P is differentiable at $x = 555$ or not.

Example 10.3.

Cole runs out of class three minutes early. He doesn't know where he's going, so his distance from Monteith 110 as a function of time is given by $d(t) = \frac{t^4}{4} - t^3 + t^2$, where $0 \leq t \leq 2.2$ is in hours. (1) When was he at his furthest from the room, and when was it that he reached this point? (2) Did he ever come back, i.e. at some $t > 0$ was $d(t) = 0$?

Solution 10.3.

1. To find when he was furthest from the classroom, we want to find the absolute maximum of $d(t)$ for t in the interval $[0, 2.2]$. Following the usual process,

$$d'(t) = t^3 - 3t^2 + 2t = 0 \iff t(t^2 - 3t + 2) = 0 \iff t = 0, t = 1, t = 2.$$

These represent all the critical points of $d(t)$, so now we test them as well as the endpoints of the interval:

$$d(0) = 0, \qquad d(1) = \frac{1}{4} - 1 + 1 = \frac{1}{4},$$
$$d(2) = \frac{16}{4} - 8 + 4 = 0, \qquad d(2.2) = 0.0484$$

This tells us **Cole was furthest from MONT 110 at time $t = 1$, at which point he was $1/4$ units away**.

2. Setting $d(t) = 0$ and solving for t, we see

$$d(t) = \frac{t^4}{4} - t^3 + t^2 = 0 \iff t^2\left(\frac{t^2}{4} - t + 1\right) = 0$$

$$\iff t = 0, t^2 - 4t + 4 = 0 \iff t = 0, t = 2.$$

This tells us **Cole returned to class at time $t = 2$ after initially being there (and leaving) at $t = 0$.** Still 1 minute left!

Example 10.4. Suppose $f(x) = \begin{cases} 2x + 5, & 0 \le x \le 2, \\ -3(x-2) + 9, & 2 \le x \le 5, \\ x - 5, & 5 \le x \le 10 \end{cases}$.

Determine the critical points of $f(x)$ as well as the absolute maximum and minimum over $[0, 10]$.

Solution 10.4. To determine critical points we find

$$f'(x) = \begin{cases} 2, & 0 < x < 2 \\ -3, & 2 < x < 5 \\ 1, & 5 < x < 10 \end{cases}.$$

Notice we haven't specified the derivative at the endpoints of the interval or at $x = 2, 5$. Using the limit definition we can show $f'(2)$ and $f'(5)$ are undefined, while the derivatives at the endpoints are always unknown (trying to take a one-sided limit is what it boils down to). Everywhere else the derivative is nonzero, thus **the critical points are $x = 2$ and $x = 5$.** Testing at these points as well as the endpoints we find

$$f(0) = 5, \qquad f(2) = 15, \qquad f(5) = 0, \qquad f(10) = 5.$$

Thus **the absolute maximum is 15 and occurs at $x = 2$, while the absolute minimum is 0 and occurs at $x = 5$.**

10.2 Day 2

Goal: Further discuss the relationship between the shape of the graph of a function and its derivative. In particular, we will use the First Derivative Test to determine whether a critical point corresponds to a local max, a local min, or neither.

On Monday we saw that the function $f(x) = x^3 - 2$ has a critical point at $x = 0$, but the function has neither a maximum nor a minimum at this point. When it comes to applications (e.g. maximizing profit, minimizing cost) we'll want to be able to identify such points and distinguish them from their maximum/minimum critical point counterparts.

Why is it that $f(x) = x^3 - 2$ doesn't have a maximum or minimum at $x = 0$? Visually, it's not a maximum because **it doesn't look like the top of a hill or the summit of a sharp peak; in terms of the derivative, this is because $f'(x) = 3x^2 \ge 0$ for all x,** so we are always going upward (never crest a peak)! Similar visual reasoning explains why $x = 0$ is not a minimum of $f(x)$: **it doesn't look like the bottom of a ravine (sharp or smooth) because it never decreases, as $f' \ge 0$,** so we never hit a lowest point. These intuitive statements can be formalized in the First Derivative Test.

First Derivative Test: Assume $f(x)$ is continuous for x in an interval I. If $f(x)$ is differentiable except possibly at $x = c$, then $f(c)$ satisfies one of the following:

(a) If $f'(x) > 0$ for $x < c$ and $f'(x) < 0$ for $x > c$, then $f(c)$ is a local maximum of f.

(b) If $f'(x) < 0$ for $x < c$ and $f'(x) > 0$ for $x > c$, then $f(c)$ is a local minimum of f.

(c) If $f'(x)$ has the same sign for $x < c$ and $x > c$, $f(c)$ is neither a local maximum nor a local minimum of f.

Remark: These statements (should) make intuitive sense. You only know you've reached a peak if you have to climb to get there and then start descending afterward; you only know you've reached the lowest point of a valley if you descended to get there and then start climbing afterward.

Example 10.5.

(a) Use the first derivative test to show that $f(x) = \frac{x^3}{3} - 3x^2 + 8x$ has a local maximum at $x = 2$ and a local minimum at $x = 4$.

(b) What is $f''(2)$? $f''(4)$?

Solution 10.5.

(a) As always when finding extrema, we first find the critical points of $f(x)$:

$$f'(x) = x^2 - 6x + 8 = 0 \iff (x-2)(x-4) = 0 \iff x = 2, 4.$$

We confirm that f has critical points at $x = 2$ and $x = 4$. According to the First Derivative Test, we will have a local max at $x = 2$ if $f'(x) > 0$ for $x < 2$ and $f'(x) < 0$ for $2 < x < 4$ (note we exclude $x \geq 4$ because we don't want to cross over the next critical point). **Choose test values in the aforementioned ranges:** $x = 0 < 2$**,** $2 < x = 3 < 4$. Plugging these into $f'(x)$ we find

$$f'(0) = 8 > 0, \qquad f'(3) = 9 - 18 + 8 = -1 < 0.$$

Thus by the First Derivative Test, $f(2) = \frac{8}{3} - 12 + 16 = \frac{20}{3}$ is a local maximum of $f(x)$.
To check that f has a local minimum at $x = 4$, the First Derivative Test says we should check to see if $f'(x) < 0$ for $2 < x < 4$ (**we've already done this from above!**) and $f'(x) > 0$ for $x > 4$. **Plugging the test value** $x = 5$ **into** $f'(x)$ we see

$$f'(5) = 25 - 30 + 8 = 3 > 0.$$

Thus the First Derivative Test implies $f(4) = \frac{64}{3} - 48 + 32 = \frac{64-48}{3} = \frac{16}{3}$ is a local minimum of f.

(b) Taking the second derivative,

$$f''(x) = 2x - 6 \implies f''(2) = 4 - 6 = -2, \quad f''(4) = 8 - 6 = 2.$$

Make note of the fact that $f''(2) < 0$**,** $f''(4) > 0$.

Let's do another example which emphasizes finding where f is increasing/decreasing, something which is inherently involved when using the First Derivative Test but often overlooked.

Example 10.6. Let $f(x) = 3x^4 - 4x^3$.

(a) Determine the critical points of $f(x)$ (there should be 2 of them).

(b) Determine where $f(x)$ is increasing and where $f(x)$ is decreasing. Express your answers using interval notation.

(c) Based on parts 1 and 2, use the First Derivative Test to determine all places where f has a local maximum or local minimum.

Solution 10.6.

(a) To determine critical points, we first find $f'(x)$:

$$f'(x) = \frac{d}{dx}(3x^4 - 4x^3) = 12x^3 - 12x^2.$$

Next, we find the critical points by setting $f'(x) = 0$ and solving for x (**in this instance we don't have to worry about looking for where $f'(x)$ is undefined because it's defined everywhere**):

$$f'(x) = 0 \iff 12x^3 - 12x^2 = 0 \iff 12x^2(x-1) = 0$$
$$\iff x = 0,\ x = 1.$$

This tells us the critical points of $f(x)$ are $x = 0$ and $x = 1$.

(b) To find where f is increasing/decreasing, we **choose test values of x which are away from the critical points and plug them into the derivative**. If the derivative is positive there, then f must be increasing on the entire interval where the test point was contained between critical points; similarly, if $f'(x) < 0$ then the function is decreasing on the entire interval where the test point was. **Helpful tip: make a number line with all your critical points and test points labeled!** My test points will be $x = -1$, $x = 1/2$, and $x = 2$:

$$f'(-1) = 12(-1) - 12 = -24 < 0;$$
$$f'(1/2) = 12(1/8) - 12(1/4) = 1.5 - 3 = -1.5 < 0;$$
$$f'(2) = 12(8) - 12(4) = 96 - 48 = 48 > 0.$$

This tells us $f(x)$ **is increasing on** $(1, \infty)$ and $f(x)$ **is decreasing on** $(-\infty, 0) \cup (0, 1)$.

(c) Since the derivative of f only changes sign at $x = 1$, this is the only possible place where it can have a local max or local min. In particular, since the derivative changes from negative to positive the First Derivative Test tells us that $f(x)$ **has a local minimum at** $x = 1$.

10.3 Day 3

Goal:



(Complete our coverage of results which tell us information about extrema/graph shape with the Second Derivative Test and concavity/convexity.)

On Wednesday we noted in a couple places that when a twice differentiable function $f(x)$ had a local maximum at $x = c$, $f''(c) < 0$, while if f had a local minimum at $x = c$, $f''(c) > 0$. It turns out this is more than a coincidence!

Definition 10.2. Suppose $f''(c)$ is defined. We say f **is concave at** c or f **is concave down at** c if $f''(c) < 0$. We say f **is convex at** c or f **is concave up at** c at if $f''(c) > 0$. Lastly, if $f''(c) = 0$ we call c an **inflection point** of f.

Second derivatives (when they exist) are useful in characterizing or identifying maxima and minima. The **classic example of a function which is concave up everywhere is $f(x) = x^2$, since $f''(x) = 2 > 0$.** This function has exactly one critical point at $x = 0$ which is a local minimum. The **classic example of a function which is concave down everywhere is $f(x) = -x^2$, since $f''(x) = -2 < 0$.** This function has exactly one critical point at $x = 9$ which corresponds to a local maximum. The Second Derivative Test, which we will now state, says **as long as a function $f(x)$ resembles x^2 or $-x^2$ near $x = c$,** (in the sense

that it has the same concavity/convexity) **then f will have the same graph behavior as the "classic example".** 🦇🦇

Second Derivative Test: Suppose $f'(c) = 0$ and $f''(x)$ is continuous over an interval containing c. Then ...

1. If $f''(c) > 0$, $f(c)$ is a local maximum of f.
2. If $f''(c) < 0$, $f(c)$ is a local minimum of f.
3. If $f''(c) = 0$, gain no information. 💀

Here's a few problems which are relevant either to the First Derivative Test, the Second Derivative Test, both, or some related/involved topics. **A suggestion: as you go through each problem, try to explain in words what the process is that you're going through. Justify each step, as this will help you understand what's happening better**. I've done this in the solutions (highlighted).

Example 10.7. Let $f(x) = x^3 - 12x + 10$.

1. Determine all critical points of $f(x)$.
2. Find the values of x where $f(x)$ is strictly increasing and strictly decreasing. Express your answer in interval notation. (**Hint/Reminder: We know $f'(x) = 0$ at critical points here; pick test points $x = a$ to the left and right of every critical point and see if $f'(a) > 0$ or $f'(a) < 0$.**)
3. Determine all local maxima and minima of $f(x)$ using the First Derivative Test.

Solution 10.7.

1. To determine critical points, we **find $f'(x)$, set equal to 0 and solve for x by factoring**:

$$f'(x) = 3x^2 - 12 = 0 \iff x^2 - 4 = 0 \iff (x+2)(x-2) = 0$$
$$\iff x = 2, -2.$$

2. As suggested by the hint, **pick test points to either side of -2 and 2**. We can choose $x = -3, x = 0$, and $x = 3$ and test in $f'(x)$:

$$f'(-3) = 3(-3)^2 - 12 = 27 - 12 = 15 > 0; \qquad f'(0) = -12 < 0;$$

$$f'(3) = 3(3)^2 - 12 = 15 > 0.$$

This tells us that f **is strictly increasing on the interval strictly to the left of -2, i.e.** $(-\infty, -2)$**, and the interval strictly to the right of** 2**, i.e.** $(2, \infty)$. By similar reasoning, f **is strictly decreasing on the interval strictly between -2 and 2, i.e.** $(-2, 2)$. More succinctly written,

$$\text{Increasing: } (-\infty, -2) \cup (2, \infty)$$

$$\text{Decreasing : } (-2, 2).$$

3. We notice first that the **sign of f' changes from positive to negative at** -2; by the First Derivative Test, this implies f **has a local maximum at** $x = -2$. Next, observe that **sign of f' changes from negative to positive at** 2; by the First Derivative Test, this implies f **has a local minimum at** $x = -2$ 👻.

Example 10.8. Suppose $f(x) = x^5 - 3x^3$.

1. Find all critical points of $f(x)$.
2. Use the Second Derivative Test (when possible) to determine all local maxima and minima. Where the Second Derivative Test is inconclusive, use the First Derivative Test!

Solution 10.8.

1. To find critical points we **find $f'(x)$, set equal to 0 and solve for x by factoring**:

$$f'(x) = 5x^4 - 9x^2 = 0 \iff 5x^2\left(x^2 - \frac{9}{5}\right) = 0$$

$$\iff x = 0, \quad x^2 - \frac{9}{5} = 0 \iff x = 0, \quad x = \pm\frac{3}{\sqrt{5}}.$$

   This tells us the critical points are $x = 0, \frac{-3}{\sqrt{5}}, \frac{3}{\sqrt{5}}$.

2. To use the Second Derivative Test, we **compute $f''(x)$ and evaluate at each critical point**:

$$f''(x) = 20x^3 - 18x; \qquad f''(0) = 0, \quad f''\left(\frac{3}{\sqrt{5}}\right) = 24.149... > 0, \quad f''\left(-\frac{3}{\sqrt{5}}\right) = -24.149... < 0.$$

   The Second Derivative Test then tells us f **has a local minimum at $x = \frac{3}{\sqrt{5}}$, a local maximum at $x = -\frac{3}{\sqrt{5}}$, but is indeterminate at $x = 0$.** This means we need to use the First Derivative Test. Choosing test points $x = -1$ and $x = 1$ we see

$$f'(-1) = 5(-1)^4 - 9(-1)^2 = 5 - 9 = -4 < 0, \qquad f'(1) = 5 - 9 = -4 < 0.$$

   This tells us f **has neither a local maximum nor a local minimum at** $x = 0$.

Example 10.9. Let $f(x) = -3x^3 + 3x^2 - 1$. Determine all x-values where $f(x)$ is concave up, concave down, or has an inflection point. Express in interval notation.

Solution 10.9. We need to **determine where $f''(x) > 0, < 0,$ or $= 0$.**

$$f''(x) = \frac{d}{dx}(f'(x)) = \frac{d}{dx}(-9x^2 + 6x) = -18x + 6. \ \text{👻}$$

This is a line with negative slope; it will be equal to 0 at exactly one place, namely $-18x + 6 = 0 \implies x = \frac{1}{3}$. This tells us f **has one inflection point at** $x = \frac{1}{3}$. Then since $-18x + 6 > 0 \iff x < \frac{1}{3}$ and $-18x + 6 < 0 \iff x > \frac{1}{3}$, f **is concave up on** $-\infty, 1/3)$ **and concave down on** $(1/3, \infty)$.

11 Week 11

Outline

Days: Monday (April 6), Wednesday (April 8), Friday (April 10)
Topics: *Graph shape, second derivative, horizontal asymptotes, optimization.*
Goals:

- Be able to utilize the second derivative test to classify critical points as maxima/minima.
- Sketch a possible graph of $f(x)$ based on information about $f'(x)$ and $f''(x)$
- Evaluate $\lim_{x\to\pm\infty} f(x)$ for polynomials, rational functions, and more.
- Solve applied optimization problems with and without constraints.

Album of the Week: *Fame Monster* by *Lady Gaga (2009)*

11.1 Day 1

Goal: Solidify our understanding of graph shapes, including (but not strictly limited to): Extreme Value Theorem, local maxima/minima, global maxima/minima, increasing, decreasing, critical points, Fermat's Theorem, concave up, concave down, inflection points, First Derivative Test, Second Derivative Test.

Example 11.1.

(a) State the Extreme Value Theorem.

(b) Explain why the function $f(x) = \frac{x}{x+1}$ does not attain a maximum value over the interval $[0, 1)$.

Solution 11.1.

(a) If $f(x)$ is a continuous function on a closed, bounded interval $I = [a, b]$, then f attains its maximum and minimum values over I.

(b) The function does not attain a maximum over the interval because $f'(x) = \frac{1\cdot(x+1)-1\cdot(x)}{(x+1)^2} = \frac{1}{(x+1)^2}$ is not zero or undefined for $0 \le x < 1$. This means f has no critical points in the given interval, so it can only attain a maximum at x=0, since the other endpoint is not included. However,

$$f(0) = 0, \qquad \lim_{x\to 1^-} f(x) = \frac{1}{2}$$

so the maximum *would* occur at $x = 1$.

Example 11.2. Suppose we know $f(x) = x^2e^x$. Determine the maximum value of $f(x)$ on the interval $[-1, 1]$ using the First Derivative Test. (Hint: $e^x > 0$ for every x.)

Solution 11.2. We first find the critical points of the function.

$$f'(x) = 2xe^x + x^2e^x = 0 \iff x(x+2) = 0,$$

where we divide by e^x since $e^x \neq 0$. Thus the critical points are $x = 0$, $x = -2$; however, we note that $x = -2$ is not in the interval $[-1, 1]$, so we can ignore it in the foregoing calculations. To use the First Derivative Test, we use test values $x = -1/2, 1/2$.

$$f'(-1/2) = -e^{-1} + \frac{1}{4}e^{-1} < 0, \qquad f'(1/2) = e^{-1} + \frac{1}{4}e^{-1} > 0$$

This shows $f(1)$ is a local minimum of $f(x)$, not a max. Checking the endpoints

$$f(-1) = e^{-1}, \quad f(1) = e.$$

Since $e > 1 > e^{-1}$, $f(1) = e$ is the maximum value of $f(x)$ over $[-1, 1]$.

Example 11.3. We are given that

$$f''(x) = -6(x-10)(x+2).$$

(a) What are the inflection points of f?

(b) Where is f concave up and where is f concave down? Express your answer in interval notation.

(c) Based on (a) and (b), determine all local maxima and minima of $f(x)$.

Solution 11.3.

(a) The places where $f''(x) = 0$ are $x = -2, 10$. Keep in mind though that inflection points are where the second derivative *changes sign*! We test $f''(x)$ at $x = -3, 0$, and 11.

$$f''(-3) = -6(-13)(-1) < 0$$
$$f''(0) = -6(-10)(2) > 0$$
$$f''(11) = -6(1)(13) < 0$$

Thus $x = -2$ and $x = 10$ are both inflection points of $f(x)$.

(b) Based on work done in (a), $f(x)$ is concave up on $(-2, 10)$ (the interval between the critical points $x = -2$ and $x = 10$ where $f'' > 0$). On the other side of things, f is concave down on $(-\infty, -2) \cup (10, \infty)$.

(c) From the given information, we cannot determine where f has maxima/minima!! To employ the Second Derivative Test, we **need to know we have critical points**.

Example 11.4. A function f satisfies the following properties:

- Has y-intercept of 3 and x-intercepts at $x = -7$ and $x = 6$.
- The function is decreasing on $(-\infty, -4) \cup (0, 3)$ and increasing on $(-4, 0) \cup (3, \infty)$
- The function is concave up on $(-\infty, -2) \cup (1, \infty)$, concave down on $(-2, 1)$, and has inflection points on $x = -2$ and $x = 1$.
- $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = 10$ (** Note: this type of infinite limit has not been covered in detail yet, but will be this week).

Sketch a graph of $f(x)$ based on the information above.

Solution 11.4.

11.2 Day 2

Goal: Be able to identify/calculate horizontal asymptotes of functions, both visually and algebraically.

11.2.1 Limits at Infinity

We want to know what it means for a function f to have a finite limit at infinity. In particular, we will emphasize the limits of rational functions as $x \to \pm\infty$. The best place to start is with an example and the use of a table of values.

Example 11.5. Suppose $f(x) = \frac{2x^2+3x+17}{x^2-5x-25}$. How does the function behave as the input x becomes arbitrarily large and positive? In other words, what is $\lim_{x\to\infty} \frac{x^2+3x+17}{2x^2-5x-25}$?

Solution 11.5. One way to find out is to use our calculator and plug in increasingly large values of x, like in the following table:

x	$f(x)$
10	9.88
100	2.1443
1000	2.0131
10000	2.0013
100000	2.0001

Table 4: Table of values for $f(x) = \frac{2x^2+3x+17}{x^2-5x-25}$ as $x \to \infty$.

As x increases, we can see $f(x)$ is becoming closer and closer to 2. In such a case we say **the limit as x tends to infinity ($x \to \infty$) of $f(x)$ is 2**.

In a similar fashion we can determine how the function behaves as $x \to -\infty$, i.e. the input x be comes arbitrarily large in magnitude and negative. We find the following table of values:

x	$f(x)$
-10	1.496
-100	1.8823
-1000	1.9871
-10000	1.9987
-100000	1.9999

Table 5: Table of values for $f(x) = \frac{2x^2+3x+17}{x^2-5x-25}$ as $x \to -\infty$.

Just like before we see that as $x \to -\infty$, the outputs $f(x)$ become approach a value of 2. We say **the limit as x tends to minus infinity ($x \to -\infty$) of $f(x)$ is 2**.

In this first example, $\lim_{x\to\pm\infty} f(x)$ were both finite values. We introduce the following definitions/terminology to describe such a limit in general.

Definition 11.1.

1. If the values of $f(x)$ approach L as x becomes arbitrarily large, positive, we say f **has limit L at infinity**. If the values of $f(x)$ approach L as x becomes arbitrarily large, negative, we say f **has limit L at negative infinity**.

2. If $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$, the horizontal (constant) line $y = L$ is a **horizontal asymptote of** f. (Note: $\lim_{x\to\infty} f(x)$ need not be equal to $\lim_{x\to-\infty} f(x)$ in general.)

In the first few weeks of the semester we looked at the 'end behavior' of polynomials. Since end behavior is really just $\lim_{x\to\pm\infty}$, we can now express the same facts in a slightly more formal way, but this is really just a refresher.

Theorem: Given a polynomial $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, the end behavior of f is the same as the end behavior of the power function $P(x) = a_n x^n$. In symbols,

$$\lim_{x\to\pm\infty}(a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0) = \lim_{x\to\pm\infty} a_n x^n.$$

Theorem: The end behavior of a power function $P(x) = cx^n$ where n is an integer is determined by the sign of c and the parity (evenness or oddness) of n. This is expressed in the following cases.

1. If n is even and $c > 0$, $\lim_{x\to\pm\infty} P(x) = +\infty$
2. If n is even and $c < 0$, $\lim_{x\to\pm\infty} P(x) = -\infty$
3. If n is odd and $c > 0$, $\lim_{x\to-\infty} P(x) = -\infty$ and $\lim_{x\to\infty} P(x) = +\infty$
4. If n is odd and $c < 0$, $\lim_{x\to-\infty} P(x) = +\infty$ and $\lim_{x\to\infty} P(x) = -\infty$.

A couple of problems to familiarize ourselves with this once more:

Example 11.6.

(a) Calculate $\lim_{x\to\infty} -2x^3 + 4x - 1000$.

(b) Calculate $\lim_{x\to-\infty} x^{100} - x^{101} + 10x^{102}$.

Solution 11.6.

(a) Since the polynomial has an odd degree and negative leading coefficient, the limit at infinity evaluates to $-\infty$.

(b) Since the degree of the polynomial is even (102) and it has positive leading coefficient, the limit at $-\infty$ is $+\infty$.

11.2.2 End Behavior of Rational Functions

We now move away from polynomials and power functions and toward rational functions. Keeping in mind the first example today, we might wonder if there is a rule for when $\lim_{x\to\pm\infty} \frac{p(x)}{q(x)}$ is finite. Indeed, **the degree of the numerator and degree of the denominator play a key role**.

Theorem: Suppose $f(x) = \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0}$ is a rational function. Then the end behavior of f falls into one of the following cases.

1. If the degree of the numerator equals the degree of the denominator, i.e. $n = m$, then $\lim_{x\to\pm\infty} f(x) = \frac{a_n}{b_m}$ (the ratio of the coefficients on the terms with highest power, the function has horizontal asymptote $y = a_n/b_m$).
2. If the degree of the numerator is less than the degree of the denominator, i.e. $n < m$, then $\lim_{x\to\pm\infty} f(x) = 0$ (the function has horizontal asymptote of $y = 0$).

3. If the degree of the numerator is greater than the degree of the denominator, i.e. $n > m$, then f has no horizontal asymptotes and $\lim_{x\to-\infty} f(x) = \pm\infty$ and $\lim_{x\to\infty} f(x) = \pm\infty$. The values of the limits depend on the signs of the leading terms.

Strategies for Determining End Behavior of Rational Functions:

Method A

(1) Determine the degree of the numerator and the degree of the denominator; **(2)** Divide every term in numerator and denominator by the LOWER of these two degrees; **(3)** Take the limit as $x \to \pm\infty$, as each term will be in a more manageable form. *See the examples for further reference.*

Method B

(1) Identify the highest order terms in the numerator and denominator; **(2)** Ignore all lower order terms (this should be done carefully); **(3)** Take the limit as $x \to \pm\infty$.

11.3 Day 3

Goal: Work through examples of determining end behavior of functions (rational and otherwise) and using that information to sketch a graph.

First, an important note about inflection points: **Inflection points are not just where $f''(x) = 0$! They are points where the sign of the second derivative changes.**

Definition 11.2. A function $f(x)$ has an **inflection point** at $x = c$ if for some $r > 0$, $f''(x) > 0$ (respectively $f''(x) < 0$) for x in the interval $(x - r, x)$ and $f''(x) < 0$ (respectively $f''(x) > 0$) for x in the interval $(x, x + r)$.

Remarks: Some things to notice about the definition...

- $f''(c)$ may be equal to 0 OR undefined.
- If $x = c$ is an inflection point, we can equivalently say it is a critical point of $f'(x)$ corresponding to either a max or min of the derivative (since the sign of f'' is required to change).

Now returning to our regularly scheduled program: determining end behavior.

Example 11.7. Let $f(x) = \frac{4x^4-11x^5+4x-10}{x^2+x^3+x^4}$. Determine the end behavior of $f(x)$.

Solution 11.7.

Since the degree of the numerator (5) is greater than the degree of the denominator (4), we know the limits at infinity will be $\pm\infty$. Ignoring all lower order terms, we see

$$\lim_{x\to\infty} \frac{4x^4 - 11x^5 + 4x - 10}{x^2 + x^3 + x^4} = \lim_{x\to\infty} \frac{-11x^5}{x^4}$$
$$= \lim_{x\to\infty} -11x = -\infty.$$

On the other side of things we have by the same reasoning

$$\lim_{x\to-\infty} \frac{4x^4 - 11x^5 + 4x - 10}{x^2 + x^3 + x^4} = \lim_{x\to-\infty} -11x = +\infty.$$

Notice here we followed Method B. If we had followed Method A, we would see

$$\frac{\frac{1}{x^4}}{\frac{1}{x^4}}\left(\frac{4x^4-11x^5+4x-10}{x^2+x^3+x^4}\right)=\frac{4-11x+\frac{4}{x^3}-\frac{10}{x^4}}{\frac{1}{x^2}+\frac{1}{x}+1}.$$

Now taking the limits as above, we again see the behavior of $-11x$ emerge.

Let's apply similar reasoning to a function involving a radical.

Example 11.8. Suppose $f(x)=\frac{3x+5}{\sqrt{9x^2+4}}$. Determine $\lim_{x\to\pm\infty} f(x)$.

Solution 11.8.

In this case, what is the highest power in the denominator? For very large values of x (in magnitude), $\sqrt{9x^2+4}\approx\sqrt{9x^2}=3|x|$. Thus the highest order term in the denominator behaves like $3|x|$, so we divide by $|x|$:

$$\frac{3x+5}{\sqrt{9x^2+4}}=\frac{\frac{1}{|x|}}{\frac{1}{|x|}}\left(\frac{3x+5}{\sqrt{9x^2+4}}\right)=\frac{\frac{3x}{|x|}+\frac{5}{|x|}}{\sqrt{\frac{9x^2}{x^2}+\frac{4}{x^2}}}=\frac{\frac{3x}{|x|}+\frac{5}{|x|}}{\sqrt{9+\frac{4}{x^2}}}.$$

Here we used $\frac{1}{|x|}\sqrt{9x^2+4}=\sqrt{\frac{1}{x^2}}\sqrt{9x^2+4}=\sqrt{\frac{1}{x^2}(9x^2+4)}$. Now taking the limit as x goes to positive infinity we obtain

$$\lim_{x\to\pm\infty}\frac{\frac{3x}{|x|}+\frac{5}{|x|}}{\sqrt{9+\frac{4}{x^2}}}=\lim_{x\to\infty}\frac{3+\frac{1}{x}}{\sqrt{9+\frac{4}{x^2}}}=1.$$

Toward negative infinity, $|x|=-x$ (since $x<0$), so

$$\lim_{x\to\pm\infty}\frac{\frac{3x}{|x|}+\frac{5}{|x|}}{\sqrt{9+\frac{4}{x^2}}}=\lim_{x\to\infty}\frac{-3-\frac{1}{x}}{\sqrt{9+\frac{4}{x^2}}}=-1.$$

Some more examples:

Example 11.9.

1. Let $f(x)=\frac{1}{x}+\frac{2x}{x-1}$. Compute $\lim_{x\to\pm\infty} f(x)$.
2. True or False: If $f(x)$ has a horizontal asymptote $y=4$, then $\lim_{x\to-\infty} f(x)=4$.
3. Sketch a graph of a function which is (i) increasing on $(-\infty,-1)\cup(-1,\infty)$, (ii) has one root at $x=-3$, (iii) has a vertical asymptote at $x=-1$, with $\lim_{x\to-1^-} f(x)=\infty$, $\lim_{x\to-1^+} f(x)=-\infty$, (iv) $f(x)$ has a limit at $-\infty$ of -2 and a limit at $+\infty$ of -1, and (v) $f(x)$ is concave up on $(-\infty,-1)$ and concave down on $(-1,\infty)$.
4. Compute $\lim_{x\to\infty}\frac{24x^5-2x+500000}{-6x^5+x^4+x^3+x^2-1000}$.

Solution 11.9.

1. Since $\frac{1}{x}\to 0$ as $x\to\pm\infty$ and $\frac{2x}{x-1}\to 2$ as $x\to\pm\infty$,
$$\lim_{x\to\pm\infty}\frac{1}{x}+\frac{2x}{x-1}=2.$$
2. False! This is not necessarily true, as it could be the case that the asymptote occurs as $x\to-\infty$.
3. Will attempt to do in class.
4. The degree of the denominator is greater than the degree of the numerator, so the limit is 0.

12 Week 12

Outline

Days: Monday (April 13), Wednesday (April 15), Friday (April 17)
Topics: *Exam 3 Review, Exam 3, Antiderivatives.*
Goals:

- Review thoroughly for Exam 3.
- Take Exam 3.
- Begin our exploration of antiderivatives/integration.

Album of the Week: *Back to Black* by *Amy Winehouse (2006)*

12.1 Day 1

Goal: Exam 3 review.

Some practice problems for the exam (as if there aren't already enough). Recall the sections being covered are

- **4.1 (Elasticity of demand)**
- **4.2 (Linear approximations)**
- **4.3 (Maxima and minima)**
- **4.5 (Derivatives and graph shape)**
- **4.6 (Limits at infinity)**

Example 12.1.

1. Suppose a product has demand function given by $x = 4e^{-p} + 2$ for $1 \leq p$ (meaning the formula applies when the price is at least 1). Calculate $E(2)$.
2. Suppose that $f(10) = 17$, $f'(10) = 3$. (i) Write a linear approximation $L(x)$ of f at $x = 10$ and (ii) Use $L(x)$ to approximate $f(10.1)$.
3. If $E(p) = 10 - 3p^2$, what price should the company charge to maximize revenue? (**Reminder/Massive Hint: Revenue is maximized when $E(p) = 1$.**)
4. Suppose a function $f(x)$ has derivative $f'(x) = 6(x-1)(x-3)$. Find all absolute maxima and minima of f.
5. We're given a function $f(x)$ such that $f''(x) = -2x(x+4)^2$. Find all inflection points of f. (**Hint: an inflection point is an input x where f'' changes sign.**)

6. Let $f(x) = \frac{-15x^5+3x^3-5x+7}{3x^5-1000000x+12}$. (i) What is the degree of the numerator? (ii) What is the degree of the denominator? (iii) What is $\lim_{x\to-\infty} f(x)$?

Solution:

1. We first need $E(p) = -\frac{p}{x}\frac{dx}{dp}$. In this instance, $\frac{dx}{dp} = \frac{d}{dp}(4e^{-p}+2) = -4e^{-p}$ so

$$E(p) = -\frac{p}{4e^{-p}+2}\left(-4e^{-p}\right) = \frac{2p}{2+e^p}.$$

Now we plug in $p = 2$:

$$\boxed{E(2) = \frac{4}{2+e^2}} = 0.42...$$

2. (i) The linearization $L(x)$ of a function $f(x)$ at a point $x = a$ is just the equation of the tangent line, $L(x) = f'(a)(x-a)+f(a)$. We see $a = 10$ (we have information about $f(10)$ and $f'(10)$) so

$$\boxed{L(x) = f'(10)(x-10)+f(10) = 3(x-10)+17.}$$

(ii) One use of the linearization at a particular point is to estimate values of $f(x)$ for x near $a = 10$. In symbols, $f(x) \approx L(x)$ for x close to 10. In this case,

$$f(10.1) \approx L(10.1) = 3(10.1-10)+17 = \boxed{17.3}$$

3. As the hint told us, to find the price which maximizes revenue we set $E(p) = 1$ and solve for p:

$$10-3p^2 = 1 \implies -3p^2 = -9 \implies p^2 = 3 \implies p = \pm\sqrt{3}.$$

However, clearly $p = -\sqrt{3}$ is not reasonable for a price so $\boxed{p = \sqrt{3}}$.

4. We know the critical points of f are $x = 1$ and $x = 3$. To see which corresponds to a max/min/neither, we test $f'(x)$ at points to either side of the critical points (applying the First Derivative Test). Our test points will be $x = 0$, $x = 2$, and $x = 4$:

$$f'(0) = 6(-1)(-3) > 0; \quad f'(2) = 6(1)(-1) < 0; \quad f'(4) = 6(3)(1) > 0.$$

The First Derivative Test then implies f has a local maximum at $x = 1$ and f has a local minimum at $x = 3$. (Use the number line to make things clearer!)

5. Since an inflection point is where f'' changes sign, $f''(c)$ must be 0 or undefined at an inflection point. Since the given f'' is defined everywhere, we can solve for where $f''(x) = 0$ first:

$$f''(x) = -2x(x+4)^2 = 0 \iff x = 0, x = -4.$$

Now we form the usual number line and test f'' at points on either side of $x = 0$ and $x = -4$ to see if f'' changes sign. Our test points will be $x = -5$, $x = -1$, and $x = 1$:

$$f''(-5) = -2(-5)(-1)^2 > 0; \quad f''(-1) = -2(-1)(3)^2 > 0; \quad f''(1) = -2(1)(4)^2 < 0.$$

This shows f'' changes sign at $x = 0$ but *not* at $x = -4$. Thus f has one inflection point at $x = 0$.

6. (i) The degree of the numerator is 5. (ii) The degree of the denominator is 5. (iii) In the case where the degree of the denominator is equal to the degree of the denominator, the limit as x tends to plus or minus infinity is the ratio of their leading coefficients:

$$\lim_{x\to-\infty} \frac{-15x^5+3x^3-5x+7}{3x^5-1000000x+12} = \frac{-15}{3} = \boxed{-5}.$$

12.2 Day 2

Goal: Take Exam 3, don't let Exam 3 take you.

12.3 Day 3

Goal: A couple of extra optimization problems. Introduce the idea of an *antiderivative*, possibly relate to integration.

Example 12.2. Cole offers tutoring at $ 60 per hour (he's not as cheap as the guy in the worksheet) and currently tutors 6 hours per week. He determines that he will be able to tutor 3 hours per week more for every $ 2 decrease in price. What price should he charge to maximize revenue?

Solution 12.1.

We can determine the revenue function of this tutoring gig, but to get there we should find the pattern of how hours tutored relates to the hourly rate:

Hourly Rate	Number of Hours Tutored
60	6
58	9
56	12
54	15
60-2x	6+3x

Table 6: Pattern of how hourly rate and hours tutored relate.

In the above table we can think of x as the number of hours (relative to 6) which Cole tutors. Since Revenue = (Number of Hours)(Hourly Rate), we find

$$R(x) = (6+3x)(60-2x) = 360+168x-6x^2.$$

This is maximized when $R'(x) = 0$, i.e.

$$R'(x) = -12x+168 = 0 \iff x = 14$$

which means Cole should set his price to be $60-28 = 32$ dollars per hour.

Example 12.3. Someone is building a fenced in area for their dog which will be a rectangle. One side will be along the side of their house, which is 30 ft. (so this side will not have fencing). If they have 100 feet of fencing, what is the maximum area they can enclose?

Solution 12.2.

We are trying to maximize area; if x and y are the side lengths of the rectangle,

$$A(x, y) = xy.$$

To reduce this to a single variable problem, we use the constraint:

$$2x + y = 100$$

where we remember we're only using the fencing for 3 sides of the enclosure. This implies $y = 100 - 2x$, therefore

$$A(x, y) = A(x) = x(100 - 2x) = -2x^2 + 100x.$$

Note that $0 \leq y \leq 30$, so because $x = -\frac{y}{2} + 50$ we get $50 \geq x \geq 35$. Now finding the maximum area:

$$A'(x) = -4x + 100 = 0 \iff x = 25.$$

Unfortunately this is not in our domain, we have to test the endpoints:

$$A(35) = 1050\,\text{ft}^2, \qquad A(50) = 0.$$

Thus the biggest the pen should have an area of 1050 square feet, with the side parallel to the house being $y = 100 - 2(35) = 100 - 70 = 30$ feet (taking up the full side of the house), while the two sides perpendicular to the house will be $x = -\frac{30}{2} + 50 = 35$ feet.

12.3.1 Antiderivatives

One natural question we've yet to answer: is there a process or operation which undoes differentiation? We'll start answering that here.

Definition 12.1. A differentiable function $F(x)$ is an **antiderivative** of $f(x)$ if $F'(x) = f(x)$.

Remark: Note that for any differentiable function f, f is an antiderivative of f' so the terminology "antiderivative" makes sense.

Example 12.4. Find an antiderivative for each of the following functions. Is your answer the only possible one?

1. $f(x) = -\frac{1}{x^2}$
2. $f(x) = 3x^2 + 2x + 1$
3. $f(x) = e^x$

Notice in the above examples that each function technically has infinitely many antiderivatives! We can add a constant C to any particular antiderivative and obtain a new one. This constant C will be called the **constant of integration**.

Definition 12.2. Given a 'nice' function f, the **indefinite integral of** f, denoted $\int f(x)dx$, is the **most general antiderivative of** f. In other words, given any antiderivative F of f, $\int f(x)dx = F(x)+C$ (shifted by a constant).

Example 12.5. Suppose the velocity of a particle moving in one dimension as a function of time is measured to be $v(t) = -5t + 10$.

(a) Can we *uniquely* determine the position of the particle $p(t)$ as a function of time based only on $v(t)$? Explain why not.

(b) Suppose we are also given that $p(0) = 10$. Now can we completely determine the position $p(t)$?

(c) Explain in words what the **initial condition** in (b) allows us to do in terms of antiderivatives.

Solution 12.3.

(a) No, we cannot determine $p(t)$ uniquely just from $v(t)$. This is because $p(t)$ has only one requirement: $p'(t) = v(t)$ (rate of change of position is velocity). As a result, all we can find is

$$p(t) = \int v(t)dt + C = -\frac{5}{2}t^2 + 10t + C$$

which contains infinitely many possibilities because of the constant C of integration.

(b) Yes, now we can fully determine $p(t)$. By plugging in $t = 0$ into the general antiderivative found in (a), we see

$$10 = p(0) = -\frac{5}{2}(0)^2 + 10(0) + C = C \implies C = 10.$$

Concretely, the initial condition forces $p(t) = -\frac{5}{2}t^2 + 10t + 10$.

(c) The initial condition determines what value the constant of integration C must take, reducing the number of possible position functions from infinitely many to at most one.

Theorem: A collection of rules for finding the most general antiderivative of a function:

$f(x)$	$\int f(x)dx$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$
$\frac{1}{x}$	$\ln(\lvert x\rvert) + C$
e^x	$e^x + C$
a^x	$\frac{a^x}{\ln(a)} + C$

Table 7: Most frequently used antiderivatives. Moving from right column to left is done just by taking derivatives.

13 Week 13

Outline

Days: Monday (April 20), Wednesday (April 22), Friday (April 24)
Topics: *Antiderivatives, estimating areas under curves, definite integral.*
Goals:

- Understand how to compute antiderivatives and their importance.
- Be able to estimate signed areas under graphs using left, right, and midpoint approximations (Riemann sums).
- Calculate exact areas using the definite integral.

Album of the Week: *To Pimp a Butterfly* by *Kendrick Lamar (2015)*

13.1 Day 1

Goal: Review of antiderivatives and connection to areas under curves. Estimation of areas via rectangles.

Example 13.1. Cole offers tutoring at $ 60 per hour (he's not as cheap as the guy in the worksheet) and currently tutors 6 hours per week. He determines that he will be able to tutor 3 hours per week more for every $ 2 decrease in price. What price should he charge to maximize revenue?

Solution 13.1.

We can determine the revenue function of this tutoring gig, but to get there we should find the pattern of how hours tutored relates to the hourly rate:

Hourly Rate	Number of Hours Tutored
60	6
58	9
56	12
54	15
60-2x	6+3x

Table 8: Pattern of how hourly rate and hours tutored relate.

In the above table we can think of x as the number of hours (relative to 6) which Cole tutors. Since Revenue = (Number of Hours)(Hourly Rate), we find

$$R(x) = (6 + 3x)(60 - 2x) = 360 + 168x - 6x^2.$$

This is maximized when $R'(x) = 0$, i.e.

$$R'(x) = -12x + 168 = 0 \iff x = 14$$

which means Cole should set his price to be $60 - 28 = 32$ dollars per hour.

Example 13.2. Someone is building a fenced in area for their dog which will be a rectangle. One side will be along the side of their house, which is 30 ft. (so this side will not have fencing). If they have 100 feet of fencing, what is the maximum area they can enclose?

Solution 13.2.

We are trying to maximize area; if x and y are the side lengths of the rectangle,

$$A(x, y) = xy.$$

To reduce this to a single variable problem, we use the constraint:

$$2x + y = 100$$

where we remember we're only using the fencing for 3 sides of the enclosure. This implies $y = 100 - 2x$, therefore

$$A(x, y) = A(x) = x(100 - 2x) = -2x^2 + 100x.$$

Note that $0 \leq y \leq 30$, so because $x = -\frac{y}{2} + 50$ we get $50 \geq x \geq 35$. Now finding the maximum area:

$$A'(x) = -4x + 100 = 0 \iff x = 25.$$

Unfortunately this is not in our domain, we have to test the endpoints:

$$A(35) = 1050\,\text{ft}^2, \qquad A(50) = 0.$$

Thus the biggest the pen should have an area of 1050 square feet, with the side parallel to the house being $y = 100 - 2(35) = 100 - 70 = 30$ feet (taking up the full side of the house), while the two sides perpendicular to the house will be $x = -\frac{30}{2} + 50 = 35$ feet.

13.1.1 Antiderivatives

Let's recall some basics about antiderivatives before we continue.

Definition 13.1. A differentiable function $F(x)$ is an **antiderivative** of $f(x)$ if $F'(x) = f(x)$.

Remark: Note that for any differentiable function f, f is an antiderivative of f' so the terminology "antiderivative" makes sense.

Example 13.3. Find an antiderivative for each of the following functions. Is your answer the only possible one?

1. $f(x) = -\frac{1}{x^2}$
2. $f(x) = 3x^2 + 2x + 1$
3. $f(x) = e^x$

Notice in the above examples that each function technically has infinitely many antiderivatives! We can add a constant C to any particular antiderivative and obtain a new one. This constant C will be called the **constant of integration**.

Definition 13.2. Given a 'nice' function f, the **indefinite integral of** f, denoted $\int f(x)dx$, is the **most general antiderivative of** f. In other words, given any antiderivative F of f, $\int f(x)dx = F(x)+C$ (shifted by a constant).

Theorem: A collection of rules for finding the most general antiderivative of a function:

$f(x)$	$\int f(x)dx$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$
$\frac{1}{x}$	$\ln(\lvert x \rvert) + C$
e^x	$e^x + C$
a^x	$\frac{a^x}{\ln(a)} + C$

Table 9: Most frequently used antiderivatives. Moving from right column to left is done just by taking derivatives.

Example 13.4. Calculate the antiderivative of each function:

1. $f(x) = 2\sqrt{x} + \frac{1}{x^2} - x^4$
2. $f(x) = e^x + e^{-2x}$

Solution 13.3. An important observation to make is that because the derivative operator is linear (it distributes over addition and scales), so is integration.

1.
$$\begin{aligned}\int f(x)dx = \int (2x^{\frac{1}{2}} + x^{-2} - x^4)dx &= 2\int x^{\frac{1}{2}}dx + \int x^{-2}dx - \int x^4 dx \\ &= 2\left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right) - x^{-1} - \frac{x^5}{5} + C.\end{aligned}$$

2.
$$\int (e^x + e^{-2x})dx = \int e^x dx + \int e^{-2x}dx = e^x - \frac{1}{2}e^{-2x} + C.$$

Notice here that $\int e^{-2x}dx \neq e^{-2x} + C$ because $\frac{d}{dx}(e^{-2x}) = -2e^{-2x}$. To account for this factor of -2, we divide by -2!

Example 13.5. Verify the following antiderivative:

$$\int (xe^x)dx = xe^x - e^x + C$$

Solution 13.4. To verify, we just take the derivative of the right side above:

$$\begin{aligned}\frac{d}{dx}(xe^x - e^x + C) = \frac{d}{dx}(xe^x) - \frac{d}{dx}(e^x) &= (e^x + xe^x) - e^x \\ &= xe^x.\end{aligned}$$

Since xe^x is the original integrand from the left side, the antiderivative is verified.

Example 13.6. Suppose we are told that a particle moves with constant velocity $v(t) = 5$ for $0 \leq t \leq 3$.

1. Determine how far the particle travels in the time frame $0 \leq t \leq 3$.
2. Determine the equation $p(t)$ for the position of the particle over time if $p(1) = 2$.
3. Determine the area under the graph of $v(t)$ above the t-axis. How does this relate to $p(3) - p(0)$?

Solution 13.5.

1. With a velocity of 5 m/s (to use reasonable units) for 3 seconds, the particle travels $(5\ \ m/s)(3\ \ s) = 15\ \ m$.

2. First, $p(t)$ is an antiderivative of $v(t)$, so it must have the form

$$p(t) = \int v(t)dt = \int 5dt = 5t + C.$$

   Given $p(1) = 2$ we further impose $2 = p(1) = 5 + C \implies C = -3$. Hence, $p(t) = 5t - 3$.

3. The graph of $v(t)$ is a straight, horizontal line at a height of 5 units and a width of 3 units; the area is the area of a rectangle which is found to be 15 units. On the other hand,

$$p(3) = 12, \quad p(0) = -3, \quad p(3) - p(0) = 12 - (-3) = 15.$$

   The area under the velocity curve from $t = 0$ to $t = 3$ is the same as $p(3) - p(0)$.

By the above example, antiderivatives are related very concretely to the area under the graph of a function f. Suppose $f(x)$ continuous and defined on a closed, bounded interval $[a, b]$. **Is it possible to determine the (signed) area contained between the graph of f and the x-axis?** It turns out that this is a question and its answer are of fundamental importance, with applications reaching far beyond this class.

The most natural way to determine the area is first by approximation, using simple shapes to roughly cover the area under the graph of f. Rectangles have especially simple areas which can be used in left, right, and midpoint approximations. We can construct these approximations in roughly three steps:

1. **Form a partition of the domain**. When we say 'take a partition of $[a, b]$,' we mean pick a collection of points $x_0 = a < x_1 < x_2 < \cdots < x_n = b$ in the interval, and in so doing chop the interval into smaller pieces (called subintervals) $[x_0, x_1], [x_1, x_2]$, etc.

2. **Estimate the area under the curve over each subinterval** $[x_i, x_{i+1}]$. This can be done in several ways, but using rectangles we have:

   - **Left endpoint approximation**: the area under f over $[x_i, x_{i+1}]$ is approximately $f(x_i)(x_{i+1} - x_i)$ for $i = 0, 1, \ldots, n - 1$.
   - **Right endpoint approximation**: the area under f over $[x_i, x_{i+1}]$ is approximately $f(x_{i+1})(x_{i+1} - x_i)$ for $i = 0, 1, \ldots, n - 1$.
   - **Midpoint approximation**: the area under f over $[x_i, x_{i+1}]$ is approximately $f\left(\frac{x_i + x_{i+1}}{2}\right)(x_{i+1} - x_i)$ for $i = 0, 1, \ldots, n - 1$. Note here that the input $\frac{x_i + x_{i+1}}{2}$ is just the halfway point between x_i and x_{i+1}.

Why do we need different kinds of approximations? Sometimes one will give an underestimate while another will give an overestimate, and this is very useful because it allows us to pin the actual area within a small range. Let's see an example.

Example 13.7. Estimate the area under $f(x) = x^2$ over the interval $[1, 9]$ using 4 rectangles of equal width in the Left Endpoint Approximation and Right Endpoint Approximation formulas.

Solution: Since we're asked to use 4 rectangles of equal width, we divide the length of the interval by the number of rectangles we want to determine their width. In this case our interval width will be $\Delta x = \frac{9-1}{4} = 2$. This tells us our partition is $\{1, 3, 5, 7, 9\}$, and so the left approximation is

$$\begin{aligned}\text{Area} \approx L_4 &= f(1)(3 - 1) + f(3)(5 - 3) + f(5)(7 - 5) + f(7)(9 - 7) \\ &= 2(1 + 9 + 25 + 49) = 2(84) = 168.\end{aligned}$$

Alternatively the right approximation is

$$\begin{aligned}\text{Area} \approx R_4 &= f(3)(3-1) + f(5)(5-3) + f(7)(7-5) + f(9)(9-7) \\ &= 2(9+25+49+81) = 2(164) = 328.\end{aligned}$$

Visually, because f is strictly increasing over $[1, 9]$, L_4 is an underestimate and R_4 is an overestimate.

A key question: since the estimates in the above example seem so far apart, how do we bring them closer together? In other words, how do we improve our estimates within the same schematic? Make the rectangles narrower, i.e. use more of them! Just by roughly sketching this idea we can see the error will decrease.

13.2 Day 2

Goal: Area estimation practice using various rules (endpoints, midpoints).

13.3 Day 3

Goal: Utilize the definition of the definite integral. State Fundamental Theorem of Calculus.

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14 Week 14

Outline

Days: Monday (December 1), Wednesday (December 3), Friday (The End)
Topics: *Fundamental Theorem of Calculus, u-substitution.*
Goals:

- Review the statement and meaning of FTC.
- Utilize FTC to calculate areas.
- Understand u-substitution as a means of undoing the chain rule.

Hozier (Special Edition)* by *Hozier (2014)

14.1 Day 1

Goal: Recall the statement of Fundamental Theorem of Calculus. Use this theorem for computation.

Prior to break, with 3 people in attendance, we discussed the most important theorem in this course.
Fundamental Theorem of Calculus: Let $f(x)$ be a function which is continuous on the interval $[a, b]$.

1. If $F(x) = \int_a^x f(t)dt$ for $a \leq x \leq b$, then $F(x)$ is an antiderivative of $f(x)$. In particular, $F(x)$ is differentiable and
$$\frac{d}{dx}F(x) = \frac{d}{dx}\left(\int_a^x f(t)dt\right) = f(x).$$
2. If $F(x)$ is any antiderivative of $f(x)$, then
$$\int_a^b f(x)dx = F(b) - F(a).$$

Remark:

- Part 1 of FTC says that finding the area under $f(x)$ is the same as finding (or produces) an antiderivative.
- Part 2 of FTC says that to find the area beneath $f(x)$ over $[a, b]$, it is enough to evaluate any antiderivative at the endpoints a and b and take the difference.

Both parts of the theorem have their uses, most obviously the second because Riemann sums are not convenient for finding areas under curves. The first part poses some interesting questions about derivatives of **integral-defined** functions.

Example 14.1. Let $F(x) = \int_1^x (t^2 - 2)dt$.

(a) What is $F'(x)$? Calculate (i) using FTC part 1, then (ii) using FTC part 2 and normal derivative rules.

(b) Write down the equation for $G(x) = F(-x^2)$ and calculate $G'(x)$. (Hint: Chain rule!)

Solution 14.1. Exercise for the reader.

Example 14.2. Let $f(x) = e^{-3x} + 4$.

(a) Evaluate $\int_{-1}^{0} f(x)dx$ using FTC part 2.

(b) Evaluate $\int_{0}^{1} f(x)dx$ using FTC part 2.

(c) Evaluate $\int_{-1}^{1} f(x)dx$ using FTC part 2.

(d) Based on (a)-(c), how can we relate $\int_a^c f(x)dx$ to $\int_a^b f(x)dx + \int_b^c f(x)dx$? Explain why this makes sense using areas.

Solution 14.2. As a convenient starting point, an antiderivative of $f(x)$ (with $C = 0$) is

$$\int (e^{-3x} + 4)dx = -\frac{1}{3}e^{-3x} + 4x.$$

(a)
$$\int_{-1}^{0} f(x)dx = \left[-\frac{1}{3}e^{-3x} + 4x\right]_{-1}^{0} = \left(-\frac{1}{3}e^{0} + 4(0)\right) - \left(-\frac{1}{3}e^{3} - 4\right)$$
$$= \frac{1}{3}e^{3} + \frac{11}{3}.$$

(b)
$$\int_{0}^{1} f(x)dx = \left[-\frac{1}{3}e^{-3x} + 4x\right]_{0}^{1} = \left(-\frac{1}{3}e^{-3} + 4\right) - \left(-\frac{1}{3}e^{0} + 4(0)\right)$$
$$= -\frac{1}{3}e^{-3} + \frac{13}{3}.$$

(c)
$$\int_{-1}^{1} f(x)dx = \left[-\frac{1}{3}e^{-3x} + 4x\right]_{-1}^{1} = \left(-\frac{1}{3}e^{-3} + 4\right) - \left(-\frac{1}{3}e^{3} - 4\right)$$
$$= \underbrace{\frac{1}{3}e^{3} - \frac{1}{3}e^{-3}}_{\text{can't combine! different exp.}} + 8.$$

(d) We see that $\int_{-1}^{1} f(x)dx = \int_{-1}^{0} f(x)dx + \int_{0}^{1} f(x)dx$. More generally, for $a \le b \le c$,

$$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$$

which geometrically says that the area under the whole curve from $x = a$ to $x = c$ can be found by adding (the area under the curve from $x = a$ to $x = b$) to (the area under the curve from $x = b$ to $x = c$). Areas can be broken down and then summed.

Example 14.3. Suppose $f(x) = -|x - 4| + 5$ is defined for $0 \le x \le 9$.

(a) Calculate $\int_0^9 f(x)dx$ using geometry.

(b) Calculate $\int_0^9 f(x)dx$ using FTC. (Hint: $-|x-4|+5 = x-4+5 = x+1$ when $x \le 4$ and $-|x-4|+5 = -x+9$ when $x \ge 4$.)

Solution 14.3. Exercise in drawing a graph and taking antiderivatives.

14.2 Day 2

Goal: Calculating antiderivatives using substitution.

Our goal in Chapter 5 of the textbook has been calculating areas under graphs, i.e. evaluating $\int_a^b f(x)dx$ where $f(x)$ is continuous on $[a,b]$. Initially we did this by Riemann sum approximation using rectangles, then we took the limit as the number of rectangles went to infinity. This process was intractable for all but the simplest functions $f(x)$, and generally not useful for computation.

Enter FTC. The biggest theorem in the class, it tells us we can avoid the difficulties of Riemann sums by finding an antiderivative $F(x)$ of $f(x)$:

$$\int_a^b f(x)dx = F(b) - F(a).$$

In many instances, this makes calculation of definite integrals (equivalently areas) relatively simple....but there's a catch: **finding an antiderivative $F(x)$ of $f(x)$ is *not* always easy or even possible!** Much like the tools/rules we used when calculating derivatives, we need to develop ways to find antiderivatives of more complicated functions.

14.2.1 Substitution

Heuristic: Given $h'(x)$, the derivative of $h(x)$, an antiderivative of $h'(x)$ is $h(x)$. This means $\int h'(x)dx = h(x) + C$, or for a definite integral $\int_a^b h'(x)dx = h(b) - h(a)$.

The above heuristic is simple, as it is just a restatement of the definition of antiderivative. However, extrapolating slightly, it can be used to obtain a more general rule for finding an antiderivative of a chain rule derivative.

Proposition 14.1 (Undoing Chain Rule). If $h(x) = f(g(x))g'(x)$ and $F(x)$ is an antiderivative of $f(x)$, then $H(x) = F(g(x))$ is an antiderivative of $h(x)$.

Proof. By definition, $H(x)$ is an antiderivative of $h(x)$ provided $H'(x) = h(x)$. We can compute the derivative:

$$H'(x) = \frac{d}{dx}(H(x)) = \frac{d}{dx}(F(g(x))) \underbrace{=}_{C.R.} F'(g(x))g'(x).$$

Since F is an antiderivative of f, $F'(\,\cdot\,) = f(\,\cdot\,)$. Using this in the above, we find

$$H'(x) = F'(g(x))g'(x) = f(g(x))g'(x) = h(x)$$

which shows H satisfies the definition of an antiderivative of h. □

Moral/Takeaway: The function $F(g(x))$ (where $F(x)$ is an antiderivative of $f(x)$) is an antiderivative of $f(g(x))g'(x)$. This tells us if we are able to identify $f(x)$, identify $g(x)$, and determine an antiderivative of $f(x)$, we can find the antiderivative of the more complicated function $f(g(x))g'(x)$ because:

$$f(g(x))g'(x) \underbrace{=}_{\text{Antiderivative}} \frac{d}{dx}(F(g(x)))$$

$$\implies \int f(g(x))g'(x)dx = \underbrace{\int \frac{d}{dx}}_{\text{Cancel by FTC}} (F(g(x)))\,dx = F(g(x)) + C.$$

Stated more formally:

Proposition 14.2 (u-Substitution). Let $u = g(x)$ be differentiable over an interval I and let $f(x)$ be continuous for x in $g(I)$. Then

$$\int f(g(x))g'(x)dx = \int f(u)du = F(u) + C$$
$$= F(g(x)) + C$$

for all x in I.

This abstract stuff is great and all, but how is this used in practice?

Steps to Use u-Sub:

1. Look at the integrand and select/identify a function $g(x)$, set $u = g(x)$. This is the part most people struggle with at first, because there's no 'rule' which applies universally for every integrand. Typically, $g(x)$ is an 'inside' function (in the sense of composition/chain rule), and/or $g'(x)$ (or a constant multiple of $g'(x)$) appears elsewhere in the integrand.

2. Substitute $u = g(x)$, $g'(x)dx = du$ (equivalently $dx = \frac{du}{g'(x)}$) in the integrand. Assuming we chose u correctly, there should be *no x's left in the integrand*, leaving us with an integral $\int f(u)du$ solely in terms of u.

3. Evaluate the new, simpler integral in terms of u. With any luck, the substitution made the integrand nice and we can find $\int f(u)du = F(u) + C$ where F is an antiderivative of f.

4. Replace $u = g(x)$. Our original problem was in terms of the variable x, so we substitute back in $g(x)$ wherever we see u, to give $F(g(x)) + C$.

Only way to get used to this is through many examples, since u-substitution can appear in many different ways.

Example 14.4. Evaluate the indefinite integrals.

(a)
$$\int \left(\frac{e^x}{e^x + 1000} \right) dx$$

(b)
$$\int x\sqrt{x^2 + 4}dx$$

(c)
$$\int \left(\frac{4x + 8}{\sqrt{x^2 + 4x}} \right) dx$$

Solution 14.4.

(a) There are two pieces of the integrand, namely the numerator and denominator of the fraction. The derivative of the numerator is precisely the numerator, so we will choose $u = e^x + 1000$. This implies $\frac{du}{dx} = e^x$, i.e. $du = e^x dx$. We see by subsitution:

$$\int \frac{e^x dx}{e^x + 1000} = \int \frac{du}{u}.$$

The integrand is far simpler in terms of u, and we are now able to evaluate after recalling our table of antiderivative rules:

$$\int \frac{du}{u} = \ln|u| + C.$$

Lastly, we return to the world of x's by replacing $u = e^x + 1000$:

$$\int \left(\frac{e^x}{e^x + 1000}\right) dx = \ln|e^x + 1000| + C.$$

(b) The piece of the integrand which most resembles an 'inside' function is x^2+4. It's derivative nearly appears elsewhere, since $\frac{d}{dx}(x^2+4) = 2x$, and x is another part of the integrand. We choose $u = x^2 + 4$, and therefore have $\frac{du}{dx} = 2x$, i.e. $dx = \frac{1}{2x}du$. By substitution:

$$\int x\sqrt{x^2 + 4}\, dx = \int x\sqrt{u}\frac{du}{2x} = \frac{1}{2}\int u^{\frac{1}{2}} du.$$

Notice how our choice of u allowed us to account for the x in front of the square root via du. We are now able to use the power rule to evaluate the integral in terms of u:

$$\frac{1}{2}\int u^{\frac{1}{2}} du = \frac{u^{\frac{3}{2}}}{3} + C.$$

Replacing $u = x^2 + 4$:

$$\int x\sqrt{x^2+4}dx = \frac{1}{3}(x^2+4)^{\frac{3}{2}} + C.$$

(c) Exercise!

14.3 Day 3

Goal: Putting a bow on the semester.

I lied, I don't actually own any bows for us to put on the semester. Instead, let's do some math.

14.3.1 u-Substitution, cont.

We saw last time that u-substitution is a **method for identifying an integrand $h(x)$ as having the form $h(x) = f(g(x))g'(x)$ and subsequently find the antiderivative of $h(x)$, which is $F(g(x))$ where $F(x)$ is an antiderivative of $f(x)$.** **The procedure for using u-sub is something you only get used to after many examples!**

Example 14.5. Evaluate the definite integral

$$\int_{-1}^{1} xe^{-2x^2}\,dx$$

through the following steps:

(a) Pick $u = -2x^2$. (Make sure you understand why this is the correct/natural choice.) Calculate $\frac{du}{dx}$ and use it to solve for dx.

(b) Rewrite the original integral in terms of u, du. Ensure that x no longer appears.

(c) Evaluate the integral from (b) in terms of u, then substitute back in $u = -2x^2$.

(d) Use FTC part 2 to evaluate the definite integral.

Solution 14.5.

(a) If $u = -2x^2$, then $du = (-4x)dx$, i.e. $dx = -\frac{1}{4x}du$. This is correct because it will account for the x which appears in front of the exponential in the integrand.

(b) Plugging in yields, ignoring bounds for now:

$$\int xe^{-2x^2}dx = \int xe^{u}\left(-\frac{du}{4x}\right) = -\frac{1}{4}\int e^{u}du.$$

(c) We now have a very tenable antiderivative to take:

$$-\frac{1}{4}\int e^{u}du = -\frac{1}{4}e^{u} = -\frac{1}{4}e^{-2x^2}.$$

Note we omit the constant of integration C because we are going to take the definite integral next.

(d) By FTC part 2, the definite integral is just the antiderivative (found in (c)) evaluated between the upper and lower bounds:

$$\int_{-1}^{1} xe^{-2x^2}dx = \left[-\frac{1}{4}e^{-2x^2}\right]_{-1}^{1} = \left(-\frac{1}{4}e^{-2}\right) - \left(-\frac{1}{4}e^{-2}\right) = 0.$$

Ah, that moment of satisfaction where we did a lot of work just to get 0 as an answer!

Could we have foreseen this? Yes, because $f(x) = xe^{-2x^2}$ is an **odd function**, meaning $f(-x) = -xe^{-2x} = -f(x)$. In such a case $\int_{-a}^{a} f(x)dx = \int_0^a f(x)dx \underbrace{-\int_0^a f(x)dx}_{=\int_0^a -f(x)dx} = 0$. In words, **the integral of any odd function over an interval $[-a, a]$ symmetric about the origin is 0**.

Example 14.6. Let $f(x) = \frac{4x-6}{x^2-3x+2}$.

1. Evaluate $\int_{-1}^{0} f(x)dx$.
2. Evaluate $\int_0^1 f(x)dx$. Is there an issue with this computation? Explain why if yes.

Solution 14.6.

1. Choose $u = x^2 - 3x + 2$, so $du = (2x - 3)dx$. This implies, since $4x - 6 = 2(2x - 3)$,

$$\int \frac{4x-6}{x^2-3x+2}dx = 2\int \frac{du}{u} = 2\ln|x^2 - 3x + 2| + C.$$

As a result,

$$\begin{aligned}\int_{-1}^{0} \frac{4x-6}{x^2-3x+2}dx &= \left[2\ln|x^2 - 3x + 2|\right]_{-1}^{0} \\ &= 2\ln(2) - 2\ln(6) = 2\ln\left(\frac{1}{3}\right).\end{aligned}$$

2. By using the same antiderivative (we're integrating the same function),

$$\begin{aligned}\int_0^1 \frac{4x-6}{x^2-3x+2}dx &= \left[2\ln|x^2 - 3x + 2|\right]_0^1 \\ &= 2\ln(0) - 2\ln(2) = -\infty.\end{aligned}$$

There is an issue, infinite area doesn't make physical sense! So what happened?

Notice: $\lim_{x\to 1^-} \frac{4x-6}{x^2-3x+2} = \lim_{x\to 1^-} \frac{2}{x-1} = -\infty$, so our integrand has an asymptote at $x = 1$.

Question: If $f(x)$ has an asymptote at $x = b$, is it always the case that $\int_a^b f(x)dx$ is infinite? Our intuition suggests yes...but our intuition sucks, so don't rely on it blindly.

Example 14.7. Show that $\int_0^1 \frac{1}{\sqrt{x}}dx$ is finite by computing its value.

Solution 14.7.

The asymptote here occurs at the lower bound, but it's the same principle. No u-sub required here, just power rule:

$$\int_0^1 x^{-1/2}dx = \left[2x^{1/2}\right]_0^1 = 2.$$

This is very easily finite, but what's the difference between this case and the previous one? **The power on x in the denominator here is $1/2 < 1$, while before it was 1, which is ≥ 1.**

Proposition 14.3. The integral $\int_0^a \frac{1}{x^p}dx$ is finite if and only if $p < 1$.

14.3.2 Review of Class Topics

What did we see this semester? Here's an itemized list of topics as a broad starting point to your studying for the final exam. **Also: I accidentally skipped one week in terms of albums!**

Un Verano Sin Ti* by *Bad Bunny

- **Chapter 1: Functions and Graphs**
  1. Simplifying expressions
  2. Domain, range of functions
  3. Types of functions: polynomials/power functions, exponentials, logarithms.
  4. Basic business functions: revenue, cost, profit, and their relationships.
- **Chapter 2: Limits**
  1. Definition of $\lim_{x\to a} f(x)$ ("x approaches a", "x near a", but $x \neq a$)
  2. One sided limits.
  3. Graphical calculation and interpretation of limits.
  4. Algebraic techniques for computing limits (e.g. multiplication by the conjugate, simplifying complex fractions, canceling common factors).
- **Chapter 3: Derivatives**
  1. Average rate of change as slope of secant line.
  2. Limit definition of derivative as limit of slopes of secant lines over interval of length h.
  3. $f'(a)$ is the slope of the tangent line to $f(x)$ at $x = a$.
  4. Rules for computation: power, product, quotient, chain, exponential, logarithm.
  5. Given graph of $f(x)$, identify graph of $f'(x)$.
  6. Marginal (revenue, cost, profit) and their interpretations.
- **Chapter 4: Applications of Derivatives**
  1. Elasticity of Demand
  2. Maxima and minima: Fermat's Theorem, critical points, global vs local.
  3. Graph shape from derivatives: increasing, decreasing, concavity
  4. Applied (Constrained) optimization problems
- **Chapter 5: Integration**
  1. Antiderivatives
  2. Approximating areas: left and right endpoint approximations
  3. Definition of the definite integral (limit of Riemann sum)
  4. Fundamental Theorem of Calculus (part 1 AND part 2)
  5. u-substitution